100 Mathematical Definitions

Textbook formulations of definitions_100_v2 · astra

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About this collection

The entries below give textbook formulations of all 100 items in `definitions_100_v2.md`, in their original order and with their original numbers. Each entry specifies its mathematical setting and gives the definition together with the local background needed to make it precise. The brief context lines indicate material assumed from earlier chapters. References to other numbered entries refer to this collection.

The source list contains concise descriptions rather than full definitions. Here hypotheses, equivalence relations, topology, and normalizations are made explicit. Where terminology has several established meanings, the intended convention is stated; material differences between related notions are retained. Auxiliary theorems are stated without proof when they justify existence, independence of choices, or an equivalent formulation. They are not additional items in the list of 100.

Unless specified otherwise, manifolds and tensor fields are smooth. Further regularity, characteristic, finiteness and compactness assumptions are given in the relevant entry. Definitions concerning a particular construction are made in the setting needed for that construction, rather than asserted in greater generality without hypotheses. The entries are intended as local passages in their respective subject textbooks, not as a sequential introductory course.

Algebra, representation theory & homotopy theory

1. Real rank of a semisimple Lie group

Context. The structure theory of real semisimple Lie groups uses Cartan decompositions to measure the noncompact directions in a group.

Definition. Let G be a real Lie group with finite-dimensional semisimple Lie algebra $\mathfrak{g}$. Choose a Cartan involution θ of $\mathfrak{g}$: this is an involutive Lie algebra automorphism for which $-B(X, \theta Y)$ is positive definite, where B is the Killing form. Write

$$\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}, \qquad \mathfrak{k} = \ker(\theta - 1), \quad \mathfrak{p} = \ker(\theta + 1).$$

A *maximal abelian subspace* of $\mathfrak{p}$ is a real linear subspace $\mathfrak{a} \subseteq \mathfrak{p}$ satisfying $[\mathfrak{a}, \mathfrak{a}] = 0$ and maximal under inclusion with this property. The **real rank** of G is

$$\operatorname{rank}_{\mathbb{R}} G = \dim_{\mathbb{R}} \mathfrak{a}.$$

The conjugacy theorem for Cartan involutions and maximal abelian subspaces implies that this integer is independent of both choices. It depends only on $\mathfrak{g}$, so it is unchanged by a covering homomorphism or a quotient by a discrete central subgroup.

When G is connected with finite centre and K is the maximal compact subgroup associated with θ, this is also the maximum dimension of a complete totally geodesic Euclidean submanifold of G/K. Such a submanifold is called a *flat*. If $\mathbf{G}$ is a connected semisimple algebraic group over $\mathbb{R}$, the real rank of $\mathbf{G}(\mathbb{R})^{0}$ equals the dimension of a maximal $\mathbb{R}$-split algebraic torus of $\mathbf{G}$.

For example, $\operatorname{rank}_{\mathbb{R}} \mathrm{SL}_n(\mathbb{R}) = n - 1$, whereas a compact semisimple group has real rank zero. Real rank is additive under direct products; requiring every noncompact simple factor to have rank at least two is stronger than requiring the total real rank to be at least two.

2. A p-block of a finite group

Context. Block theory decomposes modular representation theory into indecomposable summands and groups ordinary characters according to these summands.

Let G be a finite group and p a prime. Fix a **splitting p-modular system** $(K, \mathcal{O}, k)$: $\mathcal{O}$ is a complete discrete valuation ring with fraction field K of characteristic zero and residue field k of characteristic p, and K and k are splitting fields for the subgroups of G. Ordinary irreducible characters are taken over K, identified with complex characters through the roots of unity occurring in their values.

Definition. A **p-block**, or **Brauer block**, of G is a primitive central idempotent $b \in \mathcal{O}G$. Thus $b \neq 0$, $b^2 = b$, b commutes with every element of $\mathcal{O}G$, and b cannot be written as a sum of two nonzero orthogonal central idempotents. The corresponding **block algebra** is $\mathcal{O}Gb$.

Reduction modulo the maximal ideal of $\mathcal{O}$ gives a bijection between primitive central idempotents of $\mathcal{O}G$ and of kG. Consequently one may equivalently define the block by its reduction $\bar{b} \in kG$, with modular block algebra $kG\bar{b}$.

For $\chi \in \operatorname{Irr}(G)$, extend χ K-linearly to KG. The ordinary characters in b form the set

$$\operatorname{Irr}(G, b) = \{\chi \in \operatorname{Irr}(G) : \chi(b) \neq 0\}.$$

Equivalently, b acts as the identity on the irreducible representation affording χ; otherwise it acts as zero. In particular, $\chi(b)$ is either $\chi(1)$ or zero.

The block idempotents are pairwise orthogonal and sum to 1. They therefore partition $\mathrm{Irr}(G)$, and their reductions similarly partition the simple kG-modules. A simple modular representation belongs to b when $\bar{b}$ acts as the identity. The **principal block** is the block containing the trivial representation.

3. Defect group of a block

Context. Defect groups describe the local p-subgroup information carried by a block of a finite group.

Let G be finite, fix the splitting p-modular system of Definition 2, and let $b \in Z(\mathcal{O}G)$ be a block idempotent. The conjugation action of G on $\mathcal{O}G$ has fixed subalgebras

$$(\mathcal{O}G)^H = \{a \in \mathcal{O}G : hah^{-1} = a \text{ for every } h \in H\}$$

for $H \leq G$. For $Q \leq H$, the **relative trace** is the linear map

$$\mathrm{Tr}_Q^H : (\mathcal{O}G)^Q \longrightarrow (\mathcal{O}G)^H, \qquad a \longmapsto \sum_{h \in H/Q} hah^{-1},$$

where the sum uses one representative for each left coset. The value is independent of these representatives.

Definition. A **defect group** of b is a p-subgroup $D \leq G$ that is minimal under inclusion among the p-subgroups satisfying

$$b \in \mathrm{Tr}_D^G((\mathcal{O}G)^D).$$

Such subgroups exist and form one G-conjugacy class. If $|D| = p^d$, the integer d is the **defect** of b. The same defect groups are obtained by using kG and $\bar{b}$ in the relative-trace definition.

There is a useful equivalent description entirely in characteristic p. For a p-subgroup $Q \leq G$, the **Brauer homomorphism** is

$$\mathrm{Br}_Q : (kG)^Q \longrightarrow kC_G(Q), \qquad \sum_{g \in G} a_g g \longmapsto \sum_{g \in C_G(Q)} a_g g.$$

Then D is a defect group of b precisely when it is maximal among the p-subgroups Q with $\mathrm{Br}_Q(\bar{b}) \neq 0$. In particular, the principal block has Sylow p-subgroups as its defect groups. A block has defect zero when its defect groups are trivial.

4. Parabolic subgroups of a real semisimple or reductive group

Context. Restricted roots and the Iwasawa decomposition describe the parabolic subgroups used in the geometry and representation theory of real groups.

Let G be a connected real semisimple Lie group with finite centre. Fix a Cartan decomposition $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$, its maximal compact subgroup K, and a maximal abelian subspace $\mathfrak{a} \subseteq \mathfrak{p}$. Let $\Sigma \subset \mathfrak{a}^*$ be the restricted roots, with root spaces

$$\mathfrak{g}^\alpha = \{X : [H, X] = \alpha(H)X \text{ for every } H \in \mathfrak{a}\}.$$

Choose positive roots Σ^+ and simple roots Δ. Set

$$M = Z_K(\mathfrak{a}), \qquad A = \exp \mathfrak{a}, \qquad \mathfrak{n} = \bigoplus_{\alpha \in \Sigma^+} \mathfrak{g}^\alpha, \qquad N = \exp \mathfrak{n}.$$

Definition. The subgroup $P_0 = MAN$ is the **standard minimal parabolic subgroup**. Its conjugates are the minimal parabolic subgroups of G. A **parabolic subgroup** is a closed subgroup containing a minimal parabolic subgroup; it is **standard** if it contains P_0.

Explicitly, for $I \subseteq \Delta$, put $\Sigma_I = \Sigma \cap \operatorname{span}_{\mathbb{R}} I$ and $\mathfrak{m} = Z_{\mathfrak{k}}(\mathfrak{a})$. The standard parabolics are exactly

$$P_I = N_G(\mathfrak{q}_I), \qquad \mathfrak{q}_I = \mathfrak{m} \oplus \mathfrak{a} \oplus \bigoplus_{\alpha \in \Sigma^+ \cup \Sigma_I} \mathfrak{g}^\alpha.$$

Here $P_\emptyset = P_0$ and $P_\Delta = G$.

For the **Langlands decomposition**, let

$$\mathfrak{a}_I = \bigcap_{\alpha \in I} \ker \alpha, \quad A_I = \exp \mathfrak{a}_I, \quad N_I = \exp\left(\bigoplus_{\alpha \in \Sigma^+ \setminus \Sigma_I} \mathfrak{g}^\alpha \right).$$

The Levi subgroup $L_I = Z_G(\mathfrak{a}_I)$ has a distinguished factor M_I with $L_I = M_I A_I$, giving $P_I = M_I A_I N_I$. One can specify M_I as the intersection of the kernels of all continuous homomorphisms $L_I \to \mathbb{R}_{>0}$. Multiplication $M_I \times A_I \times N_I \to P_I$ is a diffeomorphism.

For a connected reductive algebraic group $\mathbf{G}/\mathbb{R}$ and $G = \mathbf{G}(\mathbb{R})$ or $\mathbf{G}(\mathbb{R})^0$, a real parabolic subgroup means $G \cap \mathbf{P}(\mathbb{R})$ for an $\mathbb{R}$-defined algebraic parabolic $\mathbf{P} \subseteq \mathbf{G}$. This agrees with the preceding construction in the connected semisimple case and includes the split central directions in the reductive case.

5. Infinitesimal character and Casimir eigenvalue

Context. The centre of the universal enveloping algebra supplies scalar invariants of irreducible representations of real reductive groups.

Let G be a real reductive group, K a maximal compact subgroup, $\mathfrak{g}_0 = \operatorname{Lie}(G)$, and $\mathfrak{g} = \mathfrak{g}_0 \otimes_{\mathbb{R}} \mathbb{C}$. Write $Z(\mathfrak{g}) = Z(U(\mathfrak{g}))$.

Definition. A nonzero $U(\mathfrak{g})$-module V has **infinitesimal character** $\chi : Z(\mathfrak{g}) \to \mathbb{C}$ if χ is a unital algebra homomorphism and

$$z v = \chi(z) v \qquad (z \in Z(\mathfrak{g}),\ v \in V).$$

For a smooth representation of G, use the differentiated action of $U(\mathfrak{g})$. For an admissible representation, one may equivalently use its Harish-Chandra module of K-finite smooth vectors. Every irreducible Harish-Chandra module has an infinitesimal character.

Fix a Cartan subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$, Weyl group W, and positive roots with half-sum ρ. We use the shifted Harish-Chandra isomorphism

$$\gamma : Z(\mathfrak{g}) \xrightarrow{\sim} \mathbb{C}[\mathfrak{h}^*]^W$$

normalized so that the module of highest weight μ has character $z \mapsto \gamma(z)(\mu + \rho)$. Consequently an infinitesimal character is equivalently a Weyl orbit $W\lambda \in \mathfrak{h}^*/W$, with $\chi_\lambda(z) = \gamma(z)(\lambda)$. The trivial representation has parameter $W\rho$.

If B is a nondegenerate invariant symmetric bilinear form on $\mathfrak{g}$, and (X_i) and (X^i) are dual bases for B, the **Casimir element** is

$$\Omega_B = \sum_i X_i X^i \in Z(\mathfrak{g}).$$

It is independent of the basis. On a representation with infinitesimal character χ_λ, its **Casimir eigenvalue** is

$$\chi_\lambda(\Omega_B) = \langle \lambda, \lambda \rangle_B - \langle \rho, \rho \rangle_B,$$

where the pairing on $\mathfrak{h}^*$ is dual to $B|_{\mathfrak{h}}$. The form B and the sign of the chosen Casimir must therefore be specified whenever a numerical eigenvalue is used.

6. Convergence of a spectral sequence

Context. Convergence identifies the eventual terms of a spectral sequence with successive quotients of the cohomology it computes.

Consider a cohomological spectral sequence of modules with

$$d_r : E_r^{p,q} \longrightarrow E_r^{p+r,q-r+1}.$$

Suppose each position eventually stabilizes: for each (p,q) there is an $r(p,q)$ after which both differentials entering and leaving that position vanish. The transition identifications then define its stable value $E_\infty^{p,q}$.

Definition (filtered convergence). The spectral sequence **converges to** a graded module $H^\bullet$, written

$$E_r^{p,q} \Longrightarrow H^{p+q},$$

if H^n is equipped with an exhaustive, separated decreasing filtration F^pH^n and specified isomorphisms

$$E_\infty^{p,q} \xrightarrow{\sim} \mathrm{gr}_F^p H^{p+q} = F^pH^{p+q}/F^{p+1}H^{p+q}.$$

Here **exhaustive** and **separated** mean, respectively,

$$\bigcup_p F^pH^n = H^n, \qquad \bigcap_p F^pH^n = 0.$$

We call this filtered convergence **strong** when the filtration is additionally **complete**, meaning

$$H^n \xrightarrow{\sim} \varprojlim_{p\to+\infty} H^n/F^pH^n.$$

This is a convention for the degreewise stabilized spectral sequences specified above. For nonstabilizing spectral sequences, convergence theory can require further derived-limit conditions; no definition for that setting is implicit here.

The filtration is **finite in each total degree** if, for each n, there exist integers $a_n < b_n$ with $F^{a_n}H^n = H^n$ and $F^{b_n}H^n = 0$. Such filtrations are automatically exhaustive, separated and complete, so the two convergence conventions coincide. Convergence **in a range** $I \subseteq \mathbb{Z}$ imposes the chosen conditions only for total degrees $n \in I$.

For example, a cochain complex with a decreasing filtration by subcomplexes that is finite in each degree has a spectral sequence

$$E_1^{p,q} = H^{p+q}(F^pC^\bullet/F^{p+1}C^\bullet) \Longrightarrow H^{p+q}(C^\bullet).$$

The abutment filtration is the image of $H^n(F^pC^\bullet)$ in $H^n(C^\bullet)$. Convergence determines the associated graded module; reconstructing H^n can still require solving extension problems.

7. Discrete series representations of $\mathrm{GL}_2(\mathbb{R})$

Context. In the representation theory of real groups, discrete series are characterized by square-integrability of matrix coefficients after removing the centre.

Let $G = \mathrm{GL}_2(\mathbb{R})$ and $Z = \mathbb{R}^\times I_2$. Fix a Haar measure $d\dot{g}$ on G/Z.

Definition. An irreducible unitary representation $(\pi, \mathcal{H})$ of G is a **discrete series representation**, or is **square-integrable modulo the centre**, if

$$\int_{G/Z} |\langle \pi(g)v, w\rangle|^2 \, d\dot{g} < \infty \qquad (v, w \in \mathcal{H}).$$

The integrand is well defined on G/Z because the central character is unitary. For an irreducible unitary representation it is equivalent to require this for one pair of nonzero vectors. The definition is independent of the normalization of Haar measure.

An irreducible smooth admissible representation π is **essentially discrete series** if, for some continuous character $\chi : \mathbb{R}^\times \to \mathbb{C}^\times$, the representation $\pi \otimes (\chi \circ \det)$ is the smooth representation associated with a discrete series unitary representation. The analogous definition for Harish-Chandra modules uses infinitesimal equivalence.

For each integer $k \geq 2$, let D_k denote the unitary discrete series whose central character is $zI_2 \mapsto \operatorname{sgn}(z)^k$ and whose restriction to $\mathrm{SO}(2)$ has precisely the weights

$$\pm k,\ \pm(k+2),\ \pm(k+4), \dots,$$

each with multiplicity one. Here weight m means the character of a rotation through θ given by $e^{im\theta}$. The classification gives all discrete series as $D_k \otimes |\det|^{it}$, $t \in \mathbb{R}$, and all essentially discrete series as $D_k \otimes |\det|^{s}$, $s \in \mathbb{C}$. Their **weight** is k. The representations at the limiting weight 1 are not square-integrable modulo the centre.

8. Glauberman correspondence

Context. Coprime group actions relate invariant irreducible characters to the character theory of the fixed-point subgroup.

Let a finite solvable group S act by automorphisms on a finite group Γ, and assume $\gcd(|S|, |\Gamma|) = 1$. Put

$$C_\Gamma(S) = \{g \in \Gamma : s(g) = g \text{ for every } s \in S\},$$

and let $\mathrm{Irr}_S(\Gamma)$ be the set of complex irreducible characters satisfying $\chi(s(g)) = \chi(g)$ for every $s \in S$ and $g \in \Gamma$.

Theorem and definition. The **Glauberman correspondence** is the canonical bijection

$$\mathrm{Irr}_S(\Gamma) \xrightarrow{\sim} \mathrm{Irr}(C_\Gamma(S)), \qquad \chi \longmapsto \chi^*,$$

specified by the following construction.

When S is a p-group, χ^* is the unique irreducible constituent of $\chi|_{C_\Gamma(S)}$ whose multiplicity is not divisible by p. In terms of the usual inner product of characters, it is characterized by

$$p \nmid \langle \chi|_{C_\Gamma(S)}, \chi^* \rangle_{C_\Gamma(S)}.$$

For general solvable S, choose a normal series

$$1 = S_0 \lhd S_1 \lhd \cdots \lhd S_r = S$$

with each S_i normal in S and each factor S_i/S_{i-1} a group of prime-power order. Starting with $\chi_0 = \chi$, apply the preceding prime-power construction successively to the action of S_i/S_{i-1} on $C_\Gamma(S_{i-1})$, producing $\chi_i \in \mathrm{Irr}(C_\Gamma(S_i))$. At each step the required invariance holds. Define $\chi^* = \chi_r$.

Glauberman's theorem asserts that the prime-power constituent exists uniquely, that the resulting map is bijective, and that the general construction is independent of the chosen normal series. Thus "canonical" refers to this specified correspondence, rather than to an arbitrary bijection between the two character sets. See Glauberman, *Correspondences of characters for relatively prime operator groups*.

9. Height-zero characters of a block

Context. Character heights compare ordinary character degrees with the size of a block's defect groups.

Let b be a p-block of a finite group G, as in Definition 2, and let D be a defect group of b, as in Definition 3. Write $|G|_p = p^a$ and $|D| = p^d$, where $m_p = p^{v_p(m)}$ denotes the largest power of p dividing a positive integer m.

Definition. The **height** of $\chi \in \mathrm{Irr}(G, b)$ is

$$h_b(\chi) = v_p(\chi(1)) - (a - d).$$

Equivalently,

$$\chi(1)_p = p^{h_b(\chi)} |G : D|_p.$$

The block-theoretic divisibility theorem gives $h_b(\chi) \geq 0$. Since defect groups of b are conjugate, this integer is independent of the chosen D.

A character has **height zero** if $h_b(\chi) = 0$. The set of height-zero characters of b is therefore

$$\mathrm{Irr}_0(G, b) = \{\chi \in \mathrm{Irr}(G, b) : \chi(1)_p = |G : D|_p\}.$$

It is also written $\mathrm{Irr}_0(b)$. The definition refers to ordinary irreducible characters in the block, not to irreducible Brauer characters. In the principal block, D is a Sylow p-subgroup, so height zero is equivalent to $p \nmid \chi(1)$.

10. Hochschild–Kostant–Rosenberg antisymmetrization

Context. Hochschild cohomology of a smooth commutative algebra is represented by alternating products of derivations.

Let k be a field and R a commutative k-algebra. Its Hochschild cochain complex has

$$C^n(R, R) = \mathrm{Hom}_k(R^{\otimes_k n}, R)$$

and differential

$$\begin{aligned}(bf)(a_1, \dots, a_{n+1}) = {} & a_1 f(a_2, \dots, a_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i f(a_1, \dots, a_i a_{i+1}, \dots, a_{n+1}) \\ & + (-1)^{n+1} f(a_1, \dots, a_n) a_{n+1}.\end{aligned}$$

Write $HH^n(R, R) = H^n(C^\bullet(R, R), b)$ and

$$T^n_{R/k} = \bigwedge_R^n \mathrm{Der}_k(R, R).$$

Definition. For $n \geq 0$ with $n!$ invertible in k, the **HKR antisymmetrization map** is the R-linear map

$$\mathrm{HKR}_n : T^n_{R/k} \longrightarrow Z^n C^\bullet(R, R)$$

given on decomposable polyvectors by

$$\mathrm{HKR}_n(D_1 \wedge \cdots \wedge D_n)(a_1, \dots, a_n) = \frac{1}{n!} \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \prod_{i=1}^{n} D_{\sigma(i)}(a_i).$$

For $n = 0$ this is the identity on R. Alternation makes the formula factor through the exterior power, and the Leibniz rule implies that its values are Hochschild cocycles.

HKR theorem. If R is smooth of finite type over k and $n! \in k^\times$, passage to cohomology gives an isomorphism

$$T^n_{R/k} \xrightarrow{\sim} HH^n(R, R).$$

In characteristic zero these maps in all degrees give a quasi-isomorphism from the graded module of polyvectors with zero differential to the Hochschild cochain complex. In characteristic p, the displayed normalized formula is used only in degrees $n < p$; it does not define a map in degrees for which $n!$ vanishes.

11. Split symplectic, special orthogonal and general spin groups

Context. Classical reductive groups are defined by forms and Clifford algebras, with split forms providing their standard models.

Let k be a field of characteristic different from 2; every displayed group is an algebraic group over k, specified on commutative k-algebras R.

Definition (symplectic group). For $n \geq 1$ and $J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$, the **split symplectic group** is

$$\mathrm{Sp}_{2n}(R) = \{g \in \mathrm{GL}_{2n}(R) : g^{\mathsf{T}} J g = J\}.$$

It has rank n.

Definition (special orthogonal group). On $V = k^m$, use the quadratic form

$$q_{2n}(x, y) = \sum_{i=1}^{n} x_i y_i, \qquad q_{2n+1}(x, y, z) = \sum_{i=1}^{n} x_i y_i + z^2.$$

For $m \geq 2$, the **split special orthogonal group** is

$$\mathrm{SO}_m(R) = \{g \in \mathrm{GL}(V_R) : q_R(gv) = q_R(v) \text{ for all } v,\ \det g = 1\}.$$

The equality of quadratic forms is understood as a polynomial identity. Its rank is $\lfloor m/2 \rfloor$.

Definition (spin and general spin groups). Let $C(V, q)$ be the Clifford algebra, with relations $v^2 = q(v)1$, let $C^+(V, q)$ be its even part, and let τ be reversion, the anti-automorphism fixing V. Define the **general spin group** by

$$\mathrm{GSpin}(V, q)(R) = \{x \in (C^+(V, q) \otimes_k R)^\times : xV_R x^{-1} = V_R,\ \tau(x)x \in R^\times\}.$$

Its **Clifford norm** is $\nu(x) = \tau(x)x$. The **spin group** is $\mathrm{Spin}(V, q) = \ker \nu$. For the displayed split forms, write these groups as GSpin_m and Spin_m.

Equivalently, as algebraic groups,

$$\mathrm{GSpin}_m \cong (\mathbb{G}_m \times \mathrm{Spin}_m)/\mu_2,$$

where $\epsilon \in \mu_2$ embeds as (ϵ, ϵ). This is the quotient in the fppf topology; taking rational points need not turn it into a quotient of the corresponding rational-point groups. Conjugation gives the central extension

$$1 \longrightarrow \mathbb{G}_m \longrightarrow \mathrm{GSpin}_m \longrightarrow \mathrm{SO}_m \longrightarrow 1,$$

and GSpin_m has rank $\lfloor m/2 \rfloor + 1$.

Algebraic geometry

12. The Hodge structure on H^1 of a complex abelian variety

Context. The complex structure and a polarization of an abelian variety induce a polarizable rational Hodge structure on its first cohomology.

Let A be a complex abelian variety of dimension $g \geq 1$. Analytically,

$$A(\mathbb{C}) = W/\Lambda,$$

where $W = \mathrm{Lie}(A)$ is a complex vector space of dimension g and $\Lambda \subset W$ is a lattice of rank $2g$. Integration along lattice loops identifies

$$V = H^1(A(\mathbb{C}), \mathbb{Q}) = \mathrm{Hom}_{\mathbb{Z}}(\Lambda, \mathbb{Q}), \qquad V_{\mathbb{C}} = \mathrm{Hom}_{\mathbb{R}}(W, \mathbb{C}).$$

Definition. The **weight-one Hodge structure on** $H^1(A)$ is the decomposition

$$V_{\mathbb{C}} = V^{1,0} \oplus V^{0,1}, \qquad V^{1,0} = \mathrm{Hom}_{\mathbb{C}}(W, \mathbb{C}), \quad V^{0,1} = \overline{V^{1,0}}.$$

Under the de Rham comparison, $V^{1,0}$ is the image of the holomorphic differentials $H^0(A, \Omega^1_A)$ in singular cohomology; $V^{0,1}$ is its complex conjugate. The Hodge filtration is $F^1 V_{\mathbb{C}} = V^{1,0}$, $F^0 V_{\mathbb{C}} = V_{\mathbb{C}}$, and $F^2 V_{\mathbb{C}} = 0$.

Choose an ample line bundle L on A, and put $\ell = c_1(L) \in H^2(A(\mathbb{C}), \mathbb{Q})$. A rational alternating form polarizing this Hodge structure is

$$Q_L(u, v) = \int_{A(\mathbb{C})} u \smile v \smile \ell^{g-1}.$$

It is nondegenerate, vanishes on $V^{1,0} \times V^{1,0}$ and on $V^{0,1} \times V^{0,1}$, and satisfies

$$iQ_L(\alpha, \bar{\alpha}) > 0 \qquad (0 \neq \alpha \in V^{1,0}).$$

These statements fix the sign convention for the polarization. The Hodge structure is canonical and functorial in A; a particular polarization is additional data obtained from a choice of ample class.

13. Mumford–Tate group of a rational Hodge structure

Context. A Mumford–Tate group records the algebraic symmetries forced by the Hodge decomposition and the rational structure.

Let V be a finite-dimensional rational Hodge structure of pure weight w, so

$$V_{\mathbb{C}} = \bigoplus_{p+q=w} V^{p,q}, \qquad \overline{V^{p,q}} = V^{q,p}.$$

The **Deligne torus** is $\mathbb{S} = \mathrm{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m$, with $\mathbb{S}(\mathbb{R}) = \mathbb{C}^\times$. The Hodge decomposition corresponds to the real algebraic homomorphism

$$h : \mathbb{S} \longrightarrow \mathrm{GL}(V_{\mathbb{R}})$$

whose complexified action on $V^{p,q}$ is multiplication by $z^{-p}\bar{z}^{-q}$ for $z \in \mathbb{C}^\times$.

Definition. The **Mumford–Tate group** $\mathrm{MT}(V)$ is the smallest algebraic subgroup $M \subseteq \mathrm{GL}(V)$ defined over $\mathbb{Q}$ such that h factors through $M_{\mathbb{R}}$. Equivalently, it is the Zariski closure over $\mathbb{Q}$ of $h(\mathbb{C}^\times)$ in $\mathrm{GL}(V)$.

For a complex abelian variety A, set

$$\mathrm{MT}(A) = \mathrm{MT}(H^1(A(\mathbb{C}), \mathbb{Q})),$$

using Definition 12.

The group $\mathrm{MT}(V)$ is connected. If V is polarizable, it is reductive and has the following equivalent tensor description. For $a, b \geq 0$, give

$$T^{a,b}(V) = V^{\otimes a} \otimes (V^\vee)^{\otimes b}$$

its induced Hodge structure. A **Hodge tensor** is a rational tensor $t \in T^{a,b}(V)$ of Hodge type $(0,0)$. Then $\mathrm{MT}(V)$ is precisely the subgroup fixing all such tensors in all these spaces. The tensor action includes the contragredient action on $V^\vee$.

This definition uses $\mathrm{GL}(V)$ without an additional Tate line. If tensors with Tate twists are used, one must also specify the action on that line, equivalently work with $V \oplus \mathbb{Q}(1)$. For a polarized weight-one structure (V, Q), the group lies in $\mathrm{GSp}(V, Q)$: it preserves the polarization up to its similitude character.

14. The volume form ω^n and integration on a compact Kähler manifold

Context. Complex Monge–Ampère equations use the top exterior power of a positive $(1,1)$-form as their reference measure.

Let X be a complex manifold of complex dimension $n \geq 1$, with its complex orientation. A **Hermitian form** is a smooth positive real $(1,1)$-form, written in holomorphic coordinates as

$$\omega = i \sum_{j,k=1}^{n} g_{j\bar{k}}\, dz^j \wedge d\bar{z}^k,$$

where $(g_{j\bar{k}})$ is positive definite Hermitian. It is **Kähler** when $d\omega = 0$.

Definition. The **top power** of ω is the positive real $2n$-form

$$\omega^n = \underbrace{\omega \wedge \cdots \wedge \omega}_{n \text{ factors}}.$$

Writing $z^j = x^j + iy^j$, its coordinate expression is

$$\omega^n = n!\, 2^n \det(g_{j\bar{k}})\, dx^1 \wedge dy^1 \wedge \cdots \wedge dx^n \wedge dy^n.$$

It determines a positive Radon measure μ_ω with this local density relative to Lebesgue measure. For a nonnegative measurable function u, or an integrable real or complex function u, define

$$\int_X u\, \omega^n := \int_X u\, d\mu_\omega.$$

When X is nonempty and compact, its **volume in this convention** is

$$V_\omega = \int_X \omega^n \in (0, \infty).$$

The Riemannian metric associated with ω is $g_\omega(u, v) = \omega(u, Jv)$, and its Riemannian volume form is $\omega^n/n!$. Thus the convention used here has $V_\omega = n!\,\mathrm{Vol}_{g_\omega}(X)$.

The measure construction requires positivity but not closedness, so it applies equally to Hermitian manifolds. Closedness supplies the additional cohomological invariance: on a compact Kähler manifold, $\int_X \omega^n$ depends only on the de Rham class $[\omega]$.

15. A ball quotient as a quasi-projective variety

Context. Finite-volume quotients of complex hyperbolic space carry algebraic structures, including when the defining lattice is non-arithmetic.

Let

$$\mathbb{B}^n = \{z \in \mathbb{C}^n : \|z\| < 1\}, \qquad n \geq 1.$$

Its group of holomorphic automorphisms is $G = \mathrm{PU}(1, n)$, the projective unitary group of a Hermitian form of signature $(1, n)$. A **lattice** $\Gamma \subset G$ is a discrete subgroup for which $\Gamma\backslash G$ has finite invariant Haar measure.

The analytic quotient $\Gamma\backslash\mathbb{B}^n$ has its usual quotient topology and sheaf of invariant holomorphic functions. It is a normal complex space. If Γ is torsion-free, it is a manifold and $\mathbb{B}^n \to \Gamma\backslash\mathbb{B}^n$ is its universal covering.

Algebraicity theorem. This quotient admits a normal projective compactification obtained by adjoining finitely many cusp points. For torsion-free lattices, the arithmetic case follows from Baily–Borel; the general higher-dimensional case follows from the compactification results of Siu–Yau and Mok's projective-algebraicity theorem. Dimension one is the finite-area uniformization of algebraic curves. Finite quotients give the assertion for lattices with torsion.

Definition. The **ball quotient variety** associated with Γ is the normal quasi-projective complex variety

$$S_\Gamma = \Gamma\backslash\mathbb{B}^n$$

whose analytification is the analytic quotient and whose algebraic structure is supplied by this compactification. The identification with the analytic quotient is part of the construction. This algebraic structure is canonical; in particular, replacing Γ by a subgroup of finite index induces an algebraic finite morphism between the corresponding quotients.

When Γ is cocompact, S_Γ is projective. When Γ is torsion-free, S_Γ is smooth. For a lattice with torsion, one retains both its normal quotient variety and, when differential geometry is needed, the quotient orbifold structure, which records the finite stabilizers.

16. Rational Chow groups of a Deligne–Mumford stack and degree

Context. Intersection theory on moduli stacks weights cycles by stabilizers and produces rational-valued enumerative invariants.

Let $\mathcal{M}$ be a separated Deligne–Mumford stack of finite type over a field k. An integral closed substack means a reduced irreducible closed substack. Let $Z_d(\mathcal{M})$ be the free abelian group generated by the integral closed substacks of dimension d.

For an integral closed substack $\mathcal{W}$ of dimension $d+1$ and a nonzero rational function $f \in k(\mathcal{W})^\times$, its principal divisor is

$$\mathrm{div}_{\mathcal{W}}(f) = \sum_{\mathcal{V}} \mathrm{ord}_{\mathcal{V}}(f)[\mathcal{V}],$$

summed over its integral codimension-one substacks. Orders are computed on an étale scheme atlas: pull back f and use the usual order at a codimension-one point. Étale descent makes the coefficient independent of the chosen lift. These divisors are viewed as cycles on $\mathcal{M}$.

Definition. Let $R_d(\mathcal{M})$ be the subgroup generated by these principal divisors. The **rational Chow group** is

$$A_d(\mathcal{M})_{\mathbb{Q}} = (Z_d(\mathcal{M})/R_d(\mathcal{M})) \otimes_{\mathbb{Z}} \mathbb{Q}, \qquad A_*(\mathcal{M})_{\mathbb{Q}} = \bigoplus_{d \geq 0} A_d(\mathcal{M})_{\mathbb{Q}}.$$

This is the rational cycle-theoretic Chow theory for Deligne–Mumford stacks; it agrees with the rationalization of Kresch's Chow groups.

Definition (degree). Suppose $\mathcal{M}$ is proper over k. For a closed point $x \in |\mathcal{M}|$, let $\mathcal{G}_x$ be its residual gerbe, $k(x)$ its residue field, and e_x the order of the finite automorphism group of a geometric point over x. Set

$$\deg[\mathcal{G}_x] = \frac{[k(x) : k]}{e_x}.$$

Extending linearly kills rational equivalences and defines

$$\deg = \int_{\mathcal{M}} : A_0(\mathcal{M})_{\mathbb{Q}} \longrightarrow \mathbb{Q}.$$

Over an algebraically closed field, a point with automorphism group of order e contributes $1/e$. Thus, for a finite constant group H, $\deg[BH] = 1/|H|$.

17. Cohomologically affine morphisms of algebraic stacks

Context. Exactness of pushforward replaces the usual affine-morphism condition in the theory of good moduli spaces.

Let $f : \mathcal{X} \to \mathcal{Y}$ be a quasi-compact and quasi-separated morphism of algebraic stacks. Pushforward restricts to a left exact functor

$$f_* : \mathrm{QCoh}(\mathcal{X}) \longrightarrow \mathrm{QCoh}(\mathcal{Y}).$$

Definition. The morphism f is **cohomologically affine** if this functor is exact. Explicitly, every short exact sequence of quasi-coherent $\mathcal{O}_{\mathcal{X}}$-modules

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

must give a short exact sequence

$$0 \longrightarrow f_*\mathcal{F}' \longrightarrow f_*\mathcal{F} \longrightarrow f_*\mathcal{F}'' \longrightarrow 0.$$

Equivalently, f_* must carry every surjection of quasi-coherent modules to a surjection. Quasi-compactness and quasi-separatedness are included in the convention used here. See Alper's treatment of adequate and cohomologically affine morphisms, §4.1.

For example, if H is an affine algebraic group over a field k, quasi-coherent sheaves on BH correspond to rational H-representations, and pushforward along $BH \to \operatorname{Spec} k$ is the invariants functor $V \mapsto V^H$. Thus $BH \to \operatorname{Spec} k$ is cohomologically affine precisely when H is **linearly reductive**, meaning that invariants form an exact functor.

For a morphism of schemes, cohomological affineness is equivalent to affineness. The stack definition is broader: BH for a nontrivial linearly reductive group is a basic example with nontrivial stabilizers.

18. Cup product in étale cohomology

Context. Cup products supply the multiplicative structure and duality pairings in étale cohomology of schemes and analytic spaces.

Let X be a scheme, or a rigid-analytic space viewed as an adic space with its étale site $X_{\text{ét}}$. Let Λ be a commutative coefficient ring and $\mathcal{F}, \mathcal{G}$ sheaves of Λ-modules on this site.

Definition. The **cup product** is the natural bilinear pairing

$$\smile : H^i_{\text{ét}}(X, \mathcal{F}) \otimes_\Lambda H^j_{\text{ét}}(X, \mathcal{G}) \longrightarrow H^{i+j}_{\text{ét}}(X, \mathcal{F} \otimes_\Lambda \mathcal{G}).$$

Precisely, it is induced by the canonical composite in the derived category

$$R\Gamma(X,\mathcal{F}) \otimes^{\mathbf{L}}_{\Lambda} R\Gamma(X,\mathcal{G}) \longrightarrow R\Gamma(X,\mathcal{F}\otimes^{\mathbf{L}}_{\Lambda}\mathcal{G}) \longrightarrow R\Gamma(X,\mathcal{F}\otimes_{\Lambda}\mathcal{G}).$$

Given a pairing $m:\mathcal{F}\otimes_{\Lambda}\mathcal{G}\to\mathcal{H}$, its cup product with values in $\mathcal{H}$ is obtained by composing with $H^{i+j}(m)$.

The product is associative and natural under pullback. If τ exchanges the two coefficient factors, then

$$\tau_*(\alpha\smile\beta) = (-1)^{ij}\beta\smile\alpha.$$

In particular, the cohomology of a commutative sheaf of rings is a graded-commutative ring.

If the prime p is invertible on X, let $\mathbb{F}_p(1)=\mu_p$, and let $\mathbb{F}_p(a)$ be its ath tensor power, using duals for negative a. Tensor multiplication gives

$$H^i(X,\mathbb{F}_p(a))\otimes_{\mathbb{F}_p} H^j(X,\mathbb{F}_p(b))\longrightarrow H^{i+j}(X,\mathbb{F}_p(a+b)).$$

This includes the pairings used in mod-p duality for rigid spaces over fields of characteristic zero, even when their residue characteristic is p.

19. Enriques surfaces over an algebraically closed field

Context. The numerical definition of an Enriques surface accommodates the different types that occur in characteristic two.

Let k be an algebraically closed field of arbitrary characteristic. For a smooth proper surface Y/k, its canonical line bundle is

$$\omega_Y = \bigwedge^2 \Omega^1_{Y/k}.$$

A line bundle L on Y is **numerically trivial** if $\deg(L|_C)=0$ for every integral proper curve $C\subseteq Y$. Equivalently, its intersection number with every curve is zero.

Choose a prime $\ell\neq\operatorname{char}k$. The **second Betti number** is

$$b_2(Y)=\dim_{\mathbb{Q}_\ell} H^2_{\text{ét}}(Y,\mathbb{Q}_\ell).$$

For smooth proper varieties this number is independent of ℓ.

Definition. An **Enriques surface** over k is a smooth proper connected k-scheme Y of dimension two such that ω_Y is numerically trivial and $b_2(Y)=10$. Connectedness here is equivalent to $H^0(Y,\mathcal{O}_Y)=k$; since Y is smooth, it also makes Y integral.

In characteristic different from two, the canonical line bundle of an Enriques surface is non-trivial of order two and $H^1(Y,\mathcal{O}_Y)=H^2(Y,\mathcal{O}_Y)=0$. These properties explain the familiar characteristic-zero description. In characteristic two the canonical bundle can be trivial, and the coherent cohomology can differ; the numerical definition above includes these surfaces as well. Numerical triviality is a condition on intersection numbers and does not mean that the canonical line bundle has been supplied with a trivialization.

20. Families and the moduli stack of Enriques surfaces

Context. To study Enriques surfaces over varying characteristic, one uses families of algebraic spaces and their moduli stack over $\operatorname{Spec}\mathbb{Z}$.

Definition. A **family of Enriques surfaces over a scheme** S is an algebraic space $\mathcal{Y}$ with a flat proper morphism of finite presentation

$$f:\mathcal{Y}\longrightarrow S$$

such that, for every geometric point $\bar{s} : \operatorname{Spec}\Omega \to S$ with Ω algebraically closed, the fibre

$$\mathcal{Y}_{\bar{s}} = \mathcal{Y} \times_S \operatorname{Spec}\Omega$$

is an Enriques surface in the sense of Definition 19. In particular, these fibres are schemes, although the total space is allowed to be an algebraic space. The morphism f is automatically smooth, because it is flat and locally of finite presentation with geometrically smooth fibres.

An **isomorphism of families** over S is an S-isomorphism of their total spaces. For a map $T \to S$, pullback is the family $\mathcal{Y} \times_S T \to T$.

Definition. The **moduli stack of Enriques surfaces**, denoted $\mathcal{M}_{\mathrm{Enr}}$, is the stack in groupoids on schemes over $\mathbb{Z}$, for the fppf topology, whose groupoid over S consists of these families and their S-isomorphisms. More explicitly, a morphism of objects lying over $T \to S$ is an isomorphism from the family over T to the pullback of the family over S.

For a ring R, the notation $\mathcal{M}_{\mathrm{Enr}}(R)$ means the groupoid of families over $\operatorname{Spec} R$. Saying this groupoid is empty means that no such family exists. No polarization, marking, or choice of canonical cover is included in the moduli problem.

21. Fppf cohomology with supports in a closed subset

Context. Cohomology with supports isolates the part of flat cohomology concentrated along a specified closed locus.

Let X be a scheme, $Z \subseteq |X|$ a closed subset, and $j : U = X \setminus Z \hookrightarrow X$ the complementary open immersion. Let $\mathcal{F}$ be an abelian sheaf on the fppf site of X; a commutative group scheme over X supplies such a sheaf by its functor of points.

Definition. The group of **sections supported in** Z is

$$\Gamma_Z(X, \mathcal{F}) = \ker(\mathcal{F}(X) \longrightarrow \mathcal{F}(U)).$$

This is a left exact functor on abelian fppf sheaves. Its right derived functors are the **fppf cohomology groups with supports in** Z:

$$H^i_Z(X, \mathcal{F}) = R^i\Gamma_Z(X, \mathcal{F}), \qquad i \geq 0.$$

Thus, for an injective resolution $\mathcal{F} \to I^\bullet$, one has $H^i_Z(X, \mathcal{F}) = H^i(\Gamma_Z(X, I^\bullet))$.

Equivalently, the derived sections with supports are characterized by the localization triangle

$$R\Gamma_Z(X, \mathcal{F}) \longrightarrow R\Gamma_{\mathrm{fppf}}(X, \mathcal{F}) \longrightarrow R\Gamma_{\mathrm{fppf}}(U, j^*\mathcal{F}) \longrightarrow R\Gamma_Z(X, \mathcal{F})[1].$$

In particular there is a natural long exact sequence

$$\cdots \longrightarrow H^i_Z(X, \mathcal{F}) \longrightarrow H^i_{\mathrm{fppf}}(X, \mathcal{F}) \longrightarrow H^i_{\mathrm{fppf}}(U, j^*\mathcal{F}) \longrightarrow H^{i+1}_Z(X, \mathcal{F}) \longrightarrow \cdots.$$

The definition depends only on the closed subset Z, not on any scheme structure placed on it. For a local ring $(R, \mathfrak{m})$ one writes $H^i_{\mathfrak{m}}(R, G)$ for $H^i_{\{\mathfrak{m}\}}(\operatorname{Spec} R, G)$; the complementary open is the punctured spectrum.

22. Galois action on étale cohomology of a geometric fibre

Context. Descent from a base field equips geometric étale cohomology with a natural action of its absolute Galois group.

Let X be a scheme over a field K, choose a separable closure K^{sep}, and write $G_K = \operatorname{Gal}(K^{\mathrm{sep}}/K)$. Set $X_{\overline{K}} = X \times_K K^{\mathrm{sep}}$ for the purpose of étale cohomology. For $\sigma \in G_K$, let

$$a_\sigma = \mathrm{id}_X \times \operatorname{Spec}(\sigma) : X_{\overline{K}} \longrightarrow X_{\overline{K}}.$$

Since Spec is contravariant, $a_{\sigma\tau} = a_\tau \circ a_\sigma$.

Definition. Let $\mathcal{F}$ be an abelian étale sheaf on X, with pullback $\mathcal{F}_{\overline{K}}$. The **Galois action** on geometric cohomology is

$$\rho(\sigma) : H^i_{\text{ét}}(X_{\overline{K}}, \mathcal{F}_{\overline{K}}) \xrightarrow{a_\sigma^*} H^i_{\text{ét}}(X_{\overline{K}}, a_\sigma^{-1}\mathcal{F}_{\overline{K}}) \xrightarrow{\sim} H^i_{\text{ét}}(X_{\overline{K}}, \mathcal{F}_{\overline{K}}),$$

where the last arrow is the canonical identification coming from descent of $\mathcal{F}$ to X. The two contravariances imply $\rho(\sigma\tau) = \rho(\sigma)\rho(\tau)$, so this is a left action.

For a rigid-analytic space X over a complete non-archimedean field K, put $C = \widehat{\overline{K}}$. Every $\sigma \in G_K$ extends uniquely to an isometric automorphism of C. Replace $X_{\overline{K}}$ by X_C and $\mathrm{Spec}(\sigma)$ by $\mathrm{Spa}(\sigma)$ in the construction. This defines the natural action on Huber étale cohomology $H^i_{\text{ét}}(X_C, \mathcal{F}_C)$.

In particular, over a field of characteristic zero it applies to $\mathbb{F}_p(j) = \mu_p^{\otimes j}$, with dual tensor powers for $j < 0$. The descent action on μ_p is the cyclotomic action; the Tate twist includes this coefficient action. Cup products are Galois-equivariant.

For smooth proper rigid-analytic X over a field of mixed characteristic $(0, p)$, the groups $H^i_{\text{ét}}(X_C, \mathbb{F}_p(j))$ are finite and the action is continuous for the discrete topology; see Zavyalov, *Mod-p Poincaré duality in p-adic analytic geometry*, Theorem 5.1.2. Compatible finite-coefficient actions likewise induce continuous ℓ-adic actions on their inverse limits, equipped with the inverse-limit topology, and on the resulting finite-dimensional $\mathbb{Q}_\ell$-spaces with their ℓ-adic topology.

23. Hodge-generic abelian varieties

Context. Within the moduli of polarized abelian varieties, Hodge-genericity means that the Mumford–Tate group is as large as the polarization permits.

Let $A/\mathbb{C}$ be an abelian variety of dimension $g \geq 1$, let $V = H^1(A(\mathbb{C}), \mathbb{Q})$ have the Hodge structure of Definition 12, and choose a polarization form Q on V. The **group of symplectic similitudes** is the algebraic group with

$$\mathrm{GSp}(V, Q)(R) = \{u \in \mathrm{GL}(V \otimes_{\mathbb{Q}} R) : Q(ux, uy) = cQ(x, y) \text{ for some } c \in R^\times\}$$

for every $\mathbb{Q}$-algebra R. A symplectic basis identifies it with GSp_{2g}.

Definition. The abelian variety A is **Hodge-generic**, or has **maximal Mumford–Tate group**, if

$$\mathrm{MT}(A) = \mathrm{GSp}(V, Q)$$

as algebraic subgroups of $\mathrm{GL}(V)$, with $\mathrm{MT}(A)$ as in Definition 13. Since $\mathrm{MT}(A)$ is contained in $\mathrm{GSp}(V, Q)$, this is equivalent to

$$\dim \mathrm{MT}(A) = 2g^2 + g + 1.$$

The property is independent of the chosen polarization. For an abelian variety over a subfield $F \subseteq \mathbb{C}$, use its base change to $\mathbb{C}$; in particular this defines the notion for $A/\overline{\mathbb{Q}}$ after fixing an embedding into $\mathbb{C}$.

More generally, in a specified family of Hodge structures the phrase **Hodge-generic point** means a point whose Mumford–Tate group equals that of a very general fibre of the family. For the full moduli space of polarized abelian varieties of dimension g, that generic group is GSp_{2g}, recovering the definition above. Stating the ambient family matters when it already imposes additional Hodge tensors.

24. The moduli stack of stable maps

Context. Gromov–Witten theory compactifies spaces of maps by allowing nodal source curves while retaining only finite automorphism groups.

Let X be a projective complex variety, let $g, n \geq 0$, and fix $\beta \in H_2(X(\mathbb{C}), \mathbb{Z})$.

Definition (stable map). An **n-pointed stable map of genus g and class β to** X is a tuple $(C, x_1, \dots, x_n, f)$ such that C is a connected proper nodal curve of arithmetic genus g, the x_i are distinct smooth points, $f : C \to X$ is a morphism with $f_*[C] = \beta$, and its automorphism group is finite. An automorphism is an isomorphism $u : C \to C$ fixing each x_i and satisfying $f \circ u = f$.

The stability condition has the following explicit form. For every irreducible component E contracted by f, let h be the genus of its normalization and m the number of points on that normalization lying over markings or nodes. Then

$$2h - 2 + m > 0.$$

A self-node contributes two points. Thus a contracted rational component requires at least three special points, and a contracted component of geometric genus one requires at least one.

Definition (moduli stack). The stack

$$\overline{\mathcal{M}}_{g,n}(X, \beta)$$

has, over a complex scheme S, objects consisting of a flat proper family of nodal curves $\pi : C \to S$, disjoint sections x_i through the relative smooth locus, and a morphism $f : C \to X$, whose geometric fibres are stable maps of genus g and class β. Arrows over S are isomorphisms of the marked curves commuting with f; pullback along $T \to S$ is by fibre product. The homology-class condition denotes the corresponding union of connected components of the stable-map stack, where the class of a complex fibre is locally constant.

This is a proper Deligne–Mumford stack of finite type, with projective coarse moduli space. The characteristic-zero hypothesis ensures that the finite automorphism group schemes are unramified. Over a general characteristic-zero field, one may instead fix a numerical curve class, requiring $\deg f^*L = \beta(L)$ for every line bundle L on X. Fixing only $\deg f^*\mathcal{O}_X(1) = d$ gives the degree version $\overline{\mathcal{M}}_{g,n}(X, d)$.

The evaluation morphism at the ith marking is $\mathrm{ev}_i(C, x_1, \dots, x_n, f) = f(x_i)$.

25. Nash blowup of a variety

Context. The Nash blowup replaces a singular variety by a space that remembers limiting tangent spaces.

Let k be a perfect field and let X be a reduced separated k-scheme of finite type, of pure dimension d. Its smooth locus $U = X_{\mathrm{sm}}$ is dense, and $\Omega^1_{X/k}|_U$ is locally free of rank d.

For a coherent sheaf $\mathcal{F}$ on X, write $\mathrm{Gr}_d(\mathcal{F})$ for the relative Grassmannian representing locally free rank-d quotients of pullbacks of $\mathcal{F}$. Thus a point over $h : T \to X$ is a surjection $h^*\mathcal{F} \twoheadrightarrow Q$, where Q is locally free of rank d, considered up to isomorphism of the quotient. It is projective over X.

Definition. Over U, the identity quotient of $\Omega^1_{U/k}$ defines a section

$$s : U \longrightarrow \mathrm{Gr}_d(\Omega^1_{X/k}).$$

The **Nash blowup** is the reduced closure

$$\mathrm{Nash}(X) = \overline{s(U)} \subseteq \mathrm{Gr}_d(\Omega^1_{X/k}),$$

with its projection $\nu : \mathrm{Nash}(X) \to X$. It is a projective birational morphism and an isomorphism over U.

If $X \subseteq \mathbb{A}^N_k$ is closed, the Gauss map on its smooth locus sends x to its tangent d-plane $T_xX \subseteq k^N$. In this presentation the Nash blowup is the reduced closure of the graph

$$\{(x, T_xX) : x \in U\} \subseteq X \times \mathrm{Gr}(d, N).$$

The intrinsic construction shows that it is independent of the embedding.

The universal quotient on the Grassmannian gives a locally free rank-d quotient of $\nu^*\Omega^1_{X/k}$. More generally, if $h : Y \to X$ is a morphism from a reduced scheme with $h^{-1}(U)$ dense and $h^*\Omega^1_{X/k}$ admits such a quotient, h factors uniquely through ν. The condition is that there be a locally free quotient; the full pullback of the sheaf of differentials need not itself be locally free.

26. Normalized Nash blowup

Context. Normalizing after each Nash modification produces the version of Nash iteration used for normal varieties and in positive characteristic.

Let X be as in Definition 25, and let $\nu : \mathrm{Nash}(X) \to X$ be its Nash blowup. Since $\mathrm{Nash}(X)$ is reduced and of finite type over a field, its normalization

$$\eta : \mathrm{Nash}(X)^\nu \longrightarrow \mathrm{Nash}(X)$$

is finite. For a reducible reduced scheme, normalization means the disjoint union of the normalizations of its irreducible components in their respective function fields.

Definition. The **normalized Nash blowup** of X is

$$\mathrm{NNash}(X) = \mathrm{Nash}(X)^\nu,$$

together with the composite morphism

$$\mathrm{NNash}(X) \xrightarrow{\eta} \mathrm{Nash}(X) \xrightarrow{\nu} X.$$

It is projective and birational, its source is normal, and it is an isomorphism over the smooth locus of X.

Definition (iteration). Set $X_0 = X$ and recursively set

$$X_{r+1} = \mathrm{NNash}(X_r), \qquad r \geq 0.$$

The morphism $X_r \to X$ is the composition of these successive modifications. At each stage the Nash blowup uses the differentials of the current variety X_r, and normalization is performed before proceeding to the next stage. This distinguishes the construction from taking several ordinary Nash blowups and normalizing only once at the end.

Over the perfect field k, the iteration **resolves the singularities of** X **after** r **steps** if X_r is smooth; the resulting composite is then a proper birational morphism from a smooth variety. Eventual smoothness is an additional assertion about the iteration, not part of the definition.

27. Perfect obstruction theories and virtual fundamental classes

Context. A virtual fundamental class uses deformation and obstruction data to give a moduli space a cycle of its expected dimension.

Let M be a separated Deligne–Mumford stack of finite type over a field k of characteristic zero, and let $L_{M/k}$ be its cotangent complex, with cohomological indexing.

Definition (perfect obstruction theory). A **perfect obstruction theory** is a morphism

$$\phi : E \longrightarrow L_{M/k}$$

in the derived category of $\mathcal{O}_M$-modules such that E is perfect of Tor-amplitude $[-1, 0]$, $h^0(\phi)$ is an isomorphism, and $h^{-1}(\phi)$ is surjective. Thus étale locally E is represented by a two-term complex of vector bundles $[E^{-1} \to E^0]$. Its **virtual dimension** is the locally constant integer

$$v = \operatorname{rank} E^0 - \operatorname{rank} E^{-1}.$$

We discuss one component on which v is constant. Its obstruction sheaf is $h^1(E^\vee)$.

Construction of the intrinsic normal cone. On an étale chart $U \to M$ embedded as a closed subscheme of a smooth scheme W, let

$$C_{U/W} = \operatorname{Spec}_U \bigoplus_{r \geq 0} I^r / I^{r+1}$$

be the normal cone. The quotient cone stacks $[C_{U/W} / T_W|_U]$ glue, independently of the embeddings, to the **intrinsic normal cone** $\mathfrak{C}_M$, of pure dimension zero. The obstruction theory embeds it as a closed subcone stack of

$$\mathfrak{E} = h^1 / h^0(E^\vee),$$

which locally is the vector bundle stack $[(E^{-1})^\vee / (E^0)^\vee]$, with the additive action given by the dual differential.

Definition (virtual fundamental class). The **virtual fundamental class associated with** ϕ is

$$[M, \phi]^{\mathrm{vir}} = 0^!_{\mathfrak{E}}[\mathfrak{C}_M] \in A_v(M)_{\mathbb{Q}}.$$

Here intersection theory on vector bundle stacks supplies $0^!_{\mathfrak{E}}$ as the inverse of flat pullback along $\mathfrak{E} \to M$, whose relative dimension is $-v$.

Concretely, if $E = [E^{-1} \to E^0]$ globally, put $E_1 = (E^{-1})^\vee$ and

$$C_1 = \mathfrak{C}_M \times_{\mathfrak{E}} E_1 \subseteq E_1.$$

Then $[M, \phi]^{\mathrm{vir}} = 0^!_{E_1}[C_1]$, using the ordinary zero-section Gysin map for the vector bundle E_1. This description is independent of the resolution. The construction is due to Behrend–Fantechi, *The intrinsic normal cone*.

For a morphism $M \to B$ of Deligne–Mumford type with B smooth of pure dimension b, use $L_{M/B}$ and the relative intrinsic normal cone instead; the virtual dimension is $\operatorname{rank} E + b$. For stable maps to a smooth projective variety X, the universal curve $\pi : C \to M$ and universal map $f : C \to X$ give the relative theory $(R\pi_* f^* T_X)^\vee$ over the stack of prestable marked curves. Its virtual dimension is

$$(1 - g)(\dim X - 3) + n + \int_\beta c_1(T_X).$$

If M is proper and $v = 0$, Definition 16 gives the virtual count $\deg [M, \phi]^{\mathrm{vir}} \in \mathbb{Q}$.

28. Points of an algebraic stack

Context. The underlying topological space of a stack remembers where objects lie while forgetting their automorphisms and their particular fields of definition.

Let $\mathcal{X}$ be an algebraic stack.

Definition. A **point of** $\mathcal{X}$ is an equivalence class of morphisms $x : \operatorname{Spec} K \to \mathcal{X}$, where K ranges over fields. Two such morphisms, over K and L, are equivalent if there is a field Ω with embeddings $K \hookrightarrow \Omega$ and $L \hookrightarrow \Omega$ such that their pullbacks to $\operatorname{Spec} \Omega$ are isomorphic objects of $\mathcal{X}(\Omega)$. Denote the set of points by $|\mathcal{X}|$.

Choose a smooth surjective atlas $p : U \to \mathcal{X}$ from a scheme. Give $|\mathcal{X}|$ the **quotient topology**: a subset $V \subseteq |\mathcal{X}|$ is open precisely when $|p|^{-1}(V)$ is open in $|U|$. This topology is independent of the atlas. If $R = U \times_{\mathcal{X}} U$, the two projections identify $|\mathcal{X}|$ with the topological quotient of $|U|$ by the relation induced by $|R|$. See the Stacks Project, Points of algebraic stacks.

A point $x \in |\mathcal{X}|$ is **closed** if the singleton $\{x\}$ is closed in this topology. For a stack over a field k, a **k-point** means an object of the groupoid $\mathcal{X}(k)$, equivalently a morphism $\operatorname{Spec} k \to \mathcal{X}$ over k. Its automorphism group is retained in this groupoid but forgotten in $|\mathcal{X}|$.

Even when k is algebraically closed, a k-point need not give a closed topological point. For example, the object determined by $1 \in \mathbb{A}^1(k)$ in $[\mathbb{A}^1/\mathbb{G}_m]$, with the scaling action, represents the nonzero orbit, whose closure also contains the orbit of 0. Thus the phrases "k-point" and "closed point" express different conditions.

29. Residual gerbe at a point of an algebraic stack

Context. A residual gerbe is the stack-theoretic replacement for the spectrum of the residue field at a point, retaining its stabilizer.

Let $\mathcal{X}$ be a quasi-separated algebraic stack, locally of finite type over a field, and let $x \in |\mathcal{X}|$.

Definition. The **residual gerbe at** x is a monomorphism of algebraic stacks

$$i_x : \mathcal{G}_x \longrightarrow \mathcal{X}$$

such that $\mathcal{G}_x$ is reduced and locally Noetherian, its underlying topological space has one point, and this point maps to x. Under the stated hypotheses it exists and is unique up to equivalence over $\mathcal{X}$.

There is a field $\kappa(x)$, called the **residue field of the point**, over which $\mathcal{G}_x$ is a gerbe: it is locally nonempty for the fppf topology, and any two objects become isomorphic fppf locally. Every field-valued representative of x factors through i_x, uniquely up to the unique compatible isomorphism. For schemes, this construction is exactly $\operatorname{Spec} \kappa(x) \to X$. These definitions and existence statements are treated in the Stacks Project on residual gerbes and their existence.

Suppose now that k is algebraically closed and that x is represented by an object $\xi \in \mathcal{X}(k)$. Its **stabilizer group scheme** is

$$G_\xi = \operatorname{Aut}_{\mathcal{X}}(\xi),$$

with $G_\xi(T) = \operatorname{Aut}_{\mathcal{X}(T)}(\xi_T)$ for k-schemes T. The choice of ξ identifies the residual gerbe with the classifying stack

$$\mathcal{G}_x \simeq B G_\xi.$$

The morphism $BG_\xi \to \mathcal{X}$ sends a G_ξ-torsor to the object obtained by descent from ξ. If x is closed in $|\mathcal{X}|$, this morphism is the reduced closed substack supported at x. Over a non-algebraically closed residue field, the residual gerbe need not have an object over that field, so an identification with a classifying stack requires a choice of such an object.

30. Theta divisor of a Jacobian

Context. Effective divisors of degree one less than the genus define the canonical principal polarization of a Jacobian.

Let C be a smooth projective geometrically connected curve of genus $g \geq 1$ over an algebraically closed field k, and write $J = \mathrm{Pic}^0(C)$. The symmetric product

$$C^{(d)} = C^d / S_d$$

parametrizes effective divisors of degree d, with $C^{(0)} = \operatorname{Spec} k$. The Abel map is

$$a_d : C^{(d)} \longrightarrow \mathrm{Pic}^d(C), \qquad D \longmapsto \mathcal{O}_C(D).$$

Definition. The **theta divisor on** $\mathrm{Pic}^{g-1}(C)$ is the scheme-theoretic image

$$W_{g-1} = a_{g-1}(C^{(g-1)}).$$

It is an effective reduced Cartier divisor, with underlying set

$$\{L \in \mathrm{Pic}^{g-1}(C) : H^0(C, L) \neq 0\}.$$

To obtain a divisor on J, choose a line bundle L_0 of degree $g-1$. Pull back W_{g-1} by $J \xrightarrow{\sim} \mathrm{Pic}^{g-1}(C)$, $M \mapsto M \otimes L_0$, and denote the resulting divisor by Θ_{L_0}. Any such divisor is called a **theta divisor of the Jacobian**.

The line bundle $\mathcal{O}_J(\Theta_{L_0})$ is ample and defines the canonical principal polarization: the homomorphism

$$J \longrightarrow J^\vee, \qquad a \longmapsto t_a^* \mathcal{O}_J(\Theta_{L_0}) \otimes \mathcal{O}_J(\Theta_{L_0})^{-1}$$

is an isomorphism. Changing L_0 translates the theta divisor and leaves this polarization unchanged. If $L_0^{\otimes 2} \simeq \omega_C$, then $[-1]^*\Theta_{L_0} = \Theta_{L_0}$; such an L_0 is a theta characteristic.

For a curve over a non-algebraically closed field, W_{g-1} is defined on its degree-$(g-1)$ Picard torsor over that field. Translating it to J requires a degree-$(g-1)$ line bundle, which can always be chosen after passing to the algebraic closure. For any theta divisor Θ, the symmetric ample line bundle $\mathcal{O}_J(\Theta + [-1]^*\Theta)$ represents twice the canonical polarization.

31. Totally geodesic subvarieties of a ball quotient

Context. Complex hyperbolic geometry singles out algebraic subvarieties whose intrinsic geodesics are also ambient geodesics.

Let $\Gamma \subset \mathrm{PU}(1, n)$ be a torsion-free lattice and let $S_\Gamma = \Gamma \backslash \mathbb{B}^n$ be the smooth ball quotient of Definition 15. Give $\mathbb{B}^n$ its invariant Bergman metric and S_Γ the descended Kähler metric. Multiplying this metric by a positive constant does not affect the definition below.

For a smooth submanifold $Y \subseteq S_\Gamma$, its **second fundamental form** is

$$\mathrm{II}_Y(u, v) = (\nabla^{S_\Gamma}_u v)^\perp,$$

where u, v are tangent vector fields along Y and $\perp$ denotes orthogonal projection to its normal bundle. The submanifold is **totally geodesic** when $\mathrm{II}_Y = 0$. Equivalently, every intrinsic geodesic of Y is an ambient geodesic on every interval on which it is defined.

Definition. An irreducible closed algebraic subvariety $Z \subseteq S_\Gamma$ is a **totally geodesic subvariety** if $\dim_{\mathbb{C}} Z > 0$ and its smooth locus Z_{reg} is totally geodesic. The condition concerns the complex subvariety with the induced real Riemannian metric. It allows singularities of Z, including intersections of local branches arising from an immersed quotient of a smaller ball.

A totally geodesic subvariety is **proper** if $Z \neq S_\Gamma$. A proper totally geodesic subvariety is **maximal** if there is no totally geodesic subvariety W with

$$Z \subsetneq W \subsetneq S_\Gamma.$$

Locally upstairs in $\mathbb{B}^n$, its smooth branches are open subsets of totally geodesic complex subballs. In the projective model of the ball, these subballs are intersections with complex projective linear subspaces on which the defining Hermitian form has signature $(1, m)$.

For lattices with torsion, use the orbifold convention: choose a torsion-free finite-index subgroup $\Gamma' \subset \Gamma$ and require every irreducible component of the reduced inverse image of Z in $S_{\Gamma'}$ to be totally geodesic in the preceding sense. This condition is independent of the finite cover and avoids treating the orbifold metric as a smooth metric at stabilizer points of the coarse variety.

Analysis & PDE

32. The spaces $C^k(M)$ and $C^{k,\alpha}(M)$

Context. These are the standard Banach spaces of differentiable and Hölder functions used in global analysis and smooth dynamics.

Let M be a compact smooth manifold, $k \geq 0$ an integer and $0 < \alpha \leq 1$. Choose finitely many coordinate charts (U_i, ϕ_i) and compact sets $K_i \subset U_i$ whose interiors cover M. For $u : M \to \mathbb{F}$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, write $u_i = u \circ \phi_i^{-1}$. For a function v on a subset $E \subset \mathbb{R}^n$, set

$$[v]_{\alpha,E} = \sup_{x,y\in E,\ x\neq y} \frac{|v(x) - v(y)|}{|x - y|^\alpha}.$$

Definition. The space $C^k(M)$ consists of the functions whose coordinate representatives have continuous partial derivatives through order k. Its norm, for the chosen charts, is

$$\|u\|_{C^k} = \max_i \max_{|\beta|\leq k} \sup_{\phi_i(K_i)} |\partial^\beta u_i|.$$

The **Hölder space** $C^{k,\alpha}(M)$ consists of those $u \in C^k(M)$ for which

$$\|u\|_{C^{k,\alpha}} = \|u\|_{C^k} + \max_i \max_{|\beta|=k} [\partial^\beta u_i]_{\alpha,\phi_i(K_i)} < \infty.$$

Both spaces are Banach spaces, and changing the finite chart system gives an equivalent norm. In particular, their underlying spaces and norm topologies are intrinsic. For connected M, an equivalent norm on $C^{0,\alpha}(M)$ is

$$\|u\|_\infty + \sup_{x\neq y} \frac{|u(x) - u(y)|}{d_g(x,y)^\alpha},$$

where d_g is the distance of any smooth Riemannian metric.

For noninteger $r > 0$, the notation C^r means $C^{\lfloor r\rfloor, r-\lfloor r\rfloor}$; at integer $r = k$ it means C^k. We write $C^{k,1}$ for functions with Lipschitz derivatives of order k, keeping this distinct from C^{k+1}.

On a noncompact manifold, $C^{k,\alpha}_{\mathrm{loc}}$ requires the corresponding condition on every compact subset of every chart. Tensor fields are treated componentwise in smooth local frames. Maps between manifolds are of class $C^{k,\alpha}$ when their coordinate representatives are locally of that class; a $C^{k,\alpha}$ diffeomorphism has both itself and its inverse of this class. On a smooth manifold with boundary, the definitions use boundary charts and derivatives extending continuously to the boundary.

33. Domain of dependence and development for compressible Euler equations

Context. The geometry of sound propagation describes the part of a classical fluid solution determined by specified initial data.

Fix a smooth equation of state p. Let (ρ, v) be a C^1 solution of the isentropic Euler equations on a spacetime region $U \subset \mathbb{R} \times \mathbb{R}^n$ (periodic spatial coordinates are also allowed):

$$(\partial_t + v \cdot \nabla)\rho = -\rho \operatorname{div} v, \qquad (\partial_t + v \cdot \nabla)v = -\rho^{-1}\nabla p(\rho).$$

Assume $\rho > 0$ and $p'(\rho) > 0$, and set $c = \sqrt{p'(\rho)}$. The **acoustical metric** is

$$g = -c^2 dt^2 + \sum_{j=1}^{n} (dx^j - v^j dt)^2.$$

A future-directed causal curve, parametrized by time, is a locally Lipschitz curve $\gamma(t) = (t, x(t))$ satisfying $|\dot{x} - v(t,x)| \le c(t,x)$ almost everywhere. It is **past-inextendible in** U if it admits no continuation to earlier times as a causal curve in U.

Definition. For $S \subset U$, its **future domain of dependence relative to** U is

$$D^+_U(S) = \{q \in U : \text{every past-inextendible causal curve in } U \text{ ending at } q \text{ meets } S\}.$$

Past domains are defined with time reversed, and $D_U(S) = D^+_U(S) \cup D^-_U(S)$. The metric in this definition is the metric of the given solution.

Fix a regularity class of classical solutions and compatible data on a spacelike initial portion S_0, possibly together with a specified characteristic boundary C. Put $S = S_0 \cup C$. A **future development** consists of a spacetime region D with nonempty dense interior, containing S, and a solution on its interior extending in the chosen regularity to S, such that the solution attains the prescribed data, C is null for its acoustical metric, and every past-inextendible causal curve in D from a point of D meets S. Boundary portions are included when testing this last condition. For a pure Cauchy problem, take $C = \emptyset$.

A development extends another if its domain contains the other's domain and the solutions agree there. A development is **maximal** if it has no proper extension in the chosen class. A **maximum development**, when established by a well-posedness theorem, extends every development of the data. These definitions alone assert neither existence nor uniqueness.

For data on a proper portion of a spacelike slice, the smooth causal boundary issuing from its edge is an acoustical null hypersurface, often denoted C_0. A boundary where classical regularity breaks down is a separate obstruction to extension. The acoustical metric, and therefore the domain of dependence, depends on the solution being developed.

34. Neumann eigenvalues and eigenfunctions

Context. The Neumann spectrum arises from the energy form on $H^1(\Omega)$ in the spectral theory of elliptic boundary problems.

Let $\Omega \subset \mathbb{R}^n$ be a nonempty bounded open set with Lipschitz boundary. On $L^2(\Omega;\mathbb{C})$ consider the closed nonnegative form

$$q(u,w) = \int_\Omega \nabla u \cdot \overline{\nabla w}\, dx, \qquad \mathcal{D}(q) = H^1(\Omega).$$

Definition. The **Neumann Laplacian** L_N is the self-adjoint operator associated with q. Explicitly,

$$\mathcal{D}(L_N) = \{u \in H^1(\Omega) : \exists f \in L^2(\Omega) \text{ such that } q(u,w) = \langle f, w\rangle_{L^2}\ \forall w \in H^1(\Omega)\}, \qquad L_N u = f.$$

A real number μ is a **Neumann eigenvalue** if there is a nonzero $u \in H^1(\Omega)$ with

$$\int_\Omega \nabla u \cdot \overline{\nabla w}\, dx = \mu \int_\Omega u \overline{w}\, dx \quad \text{for every } w \in H^1(\Omega).$$

Such a u is a **Neumann eigenfunction**. On a smooth domain this is the weak formulation of $-\Delta u = \mu u$ in Ω and $\partial_\nu u = 0$ on $\partial\Omega$, with ν the outward normal. The weak formulation also covers polygonal domains without imposing a normal derivative at their vertices.

Compactness of $H^1(\Omega) \hookrightarrow L^2(\Omega)$ gives an orthonormal eigenbasis and an ordered sequence

$$0 = \mu_1 \le \mu_2 \le \cdots \longrightarrow \infty,$$

with every eigenvalue repeated according to the dimension of its eigenspace. This sequence, including multiplicities, is the **Neumann spectrum**. Equivalently,

$$\mu_k = \min_{\substack{V \subset H^1(\Omega) \\ \dim V = k}} \max_{0 \neq u \in V} \frac{\int_\Omega |\nabla u|^2}{\int_\Omega |u|^2}.$$

The zero eigenspace consists of functions constant on each connected component. If Ω is connected, $\mu_2 > 0$ is its first positive eigenvalue; a **second Neumann eigenfunction** is any nonzero member of its eigenspace. Such eigenfunctions have integral zero, and μ_2 is the minimum of the displayed Rayleigh quotient among nonzero mean-zero functions in $H^1(\Omega)$.

35. Absolutely continuous spectrum

Context. Spectral types classify the scalar measures supplied by the spectral theorem for self-adjoint operators.

Let H be a possibly unbounded self-adjoint operator on a complex Hilbert space $\mathcal{H}$, with projection-valued spectral measure E_H. For $u \in \mathcal{H}$, its **scalar spectral measure** is

$$\mu_u(B) = \langle E_H(B)u, u \rangle = \|E_H(B)u\|^2, \qquad B \subset \mathbb{R} \text{ Borel}.$$

This is a finite positive measure of total mass $\|u\|^2$; the vector u need not belong to $\mathcal{D}(H)$.

Definition. The **absolutely continuous subspace** of H is

$$\mathcal{H}_{\mathrm{ac}}(H) = \{ u \in \mathcal{H} : \mu_u \ll d\lambda \}.$$

Thus $u \in \mathcal{H}_{\mathrm{ac}}(H)$ precisely when $E_H(N)u = 0$ for every Lebesgue-null Borel set N. This is a closed subspace reducing H: the orthogonal projection onto it commutes with every spectral projection. Consequently the operator

$$H_{\mathrm{ac}} u = Hu, \qquad \mathcal{D}(H_{\mathrm{ac}}) = \mathcal{D}(H) \cap \mathcal{H}_{\mathrm{ac}}(H)$$

is self-adjoint on $\mathcal{H}_{\mathrm{ac}}(H)$.

The **absolutely continuous spectrum** is the closed set

$$\sigma_{\mathrm{ac}}(H) = \sigma(H_{\mathrm{ac}}).$$

We take it to be empty if $\mathcal{H}_{\mathrm{ac}}(H) = \{0\}$. Equivalently, $\lambda \in \sigma_{\mathrm{ac}}(H)$ if for every $\varepsilon > 0$ the restriction of $E_H((\lambda - \varepsilon, \lambda + \varepsilon))$ to $\mathcal{H}_{\mathrm{ac}}(H)$ is nonzero.

The operator has **purely absolutely continuous spectrum** if $\mathcal{H}_{\mathrm{ac}}(H) = \mathcal{H}$. Its spectrum is purely absolutely continuous on a Borel set I if $E_H(I)\mathcal{H} \subset \mathcal{H}_{\mathrm{ac}}(H)$. This latter assertion is stronger than $I \subset \sigma_{\mathrm{ac}}(H)$: an absolutely continuous spectral component can coexist at the same energies with singular spectral components.

36. Amenable trace on a C^*-algebra

Context. Amenable traces express finite-dimensional approximation in the normalized Hilbert–Schmidt norm.

Let A be a C^*-algebra, not necessarily unital. A **tracial state** is a positive linear functional $\tau : A \to \mathbb{C}$ of norm one such that $\tau(ab) = \tau(ba)$ for all $a, b \in A$. It is **faithful** if $\tau(a^*a) = 0$ implies $a = 0$.

On $M_n(\mathbb{C})$ write

$$\mathrm{tr}_n(x) = \frac{1}{n} \operatorname{Tr}(x), \qquad \|x\|_{2,n} = \left(\mathrm{tr}_n(x^*x)\right)^{1/2}.$$

A linear map $\phi : A \to M_n(\mathbb{C})$ is **completely positive and contractive** if every matrix amplification $\mathrm{id}_{M_k} \otimes \phi$ is positive and $\|\phi\| \le 1$.

Definition. A tracial state τ is **amenable** if there are a directed set I, positive integers n_i, and completely positive contractive maps $\phi_i : A \to M_{n_i}(\mathbb{C})$ such that, for every $a, b \in A$,

$$\lim_i \|\phi_i(ab) - \phi_i(a)\phi_i(b)\|_{2,n_i} = 0, \qquad \lim_i \operatorname{tr}_{n_i}(\phi_i(a)) = \tau(a).$$

Equivalently, for every finite $\mathcal{F} \subset A$ and $\varepsilon > 0$, there are $n \ge 1$ and a completely positive contractive $\phi : A \to M_n(\mathbb{C})$ such that

$$\|\phi(ab) - \phi(a)\phi(b)\|_{2,n} < \varepsilon, \qquad |\operatorname{tr}_n(\phi(a)) - \tau(a)| < \varepsilon \quad (a, b \in \mathcal{F}).$$

For separable A the net can be replaced by a sequence. For unital A the approximating maps can be chosen unital. A **faithful amenable trace** satisfies both faithfulness and the approximation condition. The normalization $\operatorname{tr}_n = n^{-1}\operatorname{Tr}$ is part of the definition, as the matrix size varies with the approximation.

37. Caloric gauge and the manifold of caloric connections

Context. The Yang–Mills heat flow selects a gauge and a nonlinear space of initial data for the hyperbolic Yang–Mills equation on $\mathbb{R}^{1+4}$.

Fix a compact Lie group G with an invariant inner product on its Lie algebra $\mathfrak{g}$. A spatial connection is $a = a_j dx^j \in \dot{H}^1(\mathbb{R}^4; \mathfrak{g} \otimes T^*\mathbb{R}^4)$, where $\dot{H}^1$ is realized as a subspace of L^4. For a connection A, put

$$D_j = \partial_j + [A_j, \cdot], \qquad F_{jk} = \partial_j A_k - \partial_k A_j + [A_j, A_k].$$

Its **Yang–Mills heat flow in the gauge** $A_s = 0$ is

$$\partial_s A_j = D^\ell F_{\ell j}, \qquad A(0) = a.$$

Here the energy-class flow means the unique maximal solution in $C([0, s_*); \dot{H}^1)$ supplied by local well-posedness, obtained as the limit of regular solutions for regular approximating data on each compact subinterval of existence.

Definition. A connection a is **caloric** if this heat flow exists for every $s \ge 0$, has finite **caloric size**

$$Q(a) = \int_0^\infty \int_{\mathbb{R}^4} |F(s,x)|^3 \, dx \, ds < \infty,$$

and satisfies $\lim_{s\to\infty} \|A(s)\|_{\dot{H}^1} = 0$. Write $\mathcal{C}$ for the set of caloric connections. A spacetime connection is **in caloric gauge** if its spatial connection $A_x(t)$ belongs to $\mathcal{C}$ at every time. The heat-flow condition $A_s = 0$ alone is called the local caloric gauge.

For completeness, the ambient space in which $\mathcal{C}$ is a manifold is the Banach space

$$\mathcal{H} = \left\{ a \in \dot{H}^1 : \sum_{k \in \mathbb{Z}} \|P_k(\partial^j a_j)\|_{L^2} < \infty \right\}, \qquad \|a\|_{\mathcal{H}} = \|a\|_{\dot{H}^1} + \sum_k \|P_k(\partial^j a_j)\|_2,$$

where (P_k) is a fixed smooth homogeneous Littlewood–Paley decomposition. The caloric-manifold theorem states that $\mathcal{C}$ is a C^1 submanifold of $\mathcal{H}$.

For $a \in \mathcal{C}$, its L^2 **tangent space** is

$$T_a^{L^2}\mathcal{C} = \overline{T_a\mathcal{C} \cap L^2}^{\,L^2}.$$

Equivalently, it consists of $b \in L^2$ whose linearized heat flow along A,

$$\partial_s B_j = D^\ell(D_\ell B_j - D_j B_\ell) + [B^\ell, F_{\ell j}], \qquad B(0) = b,$$

satisfies $\|B(s)\|_2 \to 0$ as $s \to \infty$. The L^2 **tangent bundle** is

$$T^{L^2}\mathcal{C} = \{(a, b) : a \in \mathcal{C},\ b \in T^{L^2}_a \mathcal{C}\},$$

with the topology inherited from $\mathcal{H} \times L^2$; its elements are **caloric initial data**.

Every $\dot H^1$ connection with finite caloric size has a caloric representative, unique up to constant gauge conjugation. The heat-flow and manifold results used in this construction are established in Oh–Tataru, *The Yang–Mills heat flow and the caloric gauge*.

38. Fully nonlinear complex Hessian-type equations

Context. A symmetric function of the complex Hessian eigenvalues defines a class of elliptic equations containing the complex Monge–Ampère and complex Hessian equations.

Let (X, ω) be a compact Hermitian manifold of complex dimension n, without boundary, and let χ be a smooth real $(1,1)$-form. Fix the convention

$$d^c = \frac{i}{2}(\bar\partial - \partial), \qquad dd^c = i\partial\bar\partial.$$

For $\varphi \in C^2(X;\mathbb{R})$ set $\chi_\varphi = \chi + i\partial\bar\partial\varphi$. Write locally $\omega = i g_{j\bar k} dz^j \wedge d\bar z^k$ and $\chi_\varphi = i a_{j\bar k} dz^j \wedge d\bar z^k$. The **eigenvalues of χ_φ relative to ω** are the real roots, counted with multiplicity, of

$$\det(a - \lambda g) = 0.$$

Equivalently they are the eigenvalues of the g-self-adjoint endomorphism h_φ defined by χ_φ and g. Denote their unordered vector by $\lambda[h_\varphi]$.

Let $\Gamma \subset \mathbb{R}^n$ be an open permutation-invariant cone and let $f \in C^1(\Gamma)$ be symmetric and elliptic, meaning $f_j = \partial f / \partial \lambda_j > 0$ on Γ.

Definition. A real C^2 function φ is (χ, Γ)**-admissible** if $\lambda[h_\varphi](x) \in \Gamma$ at every x. Given $F \in C(X;\mathbb{R})$ and $c > 0$, it is a normalized solution of the **complex Hessian-type equation** if

$$f(\lambda[h_\varphi](x)) = c e^{F(x)} \quad (x \in X), \qquad \sup_X \varphi = 0,$$

and it is admissible. Symmetry makes the equation independent of the ordering of the eigenvalues; the normalization fixes the additive constant in φ.

Common structural hypotheses are

$$\Gamma_n \subset \Gamma \subset \Gamma_1, \qquad f(r\lambda) = r f(\lambda)\ (r > 0), \qquad \prod_{j=1}^n f_j(\lambda) \geq \gamma > 0,$$

where Γ_n is the positive orthant and $\Gamma_1 = \{\sum_j \lambda_j > 0\}$. These are additional hypotheses on the operator. One may normalize F by $\int_X e^{nF}\omega^n = \int_X \omega^n$ and treat c as an unknown constant.

For $f = (\lambda_1 \cdots \lambda_n)^{1/n}$ on Γ_n, the equation is $\chi_\varphi^n = c^n e^{nF}\omega^n$ with $\chi_\varphi > 0$. For $f = \sigma_k^{1/k}$ on $\Gamma_k = \{\sigma_1 > 0, \dots, \sigma_k > 0\}$, where σ_j is the jth elementary symmetric polynomial, it is the complex k-Hessian equation.

39. Hard sphere dynamics on the torus

Context. Hard sphere dynamics is the microscopic mechanical model underlying the kinetic theory of an elastic gas.

Let $d \geq 2$, $N \geq 1$, and $0 < \varepsilon < 1/2$. Take N labelled particles of equal mass, each representing a sphere of diameter ε, in $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$. Their interior phase space is

$$\mathcal{D}_N^\varepsilon = \{(x_1, v_1, \dots, x_N, v_N) : x_i \in \mathbb{T}^d,\ v_i \in \mathbb{R}^d,\ d_{\mathbb{T}^d}(x_i, x_j) > \varepsilon \text{ for } i \neq j\}.$$

Definition. A **regular hard sphere trajectory** is a motion with continuous positions and piecewise constant velocities, with finitely many collisions in each bounded time interval, obeying the following rules. Between collisions,

$$\dot{x}_i = v_i, \qquad \dot{v}_i = 0.$$

At a collision only one pair i, j is in contact. Let $r_{ij} \in \mathbb{R}^d$ be the shortest lift of $x_i - x_j$, which is unique at distance $\varepsilon < 1/2$, and put $\nu = r_{ij}/\varepsilon$. For incoming velocities v_i^-, v_j^- with $(v_i^- - v_j^-) \cdot \nu < 0$, the outgoing velocities are

$$v_i^+ = v_i^- - [(v_i^- - v_j^-) \cdot \nu]\nu, \qquad v_j^+ = v_j^- + [(v_i^- - v_j^-) \cdot \nu]\nu.$$

All other velocities are unchanged. At the collision time take the right-continuous, outgoing value. This rule preserves total momentum and kinetic energy and reverses the normal component of the relative velocity.

The almost-everywhere existence theorem for hard spheres states that, outside a Lebesgue-null set of initial configurations, this prescription gives a unique trajectory for all positive and negative times, with no grazing or simultaneous collisions and no finite accumulation of collision times.

The **hard sphere dynamics** is the resulting almost-everywhere defined flow Φ_t^ε, sending an initial configuration to its configuration at time t. It preserves phase-space Lebesgue measure and satisfies $\Phi_{t+s}^\varepsilon = \Phi_t^\varepsilon \circ \Phi_s^\varepsilon$ almost everywhere for each fixed s, t. Contact configurations may equivalently be included by identifying their incoming and outgoing velocities under the collision rule. For a variable particle number, the flow acts separately on each component of $\bigsqcup_{N \geq 0} \mathcal{D}_N^\varepsilon$.

40. The L^2 Laplace–Beltrami operator on a complete manifold

Context. Spectral geometry uses the self-adjoint realization of the geometric Laplacian on square-integrable functions.

Let (X, g) be a smooth complete Riemannian manifold without boundary, with volume measure $d\mathrm{vol}_g$. Adopt the nonnegative sign convention

$$\Delta_g u = -\operatorname{div}_g(\nabla_g u) = -\frac{1}{\sqrt{\det g}} \partial_i \left(\sqrt{\det g}\, g^{ij} \partial_j u \right).$$

On $L^2(X, d\mathrm{vol}_g; \mathbb{C})$ let Δ_0 have domain $C_c^\infty(X)$ and action $\Delta_0 u = \Delta_g u$. Integration by parts makes Δ_0 symmetric and nonnegative. The essential self-adjointness theorem for complete manifolds states that its closure is self-adjoint.

Definition. The **Laplace–Beltrami operator on** $L^2(X)$ is

$$\Delta_X = \overline{\Delta_0}.$$

Explicitly, $u \in \mathcal{D}(\Delta_X)$ if there are $u_m \in C_c^\infty(X)$ and $f \in L^2(X)$ with $u_m \to u$ and $\Delta_g u_m \to f$ in L^2; then $\Delta_X u = f$. Completeness also gives the description

$$\mathcal{D}(\Delta_X) = \{u \in L^2(X) : \Delta_g u \in L^2(X) \text{ in the distributional sense}\}.$$

Equivalently, Δ_X is associated with the closed form $q(u,w) = \int_X \langle \nabla u, \overline{\nabla w} \rangle_g \, d\mathrm{vol}_g$ on $H^1(X)$.

Its **spectrum** is the spectrum of this unbounded self-adjoint operator, a closed subset of $[0,\infty)$. An L^2 **eigenfunction** with eigenvalue λ is a nonzero $u \in \mathcal{D}(\Delta_X)$ satisfying $\Delta_X u = \lambda u$; its multiplicity is $\dim \ker(\Delta_X - \lambda)$. The **discrete spectrum** consists of isolated eigenvalues of finite multiplicity, and the **essential spectrum** is its complement in the spectrum.

If X is compact and connected, the spectrum is the eigenvalue sequence $0 = \lambda_0 < \lambda_1 \le \lambda_2 \le \cdots \to \infty$, counted with multiplicity. On noncompact complete manifolds the same operator definition applies, although continuous spectrum can occur.

41. Quasiregular maps

Context. Quasiregular mappings generalize orientation-preserving conformal maps by allowing uniformly bounded distortion and branching.

Let M, N be oriented Riemannian manifolds of the same dimension $n \ge 2$, without boundary. For a continuous Sobolev map $f \in W^{1,n}_{\mathrm{loc}}(M,N)$, let $Df(x) : T_xM \to T_{f(x)}N$ be its weak differential, as in Definition 47. Its operator norm is

$$|Df(x)| = \sup_{|v|_M = 1} |Df(x)v|_N,$$

and its signed Jacobian J_f is determined by

$$(Df(x))^*(\mathrm{vol}_N)_{f(x)} = J_f(x)(\mathrm{vol}_M)_x.$$

Definition. For $K \ge 1$, a map $f : M \to N$ is **K-quasiregular** if it is continuous, belongs to $W^{1,n}_{\mathrm{loc}}(M,N)$, and satisfies

$$|Df(x)|^n \le K J_f(x) \quad \text{for almost every } x \in M.$$

It is **quasiregular** if such a finite K exists. Here K bounds the outer distortion; the metrics and orientations are part of the data. The inequality implies $J_f \ge 0$ almost everywhere, and $Df = 0$ almost everywhere on $\{J_f = 0\}$.

This convention includes constant maps. When needed, nonconstancy is imposed separately. An orientation-preserving conformal local diffeomorphism is 1-quasiregular. A quasiregular homeomorphism is called **quasiconformal**. The least permissible value of K is the **outer dilatation** $K_O(f)$; some references use a distortion constant controlling both outer and inner dilatation, so numerical constants require a convention.

42. Quasiregularly elliptic manifolds

Context. Quasiregular ellipticity asks whether a manifold admits a nonconstant parametrization by Euclidean space with bounded distortion.

Definition. A connected, oriented Riemannian n-manifold N, without boundary and with $n \ge 2$, is **quasiregularly elliptic** if there exists a nonconstant quasiregular map

$$f : \mathbb{R}^n \longrightarrow N,$$

where $\mathbb{R}^n$ has its Euclidean metric and standard orientation. Explicitly, f is continuous, $f \in W^{1,n}_{\mathrm{loc}}(\mathbb{R}^n, N)$, and one constant $K < \infty$ satisfies $|Df|^n \le K J_f$ almost everywhere on all of $\mathbb{R}^n$.

For a specified $K \ge 1$, the term **K-quasiregularly elliptic** requires a map satisfying this inequality with that K. A **closed quasiregularly elliptic manifold** is additionally compact.

For closed N, quasiregular ellipticity is independent of the chosen smooth Riemannian metric: two such metrics are uniformly comparable, so changing the metric changes only the permissible distortion constant. It is also independent of the orientation, since reversing the target orientation can be compensated by precomposing f with a reflection of $\mathbb{R}^n$. For a noncompact target, the metric remains part of the definition. The flat torus is an example, through its covering map $\mathbb{R}^n \to \mathbb{R}^n/\mathbb{Z}^n$.

43. The $SU(1,1)$ nonlinear Fourier transform on the half-line

Context. The nonlinear Fourier transform of a square-integrable potential is its reflection coefficient in a half-line scattering problem.

Let $f \in L^2((0,\infty);\mathbb{R})$, and use the scattering functions of Definition 44, with the normalization

$$\partial_t G(t,z) = \begin{pmatrix} 0 & e^{-2izt} f(t) \\ e^{2izt} f(t) & 0 \end{pmatrix} G(t,z), \qquad G(0,z) = I, \qquad a = G_{22}, \quad b = G_{21}.$$

For $T < \infty$, put $r_T(z) = b(T,z)/a(T,z)$. The function $a(T,\cdot)$ has no zeros on the closed upper half-plane; r_T is holomorphic and bounded in modulus by one on $\mathbb{C}_+ = \{\operatorname{Im} z > 0\}$.

The L^2 scattering-limit theorem states that $a(T,\cdot), b(T,\cdot)$ converge locally uniformly on $\mathbb{C}_+$ to holomorphic functions a_∞, b_∞, with a_∞ nonvanishing. On $\mathbb{R}$, they converge in measure for $d\Pi(x) = dx/(1+x^2)$ to their boundary functions, and

$$|a_\infty(x)|^2 - |b_\infty(x)|^2 = 1 \quad \text{for almost every } x \in \mathbb{R}.$$

Definition. The **nonlinear Fourier transform** of f, in this reflection-coefficient convention, is

$$\mathcal{N}f(z) = \frac{b_\infty(z)}{a_\infty(z)} \quad (z \in \mathbb{C}_+),$$

together with its almost-everywhere defined nontangential boundary function on $\mathbb{R}$. Equivalently, its real-axis value is the limit in Π-measure of r_T as $T \to \infty$. Thus $\mathcal{N}f$ is a bounded holomorphic function on $\mathbb{C}_+$, and $|\mathcal{N}f(x)| < 1$ almost everywhere on $\mathbb{R}$.

For the truncated potential $f_T = f\,1_{(0,T)}$, the scattering matrix stops evolving at time T, so

$$\mathcal{N}f_T(z) = \frac{b(T,z)}{a(T,z)}.$$

The factor 2 and the sign in e^{2izt} fix the spectral normalization: the first variation at $f = 0$ is $f \mapsto \int_0^\infty f(t) e^{2izt}\,dt$. The scattering matrix itself is also called an $SU(1,1)$ nonlinear Fourier transform in some conventions; here the term denotes its quotient b/a.

The limit theorem and this normalization are described in Poltoratski, *Pointwise convergence of the non-linear Fourier transform*, §5.

44. Scattering functions of a real Dirac system

Context. Removing the free oscillation from the fundamental solution of a Dirac system produces its scattering coefficients.

Let $f \in L^1_{\mathrm{loc}}([0,\infty);\mathbb{R})$, meaning $\int_0^T |f(t)|\,dt < \infty$ for every finite T. The corresponding **real Dirac system** is

$$\Omega X'(t) = zX(t) - Q(t)X(t), \qquad \Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Q(t) = \begin{pmatrix} 0 & f(t) \\ f(t) & 0 \end{pmatrix},$$

where $z \in \mathbb{C}$ is the spectral parameter and solutions are locally absolutely continuous in t.

Definition. Let $G(t,z)$ be the unique solution of the Volterra integral equation

$$G(t,z) = I + \int_0^t \begin{pmatrix} 0 & e^{-2izs}f(s) \\ e^{2izs}f(s) & 0 \end{pmatrix} G(s,z)\,ds.$$

The **scattering functions** are

$$a(t,z) = G_{22}(t,z), \qquad b(t,z) = G_{21}(t,z).$$

For each fixed z, the integral equation is equivalent to the corresponding matrix differential equation almost everywhere in t. Existence and uniqueness follow from the linear ODE theorem for locally integrable coefficients. Picard iteration, uniformly on compact sets of z, shows that $a(t,\cdot)$ and $b(t,\cdot)$ are entire for each finite t.

Writing $h^{\#}(z) = \overline{h(\bar z)}$, one has

$$G(t,z) = \begin{pmatrix} a^{\#}(t,z) & b^{\#}(t,z) \\ b(t,z) & a(t,z) \end{pmatrix}, \qquad aa^{\#} - bb^{\#} = 1.$$

In particular $|a(t,x)|^2 - |b(t,x)|^2 = 1$ for real x, and $G(t,x) \in SU(1,1)$, where

$$SU(1,1) = \{U \in SL_2(\mathbb{C}) : U^{*}JU = J\}, \qquad J = \operatorname{diag}(1,-1).$$

To relate this to the Dirac system, let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ be its fundamental matrix with $M(0,z) = I$, and set $E = A - iC$, $\widetilde{E} = B - iD$. Then

$$a(t,z) = \tfrac{1}{2}e^{itz}(E + i\widetilde{E}), \qquad b(t,z) = \tfrac{1}{2}e^{itz}(E - i\widetilde{E}).$$

For $f = 0$, the normalization gives $a = 1$ and $b = 0$; when f vanishes beyond T, both scattering functions are constant in t for $t \geq T$.

45. Schrödinger operators with bounded real potential

Context. A bounded real potential is a bounded self-adjoint perturbation of the free Laplacian on Euclidean space.

Let $d \geq 1$ and let $V \in L^\infty(\mathbb{R}^d;\mathbb{R})$. Work on the complex Hilbert space $L^2(\mathbb{R}^d)$, with $\Delta = \sum_{j=1}^{d} \partial_j^2$.

Definition. The **Schrödinger operator with potential** V is the unbounded operator

$$H = -\Delta + V, \qquad \mathcal{D}(H) = H^2(\mathbb{R}^d), \qquad Hu = -\Delta u + Vu.$$

Derivatives are distributional. Here H^2 consists of the L^2 functions whose distributional partial derivatives of orders at most two lie in L^2. Equivalently, for the unitary Fourier transform

$$\hat u(\xi) = (2\pi)^{-d/2} \int_{\mathbb{R}^d} e^{-ix\cdot\xi} u(x)\,dx,$$

extended to L^2 by continuity, one has $u \in H^2$ exactly when $(1 + |\xi|^2)\hat{u} \in L^2$, and $\widehat{-\Delta u} = |\xi|^2 \hat{u}$.

The free Laplacian on this domain is self-adjoint, and multiplication by V is bounded and self-adjoint. Thus H is self-adjoint on the stated domain and bounded below by $\operatorname{ess\,inf} V$. Its restriction to $C_c^\infty(\mathbb{R}^d)$ is essentially self-adjoint, with closure H. These statements fix the operator, including its domain, rather than only its differential expression.

For example, a finite quasiperiodic potential has the form

$$V(x) = \sum_{m \in F} c_m \exp\left(i \left\langle \sum_{j=1}^{\ell} m_j \omega_j, x \right\rangle \right),$$

where $F \subset \mathbb{Z}^\ell$ is finite and symmetric, $\omega_j \in \mathbb{R}^d$, and $c_{-m} = \overline{c_m}$. It is real and bounded, so the definition applies. The spectrum and absolutely continuous spectrum of this operator are understood in the self-adjoint sense, as in Definition 35.

46. Schur multipliers on Schatten classes

Context. Schur multipliers act by pointwise multiplication of operator kernels, extending entrywise multiplication of matrices.

Let $H = L^2(\mathbb{R}^n)$ and let $M \in L^\infty(\mathbb{R}^n \times \mathbb{R}^n)$ be a measurable symbol. Every Hilbert–Schmidt operator $A \in S_2(H)$ has a unique kernel $k_A \in L^2(\mathbb{R}^{2n})$, up to null sets, with

$$(Au)(x) = \int_{\mathbb{R}^n} k_A(x,y) u(y)\, dy, \qquad \|A\|_{S_2} = \|k_A\|_{L^2}.$$

Definition. The **Schur multiplier with symbol** M is the bounded map on $S_2(H)$ determined by

$$k_{S_M(A)}(x,y) = M(x,y) k_A(x,y).$$

Its S_2 operator norm is $\|M\|_{L^\infty}$.

For $1 \le p < \infty$, let $S_p(H)$ be the compact operators whose singular values $(s_j(A))$ satisfy

$$\|A\|_{S_p} = \left(\sum_j s_j(A)^p \right)^{1/p} < \infty.$$

We say M is a **bounded Schur multiplier on** S_p if there is $C < \infty$ such that

$$S_M(A) \in S_p(H), \qquad \|S_M(A)\|_{S_p} \le C \|A\|_{S_p} \quad \text{for every } A \in S_2(H) \cap S_p(H).$$

Finite-rank operators lie in this intersection and are dense in S_p, so the map has a unique bounded extension $S_M : S_p(H) \to S_p(H)$. Its **multiplier norm** is the least such C, equivalently

$$\|S_M\|_{S_p \to S_p} = \sup_{\substack{A \in S_2 \cap S_p \\ 0 < \|A\|_{S_p} \le 1}} \|S_M(A)\|_{S_p},$$

where the value is $+\infty$ if the boundedness condition fails.

The same definition applies at $p = \infty$ with the explicit convention $S_\infty(H) = \mathcal{K}(H)$, the compact operators with operator norm. If instead S_∞ denotes $B(H)$, one uses the corresponding normal extension; the finite-p definition does not require that convention. On a discrete space with counting measure, the formula is simply $(S_M(A))_{ij} = M_{ij} A_{ij}$.

47. Sobolev maps between Riemannian manifolds

Context. Weak derivatives of manifold-valued maps provide the differential and Jacobian used in harmonic-map and quasiregular-map theory.

Let M, N be smooth Riemannian manifolds and $1 \le p \le \infty$. First suppose N is closed and fix a smooth isometric embedding $\iota : N \hookrightarrow \mathbb{R}^L$.

Definition. A measurable map $f : M \to N$, considered up to equality almost everywhere, belongs to $W^{1,p}_{\mathrm{loc}}(M, N)$ if $\iota \circ f \in W^{1,p}_{\mathrm{loc}}(M; \mathbb{R}^L)$, interpreted componentwise in charts of M. For a closed target this definition is independent of the chosen smooth embedding. It does not require continuity.

For an arbitrary target N and a **continuous** map f, the intrinsic chart definition is equivalent: for every pair of charts (U, ϕ) on M and (V, ψ) on N,

$$\psi \circ f \circ \phi^{-1} \in W^{1,p}_{\mathrm{loc}}(\phi(U \cap f^{-1}(V)); \mathbb{R}^{\dim N}).$$

The domain here is open because f is continuous. This is the characterization used for the continuous representatives in Definition 41. For noncompact targets and general measurable maps, a convention for target integrability must additionally be specified; the closed-target definition avoids that issue.

The **weak differential** is the almost-everywhere defined linear map

$$Df(x) : T_xM \longrightarrow T_{f(x)}N$$

characterized by $D(\iota \circ f) = d\iota \circ Df$ in the embedded definition. The ambient weak differential is tangent to $\iota(N)$ almost everywhere. For continuous maps, the equivalent coordinate formula is

$$d\psi_{f(x)} \circ Df(x) \circ (d\phi_x)^{-1} = D(\psi \circ f \circ \phi^{-1})(\phi(x)).$$

Sobolev chain rules make this independent of the charts. Its **operator norm** is $|Df(x)| = \sup_{|v|_M = 1} |Df(x)v|_N$.

If M, N are oriented and have the same dimension n, the **signed Jacobian** J_f is defined by

$$(Df(x))^*(\mathrm{vol}_N)_{f(x)} = J_f(x)(\mathrm{vol}_M)_x.$$

It is the determinant in positively oriented orthonormal bases. Hadamard's inequality gives $|J_f| \le |Df|^n$; hence $J_f \in L^1_{\mathrm{loc}}$ when $p = n$. In general $W^{1,p}_{\mathrm{loc}}(M, N)$ is a space of maps, not a vector space.

48. Topological class and instanton number of a finite-energy connection

Context. Finite-energy connections on $\mathbb{R}^4$ have a topological invariant determined by their behavior at spatial infinity.

Let G be a compact connected Lie group with Lie algebra $\mathfrak{g}$ and an invariant inner product. A **finite-energy connection** is a $\mathfrak{g}$-valued one-form $a \in H^1_{\mathrm{loc}}(\mathbb{R}^4)$ whose curvature

$$F_a = \tfrac{1}{2} F_{jk}\, dx^j \wedge dx^k, \qquad F_{jk} = \partial_j a_k - \partial_k a_j + [a_j, a_k]$$

lies in $L^2(\mathbb{R}^4)$. Gauge transformations act by

$$a \longmapsto U a U^{-1} - (dU)U^{-1};$$

at this regularity one may take $U \in H^2_{\mathrm{loc}}(\mathbb{R}^4; G)$, using a fixed faithful matrix realization of G.

Background theorem (good gauge at infinity). Every finite-energy connection has a gauge-equivalent representative of the form

$$\tilde{a} = -\eta\,(dO_\infty)O_\infty^{-1} + b, \qquad b \in \dot{H}^1(\mathbb{R}^4).$$

Here η is smooth, vanishes near 0, and equals 1 outside a compact set; $O_\infty : \mathbb{R}^4 \setminus \{0\} \to G$ is smooth and homogeneous of degree zero. The homotopy class of $O_\infty|_{S^3}$ is independent of these choices and of the admissible gauge representative. This is the good-global-gauge construction of Oh–Tataru, *The hyperbolic Yang–Mills equation for connections in an arbitrary topological class*, §§1.3–1.4.

Definition. The **topological class** of a is

$$[a] = [O_\infty|_{S^3}] \in \pi_3(G),$$

where a value at a basepoint is translated to the identity to identify free and based homotopy classes. The connection is **topologically trivial** if $[a] = 0$, equivalently if it is gauge equivalent to a connection in $\dot{H}^1(\mathbb{R}^4)$. The associated bundle over S^4 is obtained by clutching two trivial bundles using $O_\infty|_{S^3}$.

For $G = SU(r)$, $r \geq 2$, in the defining representation and with the standard orientation of $\mathbb{R}^4$, the **instanton number**, or **second Chern number**, is

$$k(a) = \frac{1}{8\pi^2}\int_{\mathbb{R}^4} \operatorname{tr}(F_a \wedge F_a) \in \mathbb{Z}.$$

The integral converges absolutely because $F_a \in L^2$. This sign convention gives the second Chern number of the clutched bundle and identifies $\pi_3(SU(r))$ with $\mathbb{Z}$.

For a general G, the characteristic number associated with the chosen inner product is

$$\chi(a) = -\int_{\mathbb{R}^4} \langle F_a \wedge F_a \rangle.$$

It depends only on $[a]$, but need not determine that class or be integer-valued. For $SU(r)$ with $\langle X, Y\rangle = -\operatorname{tr}(XY)$, one has $\chi = 8\pi^2 k$.

The clutching construction defines the topological bundle without assuming that the original connection extends smoothly across infinity. Such smooth extension follows for smooth finite-energy Yang–Mills connections from removable-singularity theory. If G is disconnected, the unframed topological invariant is instead the orbit in $\pi_3(G^0)$ under conjugation by $\pi_0(G)$; topological triviality still means the zero orbit.

Geometry & topology

49. Second fundamental form, principal curvatures and mean curvature

Context. Submanifold geometry; assume Riemannian manifolds, immersions, the Levi-Civita connection and pullback bundles.

Let $F : \Sigma^n \to (M^{n+1}, g)$ be a smooth immersion, let $h = F^*g$, and let ∇^F be the pullback of the Levi-Civita connection. On an open subset of Σ, choose a smooth unit normal ν.

Definition. The **shape operator** $S : T\Sigma \to T\Sigma$ and **second fundamental form** A relative to ν are defined by

$$dF(SX) = \nabla^F_X \nu, \qquad A(X,Y) = g(\nabla^F_X \nu, dF(Y)) = h(SX, Y).$$

The derivative of ν is tangent because ν has constant length. The form A is symmetric, so S is self-adjoint. Its eigenvalues $\kappa_1, \dots, \kappa_n$, counted with multiplicity, are the **principal curvatures**. Set

$$|A|^2 = \sum_{i,j=1}^{n} A(e_i, e_j)^2 = \sum_{i=1}^{n} \kappa_i^2, \qquad H = \operatorname{tr}_h A = \sum_{i=1}^{n} \kappa_i,$$

where (e_i) is any h-orthonormal frame. Here **mean curvature** means the trace, without division by n.

The vector-valued second fundamental form and **mean curvature vector** are

$$\mathrm{II}^{\perp}(X,Y) = (\nabla^F_X dF(Y))^{\perp} = -A(X,Y)\nu, \qquad \vec{H} = -H\nu.$$

These expressions are independent of the choice between ν and $-\nu$, as is $|A|$. They therefore make sense even when the normal line bundle is nontrivial. A hypersurface is **two-sided** if it admits a global unit normal.

Convention. The round sphere of radius r, with its outward normal, has $S = r^{-1}\mathrm{id}$ and $H = n/r$. The alternative convention $A(X,Y) = g(\nabla^F_X dF(Y), \nu)$ reverses the scalar signs. Entry 66 uses the convention fixed here.

50. Lorentzian manifold and time-oriented spacetime

Context. Lorentzian geometry; assume smooth manifolds and symmetric tensor fields.

Definition. A **Lorentzian metric** on a smooth manifold M of dimension $n+1 \geq 2$ is a smooth section g of $\mathrm{Sym}^2 T^*M$ such that each g_p is nondegenerate of signature $(-,+,\dots,+)$. The pair (M,g) is a **Lorentzian manifold**. If a regularity class such as C^k or $C^{k,\alpha}$ is specified, the components of g in smooth charts are required to have that regularity instead; nondegeneracy and signature are still required at every point.

A nonzero vector $v \in T_pM$ is **timelike**, **null**, or **spacelike** according as $g_p(v,v) < 0$, $= 0$, or > 0. It is **causal** if it is timelike or null. The timelike vectors at each point form two connected components.

A **time orientation** is a continuous choice of one of these components, called the future cone. Equivalently, it is an equivalence class of continuous timelike vector fields, where $T \sim T'$ means $g_p(T_p, T'_p) < 0$ for every p. A causal vector v is future-directed precisely when $g(v,T) < 0$ for any representative T. A Lorentzian manifold admitting such a choice is **time-orientable**.

A **spacetime** is a connected time-oriented Lorentzian manifold. A four-dimensional spacetime has dimension $3+1$; the word does not by itself impose dimension four, global hyperbolicity, or any field equation. For continuous metrics, a locally Lipschitz curve is future-directed causal if its tangent is future-directed causal almost everywhere, with nonzero tangent almost everywhere.

51. Null or characteristic hypersurface

Context. Lorentzian hypersurfaces and characteristic initial-value problems; assume entry 50 and embedded hypersurfaces.

Let (M^{n+1}, g) be Lorentzian, with g at least continuous, and let $\mathcal{N} \subset M$ be a C^1 embedded hypersurface.

Definition. The hypersurface $\mathcal{N}$ is **null** if the restriction $g_p|_{T_p\mathcal{N}}$ is degenerate at every $p \in \mathcal{N}$. Equivalently, for every nonzero conormal $\xi \in T_p^*M$ annihilating $T_p\mathcal{N}$,

$$g_p^{-1}(\xi, \xi) = 0.$$

If $\mathcal{N}$ is locally the regular level set $\{f = 0\}$, this condition is

$$g^{\mu\nu}\partial_\mu f\, \partial_\nu f = 0 \quad \text{on } \mathcal{N}, \qquad df \neq 0 \text{ there.}$$

It is independent of the defining function. The induced form is positive semidefinite with one-dimensional radical

$$\operatorname{Rad}(T_p\mathcal{N}) = T_p\mathcal{N} \cap (T_p\mathcal{N})^{\perp_g} = (T_p\mathcal{N})^{\perp_g}.$$

Thus a normal vector to a null hypersurface is also tangent to it. Locally the radical is spanned by the null field $L = (df)^\sharp$ along $\mathcal{N}$.

For a differential operator whose principal symbol is $p(x, \xi) = a(x) g_x^{-1}(\xi, \xi)$ with $a(x) \neq 0$, the characteristic-hypersurface condition $p(x, df) = 0$ is exactly this null condition. The word **characteristic** refers to the chosen operator; a general hyperbolic system can have additional characteristic families.

Local geometric fact. When g and $\mathcal{N}$ are smooth, a smooth nonvanishing radical field satisfies $\nabla_L L = \kappa L$ for a scalar function κ. Its integral curves can locally be reparametrized as null geodesics; these are the **null generators** of $\mathcal{N}$.

52. Minimal hypersurface

Context. The variational theory of area; assume immersed hypersurfaces and the curvature conventions of entry 49.

Let $F : \Sigma^n \to (M^{n+1}, g)$ be a smooth immersion, with induced metric h and mean curvature vector $\vec{H}$.

Definition. The immersion F is a **minimal hypersurface** if $\vec{H}$ vanishes identically. Equivalently, for every local unit normal, its scalar mean curvature H is zero. No global choice of unit normal is required. An embedded hypersurface is minimal when its inclusion is minimal.

First-variation characterization. Let F_t be a smooth variation of $F = F_0$ which equals F outside a compact subset of the interior of Σ, and let $V = \partial_t F_t|_{t=0}$. For a relatively compact domain $U \subset \Sigma$ containing the support of the variation in its interior,

$$\left.\frac{d}{dt}\right|_{t=0} \operatorname{Area}(F_t|_U) = -\int_U g(\vec{H}, V)\, d\mu_h.$$

Consequently, F is minimal if and only if this derivative is zero for every such variation. Area counts multiplicity for an immersion. If Σ has boundary, the permitted variations are supported away from it; free-boundary minimality requires an additional boundary condition.

Remark. Minimality is stationarity of area, not a minimizing property. For a hypersurface with singular closure points, vanishing mean curvature on its smooth part is a local condition. Requiring its associated varifold to be stationary across those points is the additional condition in entry 53.

53. Stationary varifold

Context. Geometric measure theory; assume Radon measures, rectifiable sets, approximate tangent planes and Riemannian geometry.

Let (M^N, g) be a smooth Riemannian manifold, let $U \subset M$ be open, and fix $1 \le k \le N$. Write

$$G_k(U) = \{(x, S) : x \in U,\ S \subset T_xM \text{ is an unoriented } k\text{-plane}\}.$$

Background definition. A **k-varifold** in U is a nonnegative Radon measure V on $G_k(U)$. Its **weight measure** is $\|V\| = \pi_* V$, where $\pi(x, S) = x$. For a countably k-rectifiable set E and a positive measurable multiplicity θ satisfying $\int_{E\cap K} \theta \, d\mathcal{H}^k_g < \infty$ for every compact $K \subset U$, its associated rectifiable varifold is specified by

$$\int_{G_k(U)} \varphi \, dV = \int_E \varphi(x, T_xE)\theta(x) \, d\mathcal{H}^k_g(x) \qquad (\varphi \in C_c(G_k(U))).$$

It is **integral** if θ is integer-valued almost everywhere. A smooth embedded submanifold gives this measure with $\theta = 1$, provided it has locally finite area.

Definition. For $X \in C^1_c(U; TU)$, define

$$\delta_g V(X) = \int_{G_k(U)} \operatorname{div}_S X(x) \, dV(x, S), \qquad \operatorname{div}_S X = \sum_{a=1}^{k} g(\nabla_{e_a} X, e_a),$$

where (e_a) is any orthonormal basis of S. This basis-independent linear functional is the **first variation** of V. The varifold is **stationary in** U if

$$\delta_g V(X) = 0 \qquad \text{for every } X \in C^1_c(U; TU).$$

Rectifiability and integrality are additional properties, not part of stationarity.

For the multiplicity-one varifold of a smooth submanifold closed in U and without boundary in U, the first variation is $-\int g(\vec{H}, X) \, d\mathcal{H}^k_g$. Thus stationarity recovers minimality. The test fields in the definition may meet singular points of the support, so the same definition also imposes a condition there.

54. Character variety of a surface group in a real reductive group

Context. Surface-group representations and nonabelian Hodge theory; assume reductive algebraic groups, surface groups and complete reducibility.

Let Σ be a closed connected oriented surface of genus $g \ge 2$, let $\Gamma = \pi_1(\Sigma, x)$, and let G be the real points of a reductive real algebraic group, or a union of connected components containing the identity component. This includes real forms of complex reductive algebraic groups. Write $\mathfrak{g}$ for its real Lie algebra.

The space $\operatorname{Hom}(\Gamma, G)$ carries the topology of pointwise convergence. A presentation of the surface group identifies it with

$$\left\{ (A_1, B_1, \dots, A_g, B_g) \in G^{2g} : \prod_{i=1}^{g} [A_i, B_i] = 1 \right\}.$$

This topology does not depend on the chosen generators. The group G acts by simultaneous conjugation.

Definition. A representation $\rho : \Gamma \to G$ is **reductive** if the real representation $\operatorname{Ad} \circ \rho$ on $\mathfrak{g}$ is completely reducible: every invariant linear subspace admits an invariant complement. Denote the subspace of such representations by $\operatorname{Hom}^{\mathrm{red}}(\Gamma, G)$.

The G**-character variety** of Σ is the orbit space

$$\mathcal{X}_G(\Sigma) = \mathrm{Hom}^{\mathrm{red}}(\Gamma, G)/G,$$

with the quotient topology. Its points are actual G-conjugacy classes of reductive homomorphisms.

Closed-orbit theorem. In this setting, reductivity is equivalent to closedness of the conjugacy orbit in $\mathrm{Hom}(\Gamma, G)$, and the indicated quotient is Hausdorff. For a complex reductive algebraic group, this topological space agrees with the complex points of the affine geometric invariant theory quotient. For real G the definition above specifies the topological quotient directly.

Changing the basepoint changes Γ by an isomorphism determined up to inner automorphism, and hence gives a canonical identification of these character spaces.

55. Gaussian q-cluster

Context. Weighted isoperimetric problems; assume sets of locally finite perimeter and their reduced boundaries.

Fix $n \geq 1$. The standard Gaussian probability measure on $\mathbb{R}^n$ is $d\gamma_n(x) = w_n(x)\,dx$, where $w_n(x) = (2\pi)^{-n/2} e^{-|x|^2/2}$.

Background definition. The **Gaussian perimeter** of a Borel set E is

$$P_\gamma(E) = \sup\left\{ \int_E (\operatorname{div} X - x \cdot X)\, d\gamma_n : X \in C_c^1(\mathbb{R}^n; \mathbb{R}^n),\ |X| \leq 1 \right\}.$$

Finite Gaussian perimeter implies locally finite Euclidean perimeter, and in that case

$$P_\gamma(E) = \int_{\partial^* E} w_n(x)\, d\mathcal{H}^{n-1}(x).$$

The density here is the n-dimensional Gaussian density, even though the integral is over a hypersurface.

Definition. For an integer $q \geq 2$, a **Gaussian q-cluster** is an ordered tuple $\Omega = (\Omega_1, \dots, \Omega_q)$ of pairwise disjoint Borel subsets of $\mathbb{R}^n$, each of finite Gaussian perimeter, satisfying

$$\gamma_n\left(\mathbb{R}^n \setminus \bigcup_{i=1}^{q} \Omega_i \right) = 0.$$

Its **volume vector** and **total perimeter** are

$$\gamma(\Omega) = (\gamma_n(\Omega_1), \dots, \gamma_n(\Omega_q)), \qquad P_\gamma(\Omega) = \frac{1}{2} \sum_{i=1}^{q} P_\gamma(\Omega_i).$$

The volume vector belongs to $\{v \in [0,1]^q : \sum_i v_i = 1\}$. Empty cells are allowed unless positive volumes are explicitly prescribed. Clusters are identified when corresponding cells differ by Gaussian-null sets; representatives may always be chosen pairwise disjoint.

The factor $1/2$ counts each interface once. More precisely, writing $\Sigma_{ij} = \partial^*\Omega_i \cap \partial^*\Omega_j$, the structure theorem for finite-perimeter partitions gives

$$P_\gamma(\Omega) = \sum_{i<j} \int_{\Sigma_{ij}} w_n\, d\mathcal{H}^{n-1}.$$

56. Linearized Cauchy–Riemann operator

Context. Deformation theory of pseudoholomorphic curves; assume almost complex manifolds, Riemann surfaces, connections and Sobolev spaces of sections.

Let (Σ, j) be a closed Riemann surface, let (M, J) be a smooth almost complex manifold, and let $u : \Sigma \to M$ satisfy $du \circ j = J \circ du$. For a complex bundle (E, J_E), write $\Omega^{0,1}(\Sigma, E)$ for the smooth real-linear bundle maps $A : T\Sigma \to E$ satisfying $A \circ j = -J_E \circ A$.

Definition. Fix the normalization

$$\bar{\partial}_J v = dv + J(v) \circ dv \circ j.$$

The **linearized Cauchy–Riemann operator at** u, with j fixed, is the vertical derivative of this section at its zero u:

$$D_u : \Gamma(u^*TM) \longrightarrow \Omega^{0,1}(\Sigma, u^*TM).$$

Explicitly, for any torsion-free connection ∇ on TM and $\xi \in \Gamma(u^*TM)$,

$$(D_u\xi)(X) = \nabla_X \xi + J(u)\nabla_{jX}\xi + (\nabla_\xi J)(u)\, du(jX).$$

The connections in the first two terms are pulled back by u. This expression is independent of the chosen torsion-free connection because u is J-holomorphic. Its values are J-antilinear one-forms.

It is a real-linear first-order elliptic operator and satisfies

$$D_u(f\xi) = f D_u\xi + df \otimes \xi + (df \circ j) \otimes J\xi \qquad (f \in C^\infty(\Sigma, \mathbb{R})).$$

For integers $k \geq 1$ and $1 < p < \infty$, it extends to a bounded Fredholm operator $W^{k,p}(u^*TM) \to W^{k-1,p}(\Lambda^{0,1}T^*\Sigma \otimes_{\mathbb{C}} u^*TM)$, interpreting this tensor notation as the antilinear-homomorphism bundle above.

Normalization. Many books divide $\bar{\partial}_J$ and D_u by 2. The displayed normalization follows Wendl's *Transversality and super-rigidity for multiply covered holomorphic curves*, §2.1. The operator need not be complex-linear; no variation of the domain complex structure is included here.

57. Lipschitz singular chains and their mass

Context. Filling invariants in metric geometry; assume singular homology, Lipschitz maps and the area formula.

Let (Y, d) be a metric space. Equip the standard simplex $\Delta^k \subset \mathbb{R}^{k+1}$ with its Euclidean metric.

Definition. A **Lipschitz singular k-simplex** is a Lipschitz map $\sigma : \Delta^k \to Y$. The abelian group $C_k^{\mathrm{Lip}}(Y; \mathbb{Z})$ is the free abelian group on these maps. The usual boundary

$$\partial\sigma = \sum_{j=0}^{k} (-1)^j \sigma \circ \delta_j,$$

where δ_j is the jth face inclusion, makes these groups a chain complex. Set $\partial = 0$ in degree zero. A chain with zero boundary is a **Lipschitz cycle**.

Use Hausdorff measure $\mathcal{H}^k$ normalized to equal Lebesgue measure on Euclidean k-space. The **parametrized volume** of a Lipschitz simplex is

$$\mathrm{vol}_k(\sigma) = \int_Y \#\sigma^{-1}(y)\, d\mathcal{H}^k(y).$$

This counts multiplicity. The metric area formula makes this a well-defined finite nonnegative number; the compact metric image is separable, and the multiplicity function is measurable for the completed Hausdorff measure. For $k = 0$, the volume of a simplex is 1.

Write a chain uniquely as $c = \sum_{i=1}^{m} a_i \sigma_i$, with distinct simplices and nonzero integers a_i. Its **mass** is

$$\operatorname{mass}(c) = \sum_{i=1}^{m} |a_i| \operatorname{vol}_k(\sigma_i), \qquad \operatorname{mass}(0) = 0.$$

When Y is Riemannian, the area formula gives

$$\operatorname{vol}_k(\sigma) = \int_{\Delta^k} \sqrt{\det(D\sigma_x^* D\sigma_x)}\, dx,$$

using almost-everywhere differentiability and the induced Euclidean volume on the simplex. Distinct parametrizations remain distinct generators. Thus this is the mass of a singular chain, not the mass after passing to a current and cancelling geometric overlaps.

The **filling volume** of a k-cycle z is $\inf\{\operatorname{mass}(b) : b \in C_{k+1}^{\mathrm{Lip}}(Y;\mathbb{Z}),\ \partial b = z\}$, with $\inf \emptyset = +\infty$.

58. MCG-finite representation of a surface group

Context. Dynamics of mapping class groups on local systems; assume representations, conjugacy, and the constructions in entries 59 and 61.

Let $\Sigma_{g,n}$ be the oriented surface of genus g with n punctures, let $x \in \Sigma_{g,n}$, and fix $r \geq 1$. Put

$$\mathcal{R}_r(\Sigma) = \operatorname{Hom}(\pi_1(\Sigma_{g,n}, x), \mathrm{GL}_r(\mathbb{C}))/\mathrm{GL}_r(\mathbb{C}),$$

where this is the set of ordinary conjugacy classes of homomorphisms. The mapping class group acts on it by precomposition with inverse representatives of its outer action on the fundamental group.

Definition. A representation $\rho : \pi_1(\Sigma_{g,n}, x) \to \mathrm{GL}_r(\mathbb{C})$ is **MCG-finite** if

$$\{f \cdot [\rho] : f \in \mathrm{Mod}_{g,n}\}$$

is a finite set. Equivalently, the subgroup

$$\operatorname{Stab}([\rho]) = \{f \in \mathrm{Mod}_{g,n} : f \cdot [\rho] = [\rho]\}$$

has finite index. In explicit terms, for each f in this subgroup and any representative $\phi_f \in \operatorname{Aut}(\pi_1)$ of its outer action, there is $P_f \in \mathrm{GL}_r(\mathbb{C})$ such that

$$\rho(\phi_f^{-1}(\gamma)) = P_f \rho(\gamma) P_f^{-1} \qquad \text{for all } \gamma \in \pi_1.$$

No compatibility between the different matrices P_f is required.

Replacing $\mathrm{Mod}_{g,n}$ by its finite-index pure subgroup leaves this property unchanged. The definition also makes sense for the unique rank-zero representation, which is MCG-finite.

Remark. Semisimplicity is not assumed. For nonsemisimple representations, using the affine character variety instead would identify a representation with its semisimplification and could change the property. For semisimple representations, its conjugacy class is a closed orbit, so the two formulations agree.

59. Mapping class group of a punctured surface

Context. Surface topology; assume oriented surfaces and isotopy.

Fix integers $g, n \geq 0$, a closed connected oriented surface Σ_g of genus g, and a set $P \subset \Sigma_g$ of n marked points. Write $\Sigma_{g,n} = \Sigma_g \setminus P$.

Definition. Let $\mathrm{Homeo}^+(\Sigma_g, P)$ be the group of orientation-preserving homeomorphisms of Σ_g that preserve P as a set. Two such homeomorphisms f_0, f_1 are **isotopic** if there is a continuous map $H : \Sigma_g \times [0,1] \to \Sigma_g$ such that every H_t belongs to this group and $H_0 = f_0$, $H_1 = f_1$.

The **mapping class group** of the punctured surface is

$$\mathrm{Mod}_{g,n} = \mathrm{Homeo}^+(\Sigma_g, P)/\text{isotopy}, \qquad [f][h] = [f \circ h].$$

Equivalently, it is the group of orientation-preserving homeomorphisms of $\Sigma_{g,n}$ modulo isotopy. Homeomorphisms of this finite-type punctured surface extend to its marked-point compactification, and the two descriptions give the same group. The group is also the group of path components of the homeomorphism group with its usual compact-open topology.

Each mapping class induces a permutation of P, since this permutation is constant along an isotopy. The **pure mapping class group** is

$$\mathrm{PMod}_{g,n} = \ker(\mathrm{Mod}_{g,n} \longrightarrow \mathrm{Sym}(P)).$$

The permutation map is surjective, so the pure subgroup has index $n!$.

Convention. Punctures may be permuted in $\mathrm{Mod}_{g,n}$. For a surface with boundary, the frequently used convention of fixing the boundary pointwise gives a different mapping class group. Orientation-reversing homeomorphisms are excluded here. On a smooth surface, replacing homeomorphisms and isotopies by diffeomorphisms and smooth isotopies yields the same group.

60. Normal Cauchy–Riemann operator

Context. Normal deformations of pseudoholomorphic curves; assume entry 56 and complex vector bundles.

Let $u : (\Sigma, j) \to (M^{2n}, J)$ be a nonconstant smooth J-holomorphic map from a closed connected Riemann surface.

Local background theorem. The critical points of u are isolated and hence finite. The complex line $du(T_z\Sigma)$, defined at regular points, extends uniquely to a smooth complex line subbundle $T_u \subset u^*TM$, called the **generalized tangent bundle**. The linearized operator of entry 56 preserves it:

$$D_u(\Gamma(T_u)) \subseteq \Omega^{0,1}(\Sigma, T_u).$$

This is the standard extension and tangency theorem for pseudoholomorphic curves; the extension is essential at branch points.

Definition. The **generalized normal bundle** is the complex vector bundle

$$N_u = u^*TM/T_u.$$

The **normal Cauchy–Riemann operator** is the induced real-linear first-order operator

$$D_u^N : \Gamma(N_u) \longrightarrow \Omega^{0,1}(\Sigma, N_u), \qquad D_u^N \eta = q(D_u \tilde{\eta}),$$

where $q : u^*TM \to N_u$ is the quotient and $\tilde{\eta}$ is any smooth lift of η. The tangency theorem makes this independent of the lift. A choice of a J-invariant complement identifies N_u with that complement and gives the formula $D_u^N = \pi_N D_u|_{N_u}$.

For an immersion, $T_u = du(T\Sigma)$. At a critical point the pointwise quotient by $\operatorname{im} du$ is not a vector bundle, so it must be replaced by the quotient by T_u. If $u = v \circ \varphi$, with v immersed and φ a nonconstant holomorphic branched cover, then N_u is canonically $\varphi^* N_v$.

With the normalization of entry 56, this operator satisfies the usual real Cauchy–Riemann Leibniz rule. It extends to a Fredholm map $W^{k,p}(N_u) \to W^{k-1,p}(\Lambda^{0,1}T^*\Sigma \otimes_{\mathbb{C}} N_u)$ for $k \geq 1$ and $1 < p < \infty$. A **nontrivial normal infinitesimal deformation** is a nonzero element of its kernel.

Source for the extension theorem and convention: Wendl, *Transversality and super-rigidity for multiply covered holomorphic curves*, §2.1.

61. Outer action of the mapping class group

Context. Mapping class groups and local systems; assume fundamental groups, change of basepoint and entry 59.

Fix $x \in \Sigma = \Sigma_{g,n}$ and let $\Gamma = \pi_1(\Sigma, x)$. Write $\operatorname{Out}(\Gamma) = \operatorname{Aut}(\Gamma)/\operatorname{Inn}(\Gamma)$. Path concatenation below traverses the left-hand path first.

Construction. For an orientation-preserving homeomorphism $f : \Sigma \to \Sigma$, choose a path δ from x to $f(x)$. The formula

$$f_\delta([\gamma]) = [\delta * (f \circ \gamma) * \delta^{-1}]$$

defines an automorphism of Γ. Changing δ changes f_δ by an inner automorphism. Isotopic homeomorphisms give the same outer automorphism. Moreover, a path suitable for $f \circ h$ is $\delta_f * (f \circ \delta_h)$, and with this choice $(f \circ h)_\delta = f_{\delta_f} \circ h_{\delta_h}$.

Definition. The **outer action of the mapping class group** is the homomorphism

$$\Psi : \operatorname{Mod}_{g,n} \longrightarrow \operatorname{Out}(\Gamma), \qquad [f] \longmapsto [f_\delta].$$

For any group G, it induces a left action on conjugacy classes of representations by

$$[f] \cdot [\rho] = [\rho \circ f_\delta^{-1}] \quad \text{in } \operatorname{Hom}(\Gamma, G)/G.$$

If f_δ is changed by an inner automorphism, the new representation is conjugate to the old one by an element of $\rho(\Gamma)$, so this action is well-defined. The inverse ensures the left-action identity $[fh] \cdot [\rho] = [f] \cdot ([h] \cdot [\rho])$.

Changing the basepoint gives a compatible canonical identification of the quotient sets of representations. A representative homeomorphism fixing x gives an actual automorphism f_*; without such a choice the mapping class group supplies an outer automorphism, not a specified automorphism of the based fundamental group.

62. Riemannian symmetric space of a semisimple Lie group

Context. Symmetric spaces of noncompact type; assume semisimple Lie groups, their Killing form, homogeneous manifolds and Riemannian distance.

Let G be a connected real semisimple Lie group with finite centre and let K be a maximal compact subgroup. Write $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ for the corresponding Cartan decomposition. Equivalently, $\mathfrak{p}$ is the Killing-orthogonal complement of $\mathfrak{k}$. The Killing form B is positive definite on $\mathfrak{p}$ and invariant under $\operatorname{Ad}(K)$.

Definition. The **Riemannian symmetric space associated to** G is $X = G/K$, with the unique G-invariant Riemannian metric whose value at $o = eK$ is

$$g_{X,o}(d\pi_e A, d\pi_e C) = B(A, C) \qquad (A, C \in \mathfrak{p}),$$

where $\pi : G \to G/K$. Here $d\pi_e|_{\mathfrak{p}}$ identifies $\mathfrak{p}$ with T_oX. Explicitly, the metric at gK is transported from this inner product by left translation by g; $\mathrm{Ad}(K)$-invariance makes it independent of the representative g.

The **distance** $d_X(p, q)$ is the infimum of g_X-lengths of piecewise smooth paths from p to q.

Cartan's geometric theorem. The map $\mathfrak{p} \to X$, $A \mapsto \exp(A)K$, is a diffeomorphism. The curves $t \mapsto \exp(tA)K$ are geodesics; X is complete, simply connected and has nonpositive sectional curvature. The map $\exp(A)K \mapsto \exp(-A)K$ is an isometry fixing o with derivative $-\mathrm{id}$, and its conjugates give symmetries at every point. In particular,

$$d_X(gK, hK) = \sqrt{B(A, A)} \quad \text{if } g^{-1}h = \exp(A)k,\ A \in \mathfrak{p},\ k \in K.$$

Different maximal compact subgroups give G-equivariantly isometric spaces. The stated metric fixes its normalization; rescaling simple factors is another common convention. Compact factors of G contribute a point to X.

63. Space of compatible almost complex structures

Context. Symplectic topology and genericity for holomorphic curves; assume symplectic manifolds and spaces of smooth tensor fields.

Let (M, ω) be a smooth symplectic manifold.

Definition. A smooth endomorphism field $J : TM \to TM$ is an **ω-compatible almost complex structure** if, at every $x \in M$,

$$J_x^2 = -\mathrm{id}, \qquad \omega_x(J_x v, J_x w) = \omega_x(v, w) \quad (v, w \in T_xM),$$

and

$$\omega_x(v, J_x v) > 0 \qquad (0 \neq v \in T_xM).$$

Equivalently, $J^2 = -\mathrm{id}$ and $g_J(v, w) = \omega(v, Jw)$ is a Riemannian metric. No integrability of J is required. If only $J^2 = -\mathrm{id}$ and the positivity condition are imposed, J is called **ω-tame**.

The **space of compatible almost complex structures** is

$$\mathcal{J}(M, \omega) = \{J \in \Gamma^\infty(\mathrm{End}\, TM) : J \text{ is } \omega\text{-compatible}\},$$

with the subspace topology of the compact-open C^∞ topology on smooth sections. More explicitly, an auxiliary Riemannian metric and connection define seminorms

$$p_{K,k}(A) = \max_{0 \le j \le k} \sup_{x \in K} |\nabla^j A(x)|,$$

for all compact $K \subset M$ and $k \ge 0$. These seminorms define a topology independent of the auxiliary choices.

Given $J_0 \in \mathcal{J}(M, \omega)$ and an open set $U \subset M$, define

$$\mathcal{J}(M, \omega; U, J_0) = \{J \in \mathcal{J}(M, \omega) : J|_{M \setminus U} = J_0|_{M \setminus U}\},$$

with the subspace topology. When U has compact closure, convergence in this space is uniform convergence on $\overline{U}$ of every covariant derivative of $J - J_0$; the weak and strong C^∞ topologies then agree on this space.

Compatible structures exist, and these spaces are contractible and Baire. Thus a genericity statement may be formulated using a countable intersection of open dense subsets.

64. Spin^c structure on an oriented manifold

Context. Gauge theory and low-dimensional topology; assume spin groups, principal bundles, characteristic classes and frame bundles.

For $n \geq 2$ let

$$\mathrm{Spin}^c(n) = (\mathrm{Spin}(n) \times U(1))/\{(1,1),(-1,-1)\}.$$

It has homomorphisms $\lambda^c([a,z]) = \lambda(a)$ to $\mathrm{SO}(n)$ and $\det([a,z]) = z^2$ to $U(1)$, where λ is the spin double cover. The kernel of λ^c is the central copy $\{[1,z] : z \in U(1)\}$.

Definition. Let M be an oriented smooth n-manifold with an auxiliary Riemannian metric. A Spin^c **structure** is a principal $\mathrm{Spin}^c(n)$-bundle $P \to M$ together with a map $\Phi : P \to F_{\mathrm{SO}}(TM)$ over M satisfying

$$\Phi(p \cdot a) = \Phi(p) \cdot \lambda^c(a).$$

An isomorphism of structures is a principal-bundle isomorphism commuting with these maps to the oriented orthonormal frame bundle. Write $\mathrm{Spin}^c(M)$ for the set of isomorphism classes. Its **determinant line** is the complex line bundle

$$L_P = P \times_{\det} \mathbb{C},$$

and $c_1(\mathfrak{s})$ means $c_1(L_P)$ for a representative of $\mathfrak{s}$.

Classification facts. The set is independent of the auxiliary metric. It is nonempty precisely when $w_2(TM)$ lifts to an integral class, equivalently when its integral Bockstein $W_3(TM)$ vanishes. When nonempty, it is a torsor for $H^2(M;\mathbb{Z})$: twisting by a complex line bundle E defines $\mathfrak{s} + c_1(E)$ and

$$L_{\mathfrak{s}+c_1(E)} \simeq L_{\mathfrak{s}} \otimes E^{\otimes 2}, \qquad c_1(\mathfrak{s}+a) = c_1(\mathfrak{s}) + 2a.$$

In particular the first Chern class alone need not determine the structure.

On a closed oriented three-manifold this set is equivalently the set of homology classes of nowhere-zero vector fields: two fields are homologous if they are homotopic through nowhere-zero fields outside an embedded open ball in each component. The field determines the complex oriented two-plane bundle orthogonal to it. Restriction to a boundary uses the outward-normal-first orientation and the inclusion $\mathrm{Spin}^c(n-1) \subset \mathrm{Spin}^c(n)$.

65. Strict stability of a minimal cone or hypersurface

Context. Second variation and singular minimal hypersurfaces; assume entries 49 and 52 and Sobolev quadratic forms.

Let M^d be a smooth two-sided minimal hypersurface in a Riemannian manifold (N^{d+1}, g), with unit normal ν. Its normal second-variation form is

$$Q_g(\zeta) = \int_M \left(|\nabla_M \zeta|_g^2 - (|A|_g^2 + \mathrm{Ric}_g(\nu,\nu))\zeta^2 \right) d\mu_g.$$

Usual **stability** means $Q_g(\zeta) \geq 0$ for all $\zeta \in C_c^\infty(M)$.

Definition (weighted strict stability). Fix a positive radial or distance weight r on M. The hypersurface is **strictly stable with weight** r if there is a constant $\kappa > 0$ such that

$$Q_g(\zeta) \geq \kappa \int_M \left(|\nabla_M \zeta|_g^2 + r^{-2}\zeta^2 \right) d\mu_g \qquad \text{for every } \zeta \in C_c^\infty(M).$$

For a Euclidean minimal cone, apply this to its regular part with $r(X) = |X|$ and Euclidean metric. Test functions are then compactly supported away from its singular set. This scale-sensitive definition specifies more than positivity of Q_g on each individual nonzero test function.

Definition (the precise inequality used by Simon). In $\mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^\ell$, write $X = (x, \xi, y)$ and $\tilde{r} = |x| + |\xi|$. For a smooth embedded hypersurface M disjoint from the axis $\{x = \xi = 0\}$, assume the integrals below are finite for restrictions of ambient compactly supported smooth functions. Simon's strict-stability condition is the existence of $\kappa > 0$ such that

$$\kappa \int_M \left(|\nabla_M \zeta|^2 + \tilde{r}^{-2} \zeta^2 \right) d\mathcal{H}^d \leq \int_M \left(|\nabla_M \zeta|^2 - |A_M|^2 \zeta^2 \right) d\mathcal{H}^d$$

for every $\zeta \in C_c^\infty(\mathbb{R}^{n+m+\ell})$, restricted to M. All quantities in this display use the **Euclidean** metric, even when M is minimal for a nearby metric g. Ambient test functions may meet singular points of the closure of M, so this test class is part of the definition.

This is the convention in Simon, §1, inequality (1.2). In that setting the Euclidean quadratic expression approximates the g-second variation with a controlled error; the two expressions should not be identified by definition.

66. Strictly convex boundary

Context. Riemannian manifolds with boundary and geodesic rigidity; assume the convention for the second fundamental form in entry 49.

Let (M^n, g) be a smooth Riemannian manifold with smooth boundary, $n \geq 2$. Write ν_{out} for the outward unit normal and $\nu_{\text{in}} = -\nu_{\text{out}}$.

Definition. The boundary is **strictly convex at** $p \in \partial M$ if

$$g_p(\nabla_v \nu_{\text{out}}, v) > 0 \qquad \text{for every } 0 \neq v \in T_p \partial M.$$

It is **strictly convex** if this holds at every boundary point. Thus, with entry 49's convention, the second fundamental form is positive definite for the outward normal.

Equivalently, using the other common convention for a scalar second fundamental form, one requires positivity of

$$\mathrm{II}_{\text{in}}(v, w) = g(\nabla_v \tilde{w}, \nu_{\text{in}}),$$

where $\tilde{w}$ is tangent to the boundary along the boundary. Differentiating $g(\tilde{w}, \nu_{\text{in}}) = 0$ shows that this equals $g(\nabla_v \nu_{\text{out}}, w)$. This explains the phrase "positive second fundamental form with respect to the inward normal." The formula and the normal must be specified together.

For a boundary defining function ρ, positive inside M, with $\rho = 0$ and $d\rho \neq 0$ on ∂M, one has

$$g(\nabla_v \nu_{\text{out}}, w) = -\frac{\mathrm{Hess}_g \rho(v, w)}{|d\rho|_g} \qquad (v, w \in T\partial M).$$

Hence strict convexity is equivalent to negative definiteness of the tangential Hessian of ρ. In a smooth extension of the metric across the boundary, every nonconstant geodesic tangent to a strictly convex boundary point leaves M to second order. Euclidean balls satisfy this convention.

Number theory

67. Local components of an automorphic representation

Context. This definition belongs to the tensor product theorem for automorphic representations.

Let F be a number field and G/F a connected reductive group. Fix maximal compact subgroups K_v at infinity and hyperspecial compact open subgroups $K_v \subset G(F_v)$ at almost all finite places. At finite places use smooth admissible complex representations; at infinity use admissible $(\mathfrak{g}_v, K_v)$-modules, with $\mathfrak{g}_v = \mathrm{Lie}(G(F_v)) \otimes_{\mathbb{R}} \mathbb{C}$.

Restricted tensor product. Suppose (π_v, V_v) is given at every place and, outside a finite set S_0 containing infinity, choose $0 \neq e_v \in V_v^{K_v}$. Define

$$\bigotimes_v{}' (V_v, e_v) = \varinjlim_{S \supseteq S_0} \bigotimes_{v \in S} V_v,$$

where S ranges over finite sets of places and the transition map from S to S' tensors with $\bigotimes_{v \in S' \setminus S} e_v$. Thus pure tensors have component e_v at almost every place. The finite adelic group acts componentwise, and each archimedean Lie algebra and compact group acts on its corresponding factor.

Tensor product theorem (Flath). Every irreducible admissible $(\mathfrak{g}_\infty, K_\infty) \times G(\mathbb{A}_F^\infty)$-module π admits a decomposition

$$\pi \simeq \bigotimes_v{}' \pi_v$$

with irreducible admissible factors, uniquely determined up to isomorphism. Almost all finite factors are *unramified*, meaning $\pi_v^{K_v} \neq 0$; their fixed spaces are one-dimensional, so the isomorphism class of the restricted tensor product does not depend on the chosen nonzero e_v.

Definition. The factor π_v, regarded up to isomorphism, is the **local component** of π at v. This applies in particular to an irreducible automorphic representation. For a unitary automorphic representation, one first takes its smooth K_∞-finite vectors; equivalently one uses the corresponding restricted Hilbert tensor product of unitary local representations.

68. Arithmetic lattice in a semisimple Lie group

Context. This definition belongs to a chapter on arithmetic subgroups and lattices.

Let G be a connected semisimple linear real Lie group with finite centre. A lattice is a discrete subgroup $\Gamma \subset G$ for which G/Γ has finite invariant volume. Two subgroups are *commensurable* if their intersection has finite index in both.

For a connected semisimple algebraic group $\mathbf{H}/\mathbb{Q}$, choose a faithful rational representation $\rho : \mathbf{H} \hookrightarrow \mathrm{GL}_m$ and set

$$\mathbf{H}(\mathbb{Z}) = \{ h \in \mathbf{H}(\mathbb{Q}) : \rho(h) \in \mathrm{GL}_m(\mathbb{Z}) \}.$$

Its commensurability class is independent of ρ.

Definition. A lattice $\Gamma \subset G$ is **arithmetic** if there exist such a group $\mathbf{H}$ and a continuous surjective homomorphism

$$p : \mathbf{H}(\mathbb{R})^\circ \longrightarrow G$$

with compact kernel such that Γ is commensurable with

$$p(\mathbf{H}(\mathbb{Z}) \cap \mathbf{H}(\mathbb{R})^\circ).$$

Here the superscript $\circ$ denotes the identity component in the real topology. A lattice which fails this condition is **non-arithmetic**. The definition depends only on the commensurability class of Γ and is invariant under automorphisms of G.

For a connected semisimple group with finite centre that is not assumed linear, arithmeticity is defined by requiring the image of Γ in the adjoint group $G/Z(G)$ to be arithmetic. This extends the definition to finite central covers. For a group with finitely many connected components, apply the definition to $\Gamma \cap G^\circ$ in G°.

69. Galois representation attached to a cuspidal Hecke eigenform

Context. This definition follows the construction of Galois representations associated with modular forms.

Let $f = \sum_{n \geq 1} a_n q^n$ be a normalized cuspidal newform of weight $k \geq 2$, level N, and Dirichlet character ε. Let K_f be its coefficient field and λ a finite place of K_f above the rational prime p. Regard ε as a finite-order $K_{f,\lambda}$-valued Galois character, with $\varepsilon(\mathrm{Frob}_\ell) = \varepsilon(\ell)$ for $\ell \nmid Np$.

Existence theorem (Eichler–Shimura–Deligne). There is a continuous semisimple representation of $G_{\mathbb{Q}} = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on a two-dimensional $K_{f,\lambda}$-vector space, unramified outside Np, such that

$$\det(T - \rho_{f,\lambda}(\mathrm{Frob}_\ell)) = T^2 - a_\ell T + \varepsilon(\ell)\ell^{k-1} \qquad (\ell \nmid Np).$$

Here Frob_ℓ is **arithmetic Frobenius**, inducing $x \mapsto x^\ell$ on the residue field. These characteristic polynomials determine the semisimple representation up to isomorphism.

Definition. This isomorphism class is the **λ-adic Galois representation attached to f**. Choosing a basis gives a homomorphism $\rho_{f,\lambda} : G_{\mathbb{Q}} \to \mathrm{GL}_2(K_{f,\lambda})$, determined up to conjugacy. With the same Frobenius convention,

$$\det \rho_{f,\lambda} = \varepsilon \chi_p^{k-1},$$

where χ_p is the p-adic cyclotomic character.

For a normalized cuspidal eigenform which is old at level N, the attached representation is that of its underlying primitive newform; equivalently it is characterized by the same polynomials at primes away from Np. The word "attached" specifies an isomorphism class, without choosing a lattice or a basis.

70. Abelian scheme with complex multiplication of a specified CM type

Context. This definition belongs to the theory of CM abelian schemes and their eigenspaces of invariant differentials.

Fix $\overline{\mathbb{Q}} \subset \mathbb{C}$. A **CM field** is a totally imaginary quadratic extension of a totally real number field. A **CM type** of a CM field L is a subset $\Sigma \subset J_L = \mathrm{Hom}(L, \overline{\mathbb{Q}})$ containing exactly one of $\sigma, \bar{\sigma}$ for every conjugate pair.

More generally, if L contains a CM field and K is its maximal CM subfield, a CM type of L means the inverse image of a CM type of K under $J_L \to J_K$. In either case, write $[L : \mathbb{Q}] = 2g$.

Let R be an algebra over $O_{L^{\mathrm{Gal}}}[1/d_L]$, where d_L is the discriminant and L^{Gal} the Galois closure. There is a canonical splitting

$$O_L \otimes_{\mathbb{Z}} R \simeq \prod_{\sigma \in J_L} R, \qquad a \otimes r \longmapsto (\sigma(a) r)_\sigma.$$

Definition. An **abelian scheme with CM by O_L of type Σ** over R is an abelian scheme A/R of relative dimension g, together with an injective unital homomorphism $\iota : O_L \to \mathrm{End}_R(A)$, for

which

$$\mathrm{Lie}(A/R) = \bigoplus_{\sigma \in \Sigma} \mathrm{Lie}(A/R)_{\sigma},$$

where each summand is an invertible R-module and $d\iota(a)$ acts on it by multiplication by $\sigma(a)$. All eigensummands indexed by $J_L \setminus \Sigma$ vanish.

Thus every selected embedding occurs once. On invariant differentials the pullback $\iota(a)^*$ has the same eigenvalues; it must not be confused with the contragredient group action. On the dual abelian scheme the action $a \mapsto \iota(a)^\vee$ has type $\bar{\Sigma}$.

Over bases containing ramified primes, one must specify an integral CM moduli problem; the usual determinant condition is $\det(T - d\iota(a)) = \prod_{\sigma \in \Sigma}(T - \sigma(a))$. The split-base definition above avoids that additional choice.

71. Abelian variety attached to a weight-two newform

Context. This construction belongs to a chapter on modular Jacobians and their Hecke quotients.

Let $f = \sum_{n \geq 1} a_n q^n$ be a normalized newform in $S_2(\Gamma_0(N))$, with trivial character. Write $J_0(N) = \mathrm{Jac}(X_0(N))$, and let $\mathbb{T} \subset \mathrm{End}_{\mathbb{Q}}(J_0(N))$ be the integral Hecke algebra generated by the standard Hecke correspondences. Its action on differentials agrees with the usual action on weight-two cusp forms.

The eigenvalues define a ring homomorphism

$$\lambda_f : \mathbb{T} \longrightarrow K_f = \mathbb{Q}(a_n : n \geq 1), \qquad T_n \longmapsto a_n.$$

Put $I_f = \ker \lambda_f$. For an ideal $I \subset \mathbb{T}$, the notation $IJ_0(N)$ denotes the abelian subvariety generated by the images $t(J_0(N))$, $t \in I$; equivalently it is the image of the sum map from finitely many such images for generators of I.

Definition. The **abelian variety attached to** f is the quotient over $\mathbb{Q}$

$$A_f = J_0(N)/I_f J_0(N).$$

The quotient map has connected kernel. The Hecke action descends to A_f, and yields an embedding

$$K_f \hookrightarrow \mathrm{End}_{\mathbb{Q}}(A_f) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

The modular quotient theorem gives $\dim A_f = [K_f : \mathbb{Q}]$. Galois-conjugate newforms define the same ideal I_f, hence the same quotient. For newforms with nontrivial character, the analogous construction uses $J_1(N)$ and its Hecke algebra.

72. CM automorphic representation

Context. This definition belongs to the theory of dihedral automorphic representations and Hilbert modular forms.

Let F be totally real, and let π be a cuspidal automorphic representation of $\mathrm{GL}_2(\mathbb{A}_F)$ whose components at all real places are essentially discrete series. For a Hecke character $\eta : F^\times \backslash \mathbb{A}_F^\times \to \mathbb{C}^\times$, its **twist** is the representation on the same abstract space with action

$$g \longmapsto \eta(\det g)\pi(g),$$

denoted $\pi \otimes (\eta \circ \det)$.

Definition. The representation π **has CM** if there is a nontrivial quadratic Hecke character η such that

$$\pi \simeq \pi \otimes (\eta \circ \det).$$

Otherwise it is **non-CM**. One may equivalently quantify over all nontrivial Hecke characters: equality of central characters forces every self-twist character here to satisfy $\eta^2 = 1$.

Characterization. Such a self-twist exists precisely when

$$\pi \simeq \mathrm{AI}_{E/F}(\theta)$$

for a quadratic CM extension E/F and a Hecke character θ of E with $\theta \neq \theta \circ c$, where c is the nontrivial element of $\mathrm{Gal}(E/F)$. The quadratic character is the one attached to E/F by class field theory.

For a general number field and without the discrete-series hypothesis, the same self-twist condition defines a **dihedral** representation; its inducing quadratic extension need not be CM. A quadratic self-twist on GL_n is likewise a meaningful condition, but for arbitrary n it is not by itself a definition of complex multiplication.

73. Central character of an automorphic representation

Context. This definition belongs to automorphic representations with prescribed action of the centre.

Let G/F be connected reductive, let Z_G be its centre, and let π be an irreducible automorphic representation of $G(\mathbb{A}_F)$.

Definition. The **central character** of π is the continuous character

$$\omega_\pi : Z_G(F)\backslash Z_G(\mathbb{A}_F) \longrightarrow \mathbb{C}^\times$$

characterized by

$$\pi(z)v = \omega_\pi(z)v \qquad (z \in Z_G(\mathbb{A}_F),\ v \in V_\pi).$$

In the automorphic realization with right translation, this reads $\varphi(gz) = \omega_\pi(z)\varphi(g)$. Scalar action follows from irreducibility and admissibility; triviality on $Z_G(F)$ follows from left $G(F)$-invariance of automorphic forms.

For a unitary representation, $|\omega_\pi(z)| = 1$ for every z. This is the convention for the unitary cuspidal spectrum. An arbitrary nonunitary twist can have a nonunitary central character, so unitarity is a hypothesis rather than part of the general definition.

If $\pi = \bigotimes_v' \pi_v$, its local central characters satisfy

$$\omega_\pi((z_v)_v) = \prod_v \omega_{\pi_v}(z_v).$$

Almost all factors equal 1 on a given adelic central element. For $G = \mathrm{GL}_n$, the centre is $\mathbb{G}_m$, so ω_π is a Hecke character of F.

74. Complex CM periods of a CM abelian scheme

Context. This definition compares Betti homology with invariant differentials in complex multiplication.

Let $R \subset \mathbb{C}$, and let $(A/R, \iota)$ have CM by O_L of type Σ, as in Definition 70. Assume the eigensummands of invariant differentials are free, and choose generators ω_σ for $\sigma \in \Sigma$. Thus $\iota(a)^*\omega_\sigma = \sigma(a)\omega_\sigma$.

Choose an O_L**-structure**, namely an O_L-linear isomorphism

$$\xi : O_L \xrightarrow{\sim} H_1(A(\mathbb{C}), \mathbb{Z}).$$

This is additional data: the homology lattice is generally an invertible O_L-module and need not be free.

Definition. The **complex CM period vector** attached to these choices is

$$\Omega = (\Omega_\sigma)_{\sigma\in\Sigma}, \qquad \Omega_\sigma = \int_{\xi(1)} \omega_\sigma \in \mathbb{C}^\times.$$

Under the coordinates on $\mathrm{Lie}(A_{\mathbb{C}})$ defined by the ω_σ, the period lattice is

$$\{(\Omega_\sigma \sigma(a))_{\sigma\in\Sigma} : a \in O_L\}.$$

For compatible periods of the dual, equip $A^\vee$ with the action $a \mapsto \iota(a)^\vee$ and choose generators $\omega_\tau^\vee$ for $\tau \in \bar{\Sigma}$. Use the canonical Betti pairing with values in $2\pi i\mathbb{Q}$ and specify the induced rational structure $\xi^\vee : L \xrightarrow{\sim} H_1(A^\vee(\mathbb{C}), \mathbb{Q})$ by

$$\langle \xi(x), \xi^\vee(y)\rangle = 2\pi i \,\mathrm{Tr}_{L/\mathbb{Q}}(xy).$$

Define $\Omega_\tau^\vee = \int_{\xi^\vee(1)} \omega_\tau^\vee$.

For integer multi-indices α on Σ and β on $\bar{\Sigma}$, put

$$\Omega^\alpha = \prod_{\sigma\in\Sigma} \Omega_\sigma^{\alpha_\sigma}, \qquad (\Omega^\vee)^\beta = \prod_{\tau\in\bar{\Sigma}} (\Omega_\tau^\vee)^{\beta_\tau}.$$

These are periods relative to the chosen bases and lattice structure; multiplying ω_σ by $u_\sigma \in R^\times$ multiplies Ω_σ by u_σ.

75. Critical infinity types of Hecke characters

Context. This definition fixes the sign and CM-type conventions used for critical Hecke L-values.

Let L contain a CM field, let K be its maximal CM subfield, and put $J_L = \mathrm{Hom}(L, \overline{\mathbb{Q}})$. An integral infinity type is a tuple $\mu = (\mu_\sigma)_{\sigma\in J_L} \in \mathbb{Z}^{J_L}$. Write

$$a^\mu = \prod_{\sigma\in J_L} \sigma(a)^{\mu_\sigma}.$$

Use the convention that an algebraic ideal-theoretic Hecke character of conductor dividing $\mathfrak{f}$ has type μ if $\chi((a)) = a^\mu$ whenever $a \equiv 1 \pmod{\mathfrak{f}}$ multiplicatively.

Let $\Gamma \subset O_L^\times$ have finite index. Define $\mathrm{HChar}_L(\Gamma)$ to consist of the tuples satisfying all of the following: $\mu_\sigma = \nu_{\sigma|_K}$ for a tuple $\nu \in \mathbb{Z}^{J_K}$; there is an integer w with $\nu_\tau + \nu_{\bar\tau} = w$ for every τ; and $u^\mu = 1$ for every $u \in \Gamma$.

Definition. A type $\mu \in \mathrm{HChar}_L(\Gamma)$ is **critical** if

$$\Sigma_\mu = \{\sigma \in J_L : \mu_\sigma < 0\}$$

is a CM type of L in the sense of Definition 70. Write $\mathrm{Crit}_L(\Gamma)$ for the set of critical types. Thus exactly one member of every conjugate pair has negative exponent, and the negative-support set is lifted from K.

For $\Sigma = \Sigma_\mu$ there is a unique decomposition $\mu = \beta - \alpha$, where α is supported on Σ, with $\alpha_\sigma \geq 1$, and β is supported on $\bar{\Sigma}$, with $\beta_\tau \geq 0$. In particular zero exponents belong to the nonnegative side.

For a nonzero integral ideal $\mathfrak{f}$, set $O_{\mathfrak{f}}^\times = \{u \in O_L^\times : u \equiv 1 \pmod{\mathfrak{f}}\}$. The notation in the question is $\mathrm{Crit}_L(O_{\mathfrak{f}}^\times)$. If L is itself CM, take $K = L$ throughout.

76. Cuspidal automorphic representation attached to a newform

Context. This construction passes from classical modular forms to adelic automorphic representations.

Let $f \in S_k(\Gamma_0(N), \varepsilon)$ be a normalized newform, $k \geq 2$, with convention $f|_k\gamma = \varepsilon(d)f$ for $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)$. For $g \in \mathrm{GL}_2^+(\mathbb{R})$ define

$$(f|_k g)(z) = \det(g)^{k/2}(cz + d)^{-k} f(gz).$$

Let $K_0(N) \subset \mathrm{GL}_2(\widehat{\mathbb{Z}})$ consist of matrices whose lower-left entry is divisible by N, and put $\lambda_\varepsilon(\kappa) = \varepsilon(d \bmod N)^{-1}$.

Strong approximation gives

$$\mathrm{GL}_2(\mathbb{A}_{\mathbb{Q}}) = \mathrm{GL}_2(\mathbb{Q})\bigl(\mathrm{GL}_2^+(\mathbb{R}) \times K_0(N)\bigr).$$

The **adelic lift** of f is the well-defined cusp form

$$\varphi_f(\gamma g_\infty \kappa) = (f|_k g_\infty)(i)\lambda_\varepsilon(\kappa).$$

Definition. The **cuspidal automorphic representation attached to** f, denoted π_f, is the $(\mathfrak{gl}_2(\mathbb{C}), \mathrm{O}(2)) \times \mathrm{GL}_2(\mathbb{A}_{\mathbb{Q}}^{\infty})$-module generated by φ_f under right translation and the differentiated archimedean action. The newform theorem implies that this module is irreducible. One may equivalently use its unitary Hilbert space realization.

This is the unitary normalization. If $\ell \nmid N$, its unramified Satake parameters α_ℓ, β_ℓ satisfy

$$\alpha_\ell + \beta_\ell = a_\ell \ell^{-(k-1)/2}, \qquad \alpha_\ell \beta_\ell = \varepsilon(\ell).$$

For the finite standard L-functions,

$$L(\pi_f, s) = L(f, s + (k-1)/2).$$

In particular the normalization $L(\pi_f, s - \tfrac{1}{2}) = L(f, s)$ is the weight-two case. The archimedean component is the discrete series of weight k. The central character is the unitary adelic character whose restriction to $\widehat{\mathbb{Z}}^\times$ is ε^{-1} and whose value on an unramified uniformizer at ℓ is $\varepsilon(\ell)$.

77. Emerton's completed cohomology

Context. This definition belongs to the cohomology of towers of modular curves and the p-adic Langlands programme.

Fix a prime p and a compact open subgroup $K^p \subset \mathrm{GL}_2(\mathbb{A}_{\mathbb{Q}}^{\infty,p})$. For sufficiently small compact open $K_p \subset \mathrm{GL}_2(\mathbb{Q}_p)$, let $Y_{K^pK_p}/\mathbb{Q}$ be the modular curve of level K^pK_p. When $K_p' \subset K_p$, the finite level map induces pullback on cohomology.

Definition. The **completed cohomology of tame level** K^p is

$$\widetilde{H}^i(K^p, \mathbb{Z}_p) = \varprojlim_{n \geq 1} \varinjlim_{K_p} H^i_{\text{et}}(Y_{K^pK_p, \overline{\mathbb{Q}}}, \mathbb{Z}/p^n\mathbb{Z}).$$

The direct limit uses pullback as the level shrinks; the inverse limit uses reduction of coefficients. Give each direct-limit group the discrete topology and the result its inverse-limit topology. For a finite extension $E/\mathbb{Q}_p$, put

$$\widetilde{H}^i(K^p, E) = \widetilde{H}^i(K^p, \mathbb{Z}_p) \otimes_{\mathbb{Z}_p} E.$$

One may compute the same integral module using singular cohomology of complex modular curves after fixing $\overline{\mathbb{Q}} \hookrightarrow \mathbb{C}$.

Right translations in the tower induce a continuous $\mathrm{GL}_2(\mathbb{Q}_p)$-action. The Galois actions on geometric étale cohomology induce a commuting continuous $G_{\mathbb{Q}}$-action, and Hecke correspondences away from p act compatibly. In degree one, the rational coefficient space has its standard admissible Banach representation structure.

Replacing Y_K by the compactification X_K defines the **compactified completed cohomology**; replacing ordinary cohomology by compactly supported cohomology defines another variant. These are distinct choices. For a tower of locally symmetric spaces the same double-limit construction defines completed cohomology with an action of the local group, without in general supplying a Galois action. The order of the two limits is part of the definition.

78. Good reduction of a subvariety of an abelian variety

Context. This definition belongs to integral models of embedded varieties over number fields.

Let K be a number field, let v be a finite place, and suppose that the abelian variety A/K has good reduction at v. Let $\mathcal{A}/O_{K_v}$ be its abelian-scheme model, uniquely determined together with its identification of the generic fibre with A_{K_v}. Let $Z \subset A$ be a smooth closed subvariety.

The **scheme-theoretic closure** $\mathcal{Z} \subset \mathcal{A}$ of Z_{K_v} is the closed subscheme whose ideal, on an affine open $\operatorname{Spec} B \subset \mathcal{A}$, is the inverse image in B of the ideal defining Z_{K_v} in $B \otimes_{O_{K_v}} K_v$. It is flat and proper over O_{K_v}.

Definition. The embedded subvariety $Z \subset A$ has **good reduction at** v if

$$\mathcal{Z} \longrightarrow \operatorname{Spec} O_{K_v}$$

is smooth. Since the generic fibre is smooth and the closure is flat, this is equivalent to smoothness of its special fibre over the residue field.

If A extends to an abelian scheme $\mathcal{A}/O_{K,S}$, then Z has **good reduction outside** S if its scheme-theoretic closure in $\mathcal{A}$ is smooth over $O_{K,S}$. Equivalently it has good reduction at every finite $v \notin S$.

For a hypersurface, meaning an effective Cartier divisor $H \subset A$, good reduction means that the closure is a smooth relative divisor. This condition concerns the embedding in A: it is stronger data than the existence of some smooth proper model of the abstract variety Z.

79. Minimal parabolic subgroup and Langlands decomposition in real rank one

Context. This definition belongs to the restricted-root structure of real semisimple Lie groups.

Let G be a connected semisimple real Lie group with finite centre and real rank one. Fix a Cartan involution, giving $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ and a maximal compact subgroup K. Choose a maximal abelian subspace $\mathfrak{a} \subset \mathfrak{p}$; thus $\dim \mathfrak{a} = 1$. Choose a positive system Σ^+ of restricted roots and put

$$\mathfrak{u} = \bigoplus_{\alpha \in \Sigma^+} \mathfrak{g}_\alpha, \quad A = \exp(\mathfrak{a}), \quad U = \exp(\mathfrak{u}), \quad M = Z_K(\mathfrak{a}).$$

Here $\mathfrak{g}_\alpha = \{X : [H, X] = \alpha(H)X \text{ for all } H \in \mathfrak{a}\}$.

Definition. The subgroup $P = MAU$ is the **minimal parabolic subgroup** determined by these choices. A minimal parabolic of G means any conjugate of such a subgroup. Its **Langlands decomposition** is

$$P = MAU \simeq (M \times A) \ltimes U.$$

The factor M is compact, $A \simeq \mathbb{R}_{>0}$ is the connected split factor, and U is the connected unipotent radical. Multiplication $M \times A \times U \to P$ is a diffeomorphism. The Levi factor is $MA = Z_G(A)$. The group U depends on P, while the displayed Levi decomposition also uses the Cartan involution.

In rank one, every proper parabolic subgroup is minimal. For $G = \mathrm{SO}_0(n,1)$, a minimal parabolic is the stabilizer of an isotropic line in $\mathbb{R}^{n,1}$, or equivalently of a point on the boundary of hyperbolic n-space. In this case $M \simeq \mathrm{SO}(n-1)$ and $U \simeq (\mathbb{R}^{n-1}, +)$. The notation N is often used in place of U.

80. Néron–Tate canonical height on an abelian variety

Context. This definition follows the Weil height machine and the theorem of the cube.

Let A/K be an abelian variety over a number field, and let L be a symmetric ample line bundle, so $[-1]^*L \simeq L$. Choose an absolute logarithmic Weil height $h_L : A(\overline{K}) \to \mathbb{R}$ attached to L, normalized by division by the degree of a field of definition over $\mathbb{Q}$.

The identity $[2]^*L \simeq L^{\otimes 4}$ implies that $h_L([2]x) - 4h_L(x)$ is bounded uniformly in x. Consequently the following limit exists and does not depend on the representative h_L.

Definition. The **Néron–Tate canonical height attached to** L is

$$\hat{h}_L(x) = \lim_{n \to \infty} 4^{-n} h_L([2^n]x), \qquad x \in A(\overline{K}).$$

Equivalently, it is the unique function with $\hat{h}_L - h_L$ bounded and $\hat{h}_L([2]x) = 4\hat{h}_L(x)$.

It satisfies $\hat{h}_L([m]x) = m^2 \hat{h}_L(x)$ and the parallelogram identity. Our normalization of the associated pairing is

$$\langle x, y \rangle_L = \tfrac{1}{2}\big(\hat{h}_L(x+y) - \hat{h}_L(x) - \hat{h}_L(y)\big),$$

so $\langle x, x \rangle_L = \hat{h}_L(x)$. For ample L, the height is nonnegative and vanishes exactly at torsion points over a number field.

The same construction applies over $K = k(B)$ for a regular projective curve B/k, using function-field Weil heights: on a finite extension K'/K, divisor degrees are measured over k and divided by $[K' : K]$. In that setting height zero need not imply torsion. For a Jacobian with its principal polarization, the usual canonical height is that associated to a symmetric theta line bundle; equivalently it is half the height of the line bundle representing twice that polarization.

81. Prismatic site and prismatic cohomology

Context. This definition follows the theory of bounded prisms and complete flatness.

Fix a prime p, a bounded prism (A, I), and a p-adic formal scheme X over $\mathrm{Spf}(A/I)$. A prism is a δ-ring A with a Cartier-divisor ideal I, derived (p, I)-complete, such that $p \in I + \phi_A(I)A$; bounded means that the p-power torsion in A/I is killed by a fixed power of p.

Definition of the site. An object of the **relative prismatic site** $(X/A)_\Delta$ consists of a bounded prism (B, J), a map of prisms $(A, I) \to (B, J)$, and an A/I-morphism

$$\mathrm{Spf}(B/J) \longrightarrow X.$$

Maps of prisms force $J = IB$. A morphism of site objects from (C, IC) to (B, IB) is a δ-A-algebra map $B \to C$ compatible with the displayed morphisms to X; thus the arrows are opposite to ring maps.

The topology is generated by maps $B \to C$ that are (p, IB)**-completely faithfully flat**: the derived tensor product

$$C \otimes^{\mathbf{L}}_B B/(p, IB)$$

is concentrated in degree zero and is faithfully flat over $B/(p, IB)$.

The **structure sheaf** and its quotient are

$$O_\Delta(B,J) = B, \qquad \overline{O}_\Delta(B,J) = B/J.$$

These assignments are sheaves for this topology.

Definition of cohomology. The **relative prismatic cohomology** of X is

$$R\Gamma_\Delta(X/A) = R\Gamma((X/A)_\Delta, O_\Delta) \in D(A), \qquad H^i_\Delta(X/A) = H^i(R\Gamma_\Delta(X/A)).$$

The compatible ring Frobenius maps ϕ_B induce a ϕ_A-semilinear Frobenius on this complex; its linearization is a map $A \otimes^{\mathbf{L}}_{A,\phi_A} R\Gamma_\Delta(X/A) \to R\Gamma_\Delta(X/A)$. Smoothness of X is an additional hypothesis in the principal comparison theorems, rather than a condition needed to form the site.

82. Quasi-split unitary group of a self-dual lattice

Context. This construction belongs to the integral linear algebra underlying unitary Shimura varieties.

Let F/F^+ be a CM extension with complex conjugation $a \mapsto \bar{a}$, and let $V = F^{2n}$. Equip V with the split skew-hermitian form

$$h(x,y) = \sum_{i=1}^{n} (x_i \bar{y}_{2n+1-i} - x_{2n+1-i}\bar{y}_i).$$

Its trace

$$\psi(x,y) = \mathrm{Tr}_{F/\mathbb{Q}}\, h(x,y)$$

is a nondegenerate alternating $\mathbb{Q}$-bilinear form. An O_F-lattice $L \subset V$ is **self-dual for** ψ if

$$L = \{x \in V : \psi(x, L) \subset \mathbb{Z}\}.$$

For example, $O_F^n \oplus (\mathfrak{d}_{F/\mathbb{Q}}^{-1})^n$ is self-dual with the coordinates paired in reverse order.

Definition. The **integral unitary group of** (L, ψ) is the affine group scheme $G/\mathbb{Z}$ whose points on a commutative ring R are

$$G(R) = \{g \in \mathrm{Aut}_{O_F \otimes R}(L \otimes_{\mathbb{Z}} R) : \psi_R(gx, gy) = \psi_R(x,y) \text{ for all } x, y\}.$$

Its generic fibre is $\mathrm{Res}_{F^+/\mathbb{Q}}\, U(V,h)$: an F-linear map preserving ψ preserves h by nondegeneracy of the trace pairing. The form h has a totally isotropic subspace of dimension n, so this generic fibre is quasi-split, and

$$G(\mathbb{R}) \simeq \prod_{\tau : F^+ \hookrightarrow \mathbb{R}} U(n,n).$$

This specifies the integral model even at ramified primes; quasi-splitness here describes the generic fibre. The self-duality condition is for the trace form, not for an O_F-valued perfect skew-hermitian pairing. Allowing $\psi_R(gx, gy) = c\psi_R(x,y)$ with $c \in R^\times$ instead defines the unitary similitude group. This is the lattice convention of Caraiani–Scholze, §2.1.

83. Schwartz–Hecke algebra of an adelic reductive group

Context. This definition fixes the Schwartz test-function algebra used for spectral isolation in automorphic theory.

Let G/F be connected reductive, put $G_\infty = G(F \otimes_{\mathbb{Q}} \mathbb{R})$, and fix Haar measures. Regard G_∞ as the real points of the restriction-of-scalars algebraic group. Let $\mathcal{D}_{\mathrm{alg}}(G_\infty)$ be the algebra of differential operators generated by multiplication by regular algebraic functions and by left-invariant vector fields, with complex coefficients.

Archimedean definition. A smooth function $f : G_\infty \to \mathbb{C}$ is **Schwartz** if

$$p_D(f) = \sup_{g \in G_\infty} |Df(g)| < \infty \qquad \text{for every } D \in \mathcal{D}_{\mathrm{alg}}(G_\infty).$$

Write $\mathcal{S}(G_\infty)$ for this space with its seminorm topology. This specifies the algebraic, or Nash, Schwartz convention. For example on $\mathbb{R}^\times$ it requires boundedness of $x^m (x\,d/dx)^j f$ for every $m \in \mathbb{Z}$ and $j \geq 0$. It differs from the usual Harish–Chandra L^2-Schwartz space.

Adelic definition. Set

$$\mathcal{S}(G(\mathbb{A}_F)) = \mathcal{S}(G_\infty) \otimes_{\mathbb{C}} C_c^\infty(G(\mathbb{A}_F^\infty)),$$

where the tensor product is algebraic and the finite-adelic functions are locally constant and compactly supported. Thus elements are finite sums of products of functions on the two factors. For a compact open subgroup $K \subset G(\mathbb{A}_F^\infty)$, its **level-$K$ Schwartz–Hecke algebra** consists of functions satisfying $f(k_1 g k_2) = f(g)$ for all $k_1, k_2 \in K$.

The product is convolution,

$$(f_1 * f_2)(g) = \int_{G(\mathbb{A}_F)} f_1(gh^{-1}) f_2(h)\, dh.$$

For a unitary representation, or an admissible representation with Casselman–Wallach archimedean component, the integrated action is $\pi(f)v = \int f(g)\pi(g)v\, dg$. In the right regular representation this becomes $R(f)\varphi(x) = \int f(g)\varphi(xg)\, dg$. These definitions follow the convention of Beuzart-Plessis–Liu–Zhang–Zhu, §§2.2 and 3.1.

84. Selmer sections and localization

Context. This definition belongs to the arithmetic of the étale fundamental group of a curve.

Let Y/K be a smooth projective geometrically connected curve of genus at least two over a number field. Fix a geometric base point and write

$$1 \longrightarrow \pi_1(Y_{\overline{K}}) \longrightarrow \pi_1(Y) \xrightarrow{q} G_K \longrightarrow 1.$$

The set $\mathrm{Sec}(Y/K)$ consists of continuous homomorphisms $s : G_K \to \pi_1(Y)$ with $q \circ s = \mathrm{id}$, modulo conjugation by $\pi_1(Y_{\overline{K}})$. A point $y \in Y(K)$ induces a class s_y; denote this map by κ_K.

For each place v, choose an embedding $\overline{K} \hookrightarrow \overline{K_v}$. Base change identifies the fundamental sequence of Y_{K_v} with the pullback of the displayed sequence along $G_{K_v} \to G_K$. Restriction therefore gives

$$\mathrm{res}_v : \mathrm{Sec}(Y/K) \longrightarrow \mathrm{Sec}(Y_{K_v}/K_v).$$

On conjugacy classes this map is independent of the choices under the canonical base-point identifications.

Definition. A section class s is **Selmer** if

$$\mathrm{res}_v(s) \in \kappa_{K_v}(Y(K_v)) \quad \text{for every place } v.$$

The set of such classes is denoted $\mathrm{Sec}^{\mathrm{Sel}}(Y/K)$.

Local injectivity theorem. At every finite place v, the point-section map κ_{K_v} is injective. Consequently the **localization map** at a finite place is well-defined by

$$\mathrm{loc}_v : \mathrm{Sec}^{\mathrm{Sel}}(Y/K) \longrightarrow Y(K_v), \qquad \kappa_{K_v}(\mathrm{loc}_v(s)) = \mathrm{res}_v(s).$$

For a rational point y, this sends s_y to its image in $Y(K_v)$. At real places the section class determines a connected component rather than an individual point; at complex places the local section set is a singleton. Thus a point-valued localization is asserted here only at finite places, while the Selmer condition includes every place.

85. Stable Faltings height of an abelian variety

Context. This definition belongs to Arakelov degrees of Hodge bundles.

Let A/K be an abelian variety of dimension g over a number field. Choose a finite extension K'/K over which A has semistable reduction everywhere, and let $\mathcal{A}/O_{K'}$ be its Néron model with identity section e. Such an extension exists by semistable reduction. Its **Hodge line** is

$$\omega_{\mathcal{A}} = \det(e^*\Omega^1_{\mathcal{A}/O_{K'}}).$$

For each embedding $\sigma : K' \hookrightarrow \mathbb{C}$, identify its complex fibre with $H^0(A_\sigma, \Omega^g)$. Equip it with the **Faltings norm**

$$\|\alpha\|^2_\sigma = \frac{i^{g^2}}{2^g} \int_{A_\sigma(\mathbb{C})} \alpha \wedge \bar{\alpha}.$$

The integral uses the complex orientation, so this expression is positive for $\alpha \neq 0$.

For a rank-one projective $O_{K'}$-module $\mathcal{L}$ with such norms, its arithmetic degree is

$$\widehat{\deg}(\overline{\mathcal{L}}) = \log \#(\mathcal{L}/O_{K'}s) - \sum_{\sigma : K' \hookrightarrow \mathbb{C}} \log \|s\|_\sigma,$$

where $0 \neq s \in \mathcal{L}$. The sum counts both embeddings at a complex place. The product formula makes the degree independent of s.

Definition. The **stable Faltings height** is

$$h_{\mathrm{Fal}}(A) = \frac{\widehat{\deg}(\omega_{\mathcal{A}}, \|\cdot\|)}{[K' : \mathbb{Q}]}.$$

Semistable base change implies that this is independent of the chosen semistable extension. It is invariant under finite extension of the original number field. The displayed norm fixes the normalization: replacing its square by c times the displayed square shifts the height by $-\frac{1}{2}\log c$.

There is also a function-field version. For $K = k(B)$, with B a regular projective integral curve, choose a finite semistable extension K'/K and its regular projective curve B'/k. If $\mathcal{A}/B'$ is the Néron model, define

$$h_{\mathrm{Fal},K}(A) = \frac{\deg_{B'/k} \det(e^*\Omega^1_{\mathcal{A}/B'})}{[K' : K]}.$$

Here closed points have degree $[k(v) : k]$, and the original field K remains the normalization base. For a smooth projective curve C, set $h_{\mathrm{Fal}}(C) = h_{\mathrm{Fal}}(\mathrm{Jac}(C))$.

86. Symmetric power of a Weil–Deligne representation

Context. This construction belongs to operations on local Langlands parameters.

Let F be a nonarchimedean local field with residue cardinality q, and let E be a characteristic-zero coefficient field. A Weil–Deligne representation on a finite-dimensional E-space V is a pair (r, N), where $r : W_F \to \mathrm{GL}(V)$ is a Weil representation with open kernel on inertia and $N \in \mathrm{End}_E(V)$ is nilpotent, satisfying

$$r(w) N r(w)^{-1} = |w| N.$$

Fix the convention $|\mathrm{Frob}_{\mathrm{geom}}| = q^{-1}$.

Definition. For $m \geq 0$, the **m-th symmetric power** is the Weil–Deligne representation on $\mathrm{Sym}^m_E V$

$$\mathrm{Sym}^m(r, N) = (\mathrm{Sym}^m r, N^{(m)}),$$

where

$$N^{(m)}(v_1 \cdots v_m) = \sum_{j=1}^{m} v_1 \cdots v_{j-1} (N v_j) v_{j+1} \cdots v_m.$$

For $m = 0$ the representation is trivial and the monodromy operator is zero. Equivalently, $N^{(m)} = d(\mathrm{Sym}^m)(N)$ is the differential of the algebraic group representation $\mathrm{Sym}^m : \mathrm{GL}(V) \to \mathrm{GL}(\mathrm{Sym}^m V)$.

The resulting operator is nilpotent and satisfies the same Weil–Deligne relation. It is essential to use this differential, rather than the polynomial map $v_1 \cdots v_m \mapsto N v_1 \cdots N v_m$.

When $\dim V = 2$ and $m = n - 1$, this gives an n-dimensional representation. If $N e_1 = 0$ and $N e_2 = e_1$, then

$$N^{(m)}(e_1^{m-j} e_2^{j}) = j e_1^{m-j+1} e_2^{j-1} \qquad (0 \leq j \leq m).$$

At archimedean places, the analogous symmetric-power operation simply composes the Weil-group parameter with Sym^m.

87. Tate–Shafarevich group of an abelian variety

Context. This definition belongs to the local-to-global arithmetic of torsors under abelian varieties.

Let A/K be an abelian variety over a number field, with algebraic closure $\overline{K}$. Its **Weil–Châtelet group** is

$$H^1(K, A) = H^1_{\mathrm{cont}}(G_K, A(\overline{K})),$$

where $A(\overline{K})$ has the discrete topology. It classifies isomorphism classes of torsors under A over K, with the trivial class represented by A itself. A torsor is trivial precisely when it has a K-rational point.

For every place v, base change of torsors gives a restriction homomorphism

$$\mathrm{res}_v : H^1(K, A) \longrightarrow H^1(K_v, A), \qquad H^1(K_v, A) = H^1_{\mathrm{cont}}(G_{K_v}, A(\overline{K_v})).$$

Definition. The **Tate–Shafarevich group** is

$$\mathrm{Sha}(A/K) = \ker\left(H^1(K, A) \longrightarrow \prod_v H^1(K_v, A) \right),$$

where the product runs over all finite and archimedean places of K.

Equivalently, its elements are the isomorphism classes of A-torsors X/K such that $X(K_v) \neq \emptyset$ at every place. A nonzero element is therefore an everywhere locally soluble torsor with no global rational point. This group is defined without any finiteness assumption; its finiteness is conjectural in general. At a complex place the local group is zero, but including that factor keeps the formulation uniform.

88. Weak functorial transfer to a general linear group

Context. This definition belongs to Langlands functoriality with prescribed unramified data and infinitesimal characters.

Let G/F be connected reductive and fix an L-representation $r : {}^L G \to \mathrm{GL}_N(\mathbb{C})$, algebraic on the complex dual group $\widehat{G}$. For simplicity use an L-group $\widehat{G} \rtimes \mathrm{Gal}(E/F)$ through which the pinned Galois action factors. Let π and Π be irreducible automorphic representations of $G(\mathbb{A}_F)$ and $\mathrm{GL}_N(\mathbb{A}_F)$.

At an unramified finite place, use Satake parameters defined by normalized parabolic induction. If the parameter of π_v is represented by $t_v \rtimes \mathrm{Frob}_v$, its **unramified transfer** is the unique irreducible unramified representation of $\mathrm{GL}_N(F_v)$ with Satake parameter conjugate to $r(t_v \rtimes \mathrm{Frob}_v)$.

At an archimedean place, identify an infinitesimal character using the Harish–Chandra parameter in $\mathfrak{t}/W$. We use the convention that a finite-dimensional highest-weight representation of highest weight λ has parameter $\lambda + \rho$. The map induced by r sends $[\lambda]$ to the unordered eigenvalues of $dr(\lambda)$, or equivalently to the values of the weights of r on λ, counted with multiplicities. At a complex place apply this to both factors of the complexified real Lie algebra.

Definition. The representation Π is a **weak functorial transfer of π along r** if:

1. There is a finite set S of finite places such that, for every $v \notin S$, the group and both representations are unramified and Π_v is the unramified transfer of π_v.
2. At every archimedean place the infinitesimal character of Π_v is the image under dr of that of π_v.

This convention imposes no condition at the exceptional finite places. It does not assert that Π is cuspidal. Some authors use “weak transfer” for condition 1 alone; both conditions are included here. For split classical groups, r is usually the standard representation of $\widehat{G}$.

89. Galois representation valued in the Mumford–Tate group

Context. This definition belongs to the comparison between absolute Hodge tensors and étale cohomology of abelian varieties.

Let A/E be an abelian variety over a number field with a fixed embedding $E \hookrightarrow \mathbb{C}$. Put

$$V_B = H^1(A(\mathbb{C}), \mathbb{Q}), \qquad G = \mathrm{MT}(V_B) \subset \mathrm{GL}(V_B),$$

using its weight-one Hodge structure. For each prime ℓ, the comparison isomorphism identifies

$$V_\ell = H^1_{\text{ét}}(A_{\overline{E}}, \mathbb{Q}_\ell) \simeq V_B \otimes_{\mathbb{Q}} \mathbb{Q}_\ell.$$

Let $\rho_\ell : G_E \to \mathrm{GL}(V_\ell)$ be the natural continuous Galois action.

Factorization theorem. Absolute Hodge theory for abelian varieties gives a finite extension E'/E such that, for every prime ℓ, the image of $G_{E'}$ under ρ_ℓ is contained in $G(\mathbb{Q}_\ell)$ under the comparison identification. One may choose E' so that a finite collection of Hodge tensors defining G has Galois-invariant étale realizations.

Definition. Over such a field of definition, the **Mumford–Tate-valued ℓ-adic representation** is the continuous homomorphism

$$\rho_\ell^G : G_{E'} \longrightarrow G(\mathbb{Q}_\ell)$$

whose composite with $G(\mathbb{Q}_\ell) \hookrightarrow \mathrm{GL}(V_\ell)$ is $\rho_\ell|_{G_{E'}}$. This specifies it uniquely.

We use H^1 without a Tate twist throughout: $V_\ell = \mathrm{Hom}_{\mathbb{Z}_\ell}(T_\ell A, \mathbb{Q}_\ell)$. In particular $T_\ell(A^\vee) \otimes \mathbb{Q}_\ell$ identifies with $V_\ell(1)$, not with V_ℓ.

At a place v of E' of good reduction with residue characteristic different from ℓ, the representation is unramified. Its **Frobenius conjugacy class** is the image of $\rho_\ell^G(\mathrm{Frob}_v)$ in the affine quotient $G/\!/G = \mathrm{Spec}(\mathbb{Q}[G]^G)$, using geometric Frobenius. This image records the semisimple geometric conjugacy class and is independent of the chosen Frobenius lift.

90. Nearly isotypic cuspidal subspace

Context. This definition belongs to the decomposition of the cuspidal automorphic spectrum.

Let G/F be connected reductive, let Z be the maximal F-split central torus, and fix a unitary character $\omega : Z(F)\backslash Z(\mathbb{A}_F) \to \mathbb{C}^\times$. Let

$$H = L^2_{\mathrm{cusp}}(G(F)\backslash G(\mathbb{A}_F), \omega)$$

be the cuspidal Hilbert space, with the norm taken modulo $Z(\mathbb{A}_F)$. It decomposes discretely as a Hilbert sum $H \simeq \widehat{\bigoplus}_\sigma m(\sigma) H_\sigma$ of irreducible unitary cuspidal representations, with finite multiplicities.

Fix an irreducible admissible representation $\pi = \bigotimes_v' \pi_v$. A cuspidal representation σ is **nearly equivalent** to π, written $\sigma \sim \pi$, if there is a finite set S of places, including all archimedean places, such that $\sigma_v \simeq \pi_v$ for every $v \notin S$.

Definition. The **π-nearly isotypic cuspidal subspace** is the closed sum of all irreducible subrepresentations nearly equivalent to π:

$$H_\pi = \widehat{\bigoplus}_{\sigma \sim \pi} m(\sigma) H_\sigma.$$

Equivalently it is the closure of their algebraic sum inside H, so it does not depend on a choice of decomposition and retains all multiplicities.

For a compact open $K \subset G(\mathbb{A}_F^\infty)$, define

$$L^2_{\mathrm{cusp}}(G(F)\backslash G(\mathbb{A}_F)/K, \omega)_\pi = (H_\pi)^K.$$

This space can be zero. Near equivalence requires agreement at almost all finite places, and adds no condition on archimedean components or their infinitesimal characters. Those can be imposed separately when defining a smaller spectral subspace.

Probability, combinatorics, dynamics & logic

91. Anosov flow

Context. This definition belongs to uniform hyperbolicity in smooth dynamics.

Let M be a compact smooth manifold without boundary, equipped with a Riemannian metric, and let $g : \mathbb{R} \times M \to M$ be a C^1 flow. Write $g^t(x) = g(t, x)$ and let $X(x) = \frac{d}{dt}\big|_{t=0} g^t(x)$ be its generating vector field.

Definition. The flow is **Anosov** if X is nowhere zero and there are continuous vector subbundles $E^s, E^u \subset TM$ such that

$$TM = E^s \oplus \mathbb{R}X \oplus E^u,$$

the splitting is invariant under Dg^t for every $t \in \mathbb{R}$, and there exist constants $C \geq 1$ and $\lambda > 0$ for which, for every $x \in M$ and $t \geq 0$,

$$\|Dg_x^t v\| \leq Ce^{-\lambda t}\|v\| \quad (v \in E_x^s), \qquad \|Dg_x^{-t} w\| \leq Ce^{-\lambda t}\|w\| \quad (w \in E_x^u).$$

Here invariance means $Dg_x^t(E_x^s) = E^s_{g^t x}$ and the analogous identity for E^u. The constants are uniform in both the base point and the tangent vector.

The bundles E^s and E^u are the **stable** and **unstable** bundles, and $\mathbb{R}X$ is the **flow direction**. The definition is independent of the chosen Riemannian metric, since metrics on a compact manifold have uniformly equivalent norms. In the usual nontrivial setting $\dim M \geq 2$, both stable and unstable bundles have positive rank; for a three-dimensional Anosov flow they are line bundles.

A flow is **transitive** if it has a dense orbit and **topologically mixing** if for every pair of nonempty open sets U, V there is T such that $g^t(U) \cap V \neq \emptyset$ for all $t \geq T$. These are additional properties, not part of the definition of Anosov.

92. Billiard ball map of a convex table

Context. This definition belongs to the geometry of convex billiards and area-preserving maps.

Let $\Omega \subset \mathbb{R}^2$ be a compact strictly convex region with C^2 boundary of positive curvature. Parametrize its boundary counterclockwise by arclength,

$$\gamma : \mathbb{R}/\ell\mathbb{Z} \longrightarrow \partial\Omega,$$

and write $\tau(s) = \gamma'(s)$ and $n(s)$ for the inward unit normal. The collision phase space is the cylinder $\mathcal{A} = (\mathbb{R}/\ell\mathbb{Z}) \times (0, \pi)$. A point (s, θ) records the outgoing unit velocity

$$u = \cos\theta\, \tau(s) + \sin\theta\, n(s).$$

Definition. The **billiard ball map** $T : \mathcal{A} \to \mathcal{A}$ is defined by $T(s, \theta) = (s', \theta')$ as follows. Travel from $\gamma(s)$ in direction u to the next boundary point $\gamma(s')$. Reflect the incoming velocity in the tangent line there:

$$u' = u - 2\langle u, n(s')\rangle n(s').$$

The angle $\theta' \in (0, \pi)$ is the unique angle with $u' = \cos\theta'\, \tau(s') + \sin\theta'\, n(s')$. Strict convexity makes the next collision unique. This is the law of equal incidence and reflection angles.

The map preserves the measure $\sin\theta\, ds\, d\theta$ and extends continuously to the closed cylinder by fixing the grazing states $\theta = 0, \pi$.

A **rotational invariant curve** is an embedded essential circle $\alpha \subset \mathcal{A}$ with $T(\alpha) = \alpha$. The natural lift $\widetilde{T}$ satisfies $s < s' < s + \ell$. For a point of the lifted invariant curve, its rotation number is

$$\rho(\alpha) = \lim_{j \to \infty} \frac{s_j - s_0}{j\ell}, \qquad (s_j, \theta_j) = \widetilde{T}^j(s_0, \theta_0).$$

This is independent of the starting point and fixes the normalization of rotation number used for convex billiards.

93. Billiard flow in a polygon

Context. This definition belongs to the ergodic theory of polygonal billiards.

Let $P \subset \mathbb{R}^2$ be the compact region bounded by a simple closed polygonal curve. A particle state consists of a position $x \in P$ and a unit velocity $v \in S^1$. At a point in the interior of a side with inward normal n, identify the incoming and outgoing states

$$(x, v) \sim (x, v - 2\langle v, n\rangle n).$$

Take the resulting quotient phase space. Remove states at vertices, tangent boundary states, and states whose trajectory hits a vertex in positive or negative time. The removed set has Liouville measure zero; the remaining set is invariant under the regular billiard evolution.

Definition. The **billiard flow** $(F^t)_{t \in \mathbb{R}}$ moves the particle at unit speed along straight segments in the interior,

$$(x, v) \longmapsto (x + tv, v),$$

until the first boundary collision, then replaces v by its reflection in that side and continues. Negative time reverses the same rule. On the regular quotient phase space this gives a measurable flow, satisfying $F^{t+s} = F^t \circ F^s$ for all real s, t.

Its invariant **Liouville probability measure** is induced by

$$d\mu_P(x, \theta) = \frac{dx\, d\theta}{2\pi \operatorname{Area}(P)},$$

where dx is planar Lebesgue measure and $d\theta$ is arclength on the unit circle. Boundary states have measure zero, so this is also the normalized product measure on $P^\circ \times S^1$.

Thus statements about ergodicity or weak mixing of a polygonal billiard refer to this flow and this measure. The flow is **weakly mixing** if the product flow $F^t \times F^t$ is ergodic on the product probability space. It is distinct from the collision map, which records only successive impacts with the boundary.

94. Borel complete class of countable structures

Context. This definition belongs to classification problems in invariant descriptive set theory.

For equivalence relations E on a standard Borel space X and F on a standard Borel space Y, a **Borel reduction** from E to F is a Borel map $b : X \to Y$ such that

$$x \mathrel{E} x' \quad \Longleftrightarrow \quad b(x) \mathrel{F} b(x') \qquad (x, x' \in X).$$

Write $E \leq_B F$ when such a map exists. The reduction need not be injective on individual points, but it induces an injection on equivalence classes.

Let L be a countable first-order language and let $\mathcal{K} \subset \operatorname{Mod}_L(\omega)$ be an isomorphism-invariant Borel set of structures with universe ω, as in Definition 98.

Definition. The class $\mathcal{K}$ is **Borel complete** if, for every countable language L' and every isomorphism-invariant Borel set $C \subset \mathrm{Mod}_{L'}(\omega)$, there is a Borel map $b : C \to \mathcal{K}$ satisfying

$$A \cong B \iff b(A) \cong b(B) \qquad (A, B \in C).$$

In symbols, $\cong_C \leq_B \cong_{\mathcal{K}}$ for every such C.

The Friedman–Stanley coding theorem says that the class of countable simple graphs is Borel complete. Therefore it suffices to construct a Borel reduction from graph isomorphism to isomorphism on $\mathcal{K}$. A continuous reduction in the respective logic topologies establishes the stronger continuity assertion as well. The comparison here is with isomorphism relations of countable structures; completeness among all analytic equivalence relations is a different notion.

95. Dynamical degree of a dominant rational self-map

Context. This definition belongs to the growth of intersection numbers in algebraic dynamics.

Let X be a normal integral projective variety of dimension d over a field of characteristic zero, and let $f : X \dashrightarrow X$ be dominant. Fix an ample Cartier divisor H. Let $\Gamma_f \subset X \times X$ be the closure of the graph on the domain of definition of f, with projections p_1, p_2.

For $0 \leq k \leq d$, define its k**-th degree with respect to** H by

$$\deg_{k,H}(f) = \left((p_2^*H)^k \cdot (p_1^*H)^{d-k}\right)_{\Gamma_f}.$$

This is an intersection number of Cartier divisors on the graph. It can equivalently be computed after pulling back to a resolution of the graph. In particular this formula does not presume that pullback by a rational map is functorial under composition.

Existence and independence theorem. The sequence $\deg_{k,H}(f^n)^{1/n}$ has a finite limit, independent of the ample divisor H. One ingredient is the uniform estimate

$$\deg_{k,H}(a \circ b) \leq C \deg_{k,H}(a) \deg_{k,H}(b)$$

for dominant rational maps a, b, with C depending only on X, H, k.

Definition. The k**-th dynamical degree** of f is

$$\lambda_k(f) = \lim_{n \to \infty} \deg_{k,H}(f^n)^{1/n}.$$

Here the degree is recomputed from the graph of each iterate. The endpoint values are $\lambda_0(f) = 1$ and $\lambda_d(f) = [K(X) : f^*K(X)]$. The dynamical degrees are invariant under birational conjugacy.

For a dominant polynomial self-map of affine space, its dynamical degrees mean those of the rational extension to projective space. For $X = \mathbb{P}^d$, $\lambda_1(f) = \lim_n (\deg f^n)^{1/n}$ with the ordinary homogeneous degree.

96. Equilibrium state of a Hölder potential

Context. This definition belongs to thermodynamic formalism for dynamical systems.

Let $g = (g^t)_{t \in \mathbb{R}}$ be a continuous flow on a compact metric space X, with $h_{\mathrm{top}}(g^1) < \infty$. Write $\mathcal{M}(g)$ for the Borel probability measures invariant under **every** g^t. Let $F : X \to \mathbb{R}$ be Hölder continuous, meaning that $|F(x) - F(y)| \leq C d(x,y)^\alpha$ for some $C < \infty$ and $0 < \alpha \leq 1$.

The **pressure** of F is

$$P_g(F) = \sup_{\nu \in \mathcal{M}(g)} \left(h_\nu(g^1) + \int_X F \, d\nu \right),$$

where $h_\nu(g^1)$ is Kolmogorov–Sinai entropy (Definition 97). The variational principle identifies this quantity with topological pressure defined using separated orbit segments and weights $\exp(\int_0^t F(g^s x)\,ds)$.

Definition. An **equilibrium state** or **equilibrium measure** for g and F is a measure $\nu \in \mathcal{M}(g)$ such that

$$h_\nu(g^1) + \int_X F\,d\nu = P_g(F).$$

Thus it attains the supremum defining pressure. The definition itself imposes neither existence nor uniqueness. For a transitive C^2 Anosov flow on a compact manifold, the Bowen–Ruelle theorem provides a unique equilibrium state for each Hölder potential, conventionally denoted $\nu_{g,F}$.

For a continuous map $T : X \to X$ and potential φ, the corresponding definition maximizes $h_\mu(T) + \int \varphi\,d\mu$ over T-invariant probability measures. To express flow pressure through the time-one map, the potential is $F_1(x) = \int_0^1 F(g^s x)\,ds$, and $P_g(F) = P_{g^1}(F_1)$. A measure invariant only under g^1 need not be flow-invariant.

97. Measure of maximal entropy

Context. This definition belongs to the variational principle for entropy.

Let $f : X \to X$ be continuous on a compact metric space. For an f-invariant Borel probability measure μ and a finite measurable partition $\mathcal{P}$, put

$$H_\mu(\mathcal{P}) = -\sum_{A \in \mathcal{P}} \mu(A) \log \mu(A),$$

with $0 \log 0 = 0$. The **Kolmogorov–Sinai entropy** is

$$h_\mu(f) = \sup_{\mathcal{P}} \lim_{n \to \infty} \frac{1}{n} H_\mu\left(\bigvee_{j=0}^{n-1} f^{-j}\mathcal{P}\right).$$

The limit for each partition exists by subadditivity; the supremum ranges over all finite measurable partitions. Natural logarithms are used for both measure-theoretic and topological entropy.

Variational principle. If $\mathcal{M}(f)$ denotes the invariant Borel probability measures, then

$$h_{\mathrm{top}}(f) = \sup_{\nu \in \mathcal{M}(f)} h_\nu(f).$$

Definition. A **measure of maximal entropy** is a measure $\mu \in \mathcal{M}(f)$ satisfying

$$h_\mu(f) = h_{\mathrm{top}}(f).$$

An **ergodic measure of maximal entropy** additionally satisfies $\mu(A) \in \{0, 1\}$ for every measurable set A invariant modulo μ under f.

The definition does not presume that the supremum is attained or that the maximizing measure is unique. In the usual finite-entropy setting, upper semicontinuity of $\nu \mapsto h_\nu(f)$ ensures existence because $\mathcal{M}(f)$ is weakly compact. A measure of maximal entropy is exactly an equilibrium state for the zero potential. When the entropy is finite, almost every ergodic component of a maximizing measure also maximizes entropy; consequently counting ergodic maximizing measures is different from counting all maximizing measures.

98. Polish space of countable structures

Context. This definition belongs to the topology and descriptive set theory of spaces of first-order structures.

Let L be a countable first-order language with relation symbols R of finite arity n_R and function symbols f of finite arity m_f. Constants are treated as zero-ary function symbols. Put $\omega = \{0, 1, 2, \dots\}$, and give both $2 = \{0, 1\}$ and ω the discrete topology.

Definition. The **space of L-structures on ω** is

$$\mathrm{Mod}_L(\omega) = \prod_{R} 2^{\omega^{n_R}} \times \prod_{f} \omega^{\omega^{m_f}},$$

with its product topology. A point specifies the characteristic function of each interpreted relation and the value of each interpreted function on every tuple. The topology is called the **logic topology**: a basic open set specifies finitely many such values. It is Polish, and its Borel sets give the associated standard Borel space.

The permutation group S_∞ acts continuously by relabelling the universe:

$$R^{g\cdot M}(\bar a) \iff R^M(g^{-1}\bar a), \qquad f^{g\cdot M}(\bar a) = g\big(f^M(g^{-1}\bar a)\big).$$

Two structures are isomorphic precisely when they lie in the same orbit.

A **Borel class of countable structures** is an isomorphism-invariant Borel subset of $\mathrm{Mod}_L(\omega)$, with its induced Borel structure. For example, simple graphs are obtained by imposing symmetry and irreflexivity on one binary relation, and torsion-free abelian groups by imposing the group axioms, commutativity, and $nx = 0 \Rightarrow x = 0$ for each integer $n \geq 1$.

The product above is the full space of structures for languages with functions as well as relations. Coding functions instead by their graphs requires an additional total-function condition. Structures here have universe exactly ω, hence are countably infinite; finite universes can be coded in separate spaces.

99. Scaling limit of lattice ϕ^4 fields: a reachable random field

Context. This definition belongs to the removal of lattice and volume cutoffs in Euclidean field theory.

Put $V_R = [-R, R]^d \cap \mathbb{Z}^d$. For $J \geq 0$, $\lambda > 0$, and $b \in \mathbb{R}$, the finite-volume **lattice ϕ^4 measure with free boundary** is

$$d\mu_{J,\lambda,b,R}(\phi) = Z^{-1} \exp\left(J \sum_{\substack{\{x,y\}\subset V_R \\ |x-y|_1 = 1}} \phi_x \phi_y - \sum_{x \in V_R} (\lambda \phi_x^4 - b\phi_x^2) \right) \prod_{x \in V_R} d\phi_x.$$

Each unoriented edge is counted once, and Z normalizes total mass to one. The Ising variant replaces the single-site factor by $\tfrac{1}{2}(\delta_{-1} + \delta_1)$.

For $1 \leq L \leq R$ set

$$\Sigma_{L,R} = \mathbb{E}_\mu\left[\left(\sum_{x \in V_L} \phi_x \right)^2 \right], \qquad T_{f,L,R} = \Sigma_{L,R}^{-1/2} \sum_{x \in V_R} f(x/L)\phi_x,$$

where $f \in C_c(\mathbb{R}^d; \mathbb{R})$. The variance is computed in the same measure as the observable. The continuum lattice spacing is L^{-1}.

A generalized random field is considered through a linear family of real random variables $\Phi(f)$ indexed by these test functions, up to equality of all finite-dimensional distributions; $\int f(x)\Phi(x)\,dx$ is notation for $\Phi(f)$.

Definition. The field Φ is **reachable** by these lattice constructions if there are scales $L_n \to \infty$ and, for each n, volumes $R_{n,m}$ with $R_{n,m}/L_n \to \infty$ as $m \to \infty$, together with choices of the model parameters, such that for every $r \geq 1$ and $f_1, \dots, f_r \in C_c(\mathbb{R}^d; \mathbb{R})$,

$$\lim_{n\to\infty} \lim_{m\to\infty} \mathrm{Law}(T_{f_1,L_n,R_{n,m}}, \dots, T_{f_r,L_n,R_{n,m}}) = \mathrm{Law}(\Phi(f_1), \dots, \Phi(f_r))$$

weakly as probability measures on $\mathbb{R}^r$. Each inner limit must exist, and one common choice of sequences must work for all finite families. The parameters may vary along the chosen approximation, including during removal of the volume cutoff.

This is the ordered double-limit convention $\lim_{L\to\infty} \lim_{R/L\to\infty}$ of Aizenman–Duminil-Copin, Definition 1.1. Reachability asserts convergence of smeared observables; it does not assert pointwise convergence of fields or convergence of their moments.

100. Translation-invariant ergodic Gibbs measure on lozenge tilings

Context. This definition belongs to infinite-volume random tilings and their slopes.

Let $\mathbb{T}$ have vertex set $\mathbb{Z}^2$ and edges in directions $\pm e_1, \pm e_2, \pm(e_1 + e_2)$. A **lozenge tiling** pairs every elementary triangular face with one adjacent face. Equivalently it is a set M of lattice edges such that every triangular face contains exactly one edge of M; this edge is internal to the paired lozenge. Let Ω be the resulting closed subspace of $\{0,1\}^{E(\mathbb{T})}$, with its Borel structure.

For a finite collection D of triangular faces, let E_D be the edges whose two adjacent faces lie in D. Given M, let $C_D(M)$ be the finite set of tilings agreeing with M on every edge outside E_D, and let $\mathcal{F}_{D^c}$ be the sigma-algebra recording those exterior edges.

Gibbs and ergodicity conditions. A probability measure μ on Ω is **Gibbs** if, for every such D and bounded Borel function F,

$$\mathbb{E}_\mu[F \mid \mathcal{F}_{D^c}](M) = \frac{1}{|C_D(M)|} \sum_{M' \in C_D(M)} F(M') \quad \text{for } \mu\text{-almost every } M.$$

It is **translation-invariant** if translating by any $w \in \mathbb{Z}^2$ preserves μ, and **ergodic** for translations if every translation-invariant measurable event has probability zero or one.

The normalized **height function** $H_M : \mathbb{Z}^2 \to \mathbb{Z}$ is determined by $H_M(0) = 0$ and increments

$$H_M(u + e_i) - H_M(u) = \mathbf{1}_{\{u,u+e_i\} \in M} \quad (i = 1, 2),$$

$$H_M(u + e_1 + e_2) - H_M(u) = 1 - \mathbf{1}_{\{u,u+e_1+e_2\} \in M}.$$

The matching condition makes these increments consistent around each triangle. For translation-invariant μ, its **slope** is

$$(s, t) = (\mathbb{E}_\mu[H_M(e_1)], \mathbb{E}_\mu[H_M(e_2)]).$$

The three tile densities are $s, t, 1 - s - t$.

Existence and uniqueness theorem; definition. For every interior slope $s > 0$, $t > 0$, $s + t < 1$, there is a unique translation-invariant ergodic Gibbs measure with that slope. Denote it by $\mu_{s,t}$. This notation uses the preceding height normalization; the interior-slope restriction is part of the theorem. These are the conventions for the infinite-volume measures in Aggarwal, §1.3.